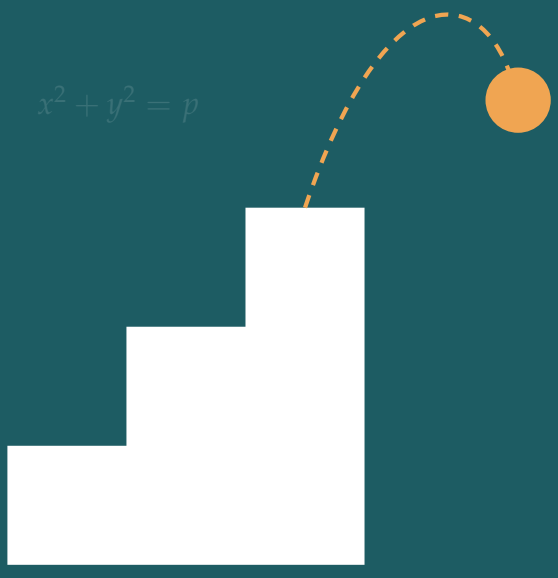
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP II AND III

2023 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP II AND III

2023 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2023 Worked Solutions

Worked solutions to all 24 questions of the 2023 STEP papers (STEP II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

First published online November 2026.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP II 2023, Question 1

Pure • Integration by Substitution

STEP II 2023, Question 1

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

The substitution $x = 1/t$ swaps $\frac{1}{2}$ and 2, so it maps the interval $[\frac{1}{2}, 2]$ onto itself (backwards). An integral over this interval can therefore be rewritten as a *different-looking* integral over the *same* interval. Sometimes the new integrand is easy to integrate directly; sometimes it is a partner of the old one, so that adding the two (or comparing them) does the work for us.

Solution

Part (i)

Put $x = 1/t$, so $\mathrm{d}x = -\frac{1}{t^2}\,\mathrm{d}t$. Since $a > 0$, every x in $[a, b]$ is positive, and so is the corresponding t. Then

$$1 + x^2 = \frac{t^2 + 1}{t^2} \quad \Longrightarrow \quad (1 + x^2)^{3/2} = \frac{(1 + t^2)^{3/2}}{t^3},$$

where we used $\sqrt{t^2} = t$ because $t > 0$. The limits $x = a$ and $x = b$ become $t = a^{-1}$ and $t = b^{-1}$. Hence

$$\int_a^b \frac{1}{(1 + x^2)^{3/2}}\,\mathrm{d}x = \int_{a^{-1}}^{b^{-1}} \frac{t^3}{(1 + t^2)^{3/2}} \cdot \left(-\frac{1}{t^2}\right)\mathrm{d}t = \int_{a^{-1}}^{b^{-1}} \frac{-t}{(1 + t^2)^{3/2}}\,\mathrm{d}t,$$

as required.

Part (ii)

(a) With $a = \frac{1}{2}$ and $b = 2$, the new limits are 2 and $\frac{1}{2}$. Swapping them absorbs the minus sign:

$$\int_{1/2}^{2} \frac{1}{(1 + x^2)^{3/2}}\,\mathrm{d}x = \int_{1/2}^{2} \frac{t}{(1 + t^2)^{3/2}}\,\mathrm{d}t.$$

The new integrand is (a multiple of) the derivative of a power of $1+t^2$: since $\frac{\mathrm{d}}{\mathrm{d}t}(1+t^2)^{-1/2} = -t(1+t^2)^{-3/2}$,

$$\int_{1/2}^{2} \frac{t}{(1+t^2)^{3/2}}\,\mathrm{d}t = \Big[-(1+t^2)^{-1/2}\Big]_{1/2}^{2} = -\frac{1}{\sqrt{5}} + \frac{1}{\sqrt{5/4}} = -\frac{1}{\sqrt{5}} + \frac{2}{\sqrt{5}} = \frac{1}{\sqrt{5}}.$$

(b) The integrand is even, so the integral over $[-2,2]$ is twice the integral over $[0,2]$. Split this as $[0,\frac{1}{2}] \cup [\frac{1}{2},2]$; the second piece is part (a).

? **Ask yourself** Part (i) needs $a > 0$. How can we handle the piece starting at 0?

Work on $[a,\frac{1}{2}]$ with $0 < a < \frac{1}{2}$, where part (i) does apply, and then let $a \to 0^+$. Part (i) with $b = \frac{1}{2}$ gives

$$\int_{a}^{1/2} \frac{1}{(1+x^2)^{3/2}}\,\mathrm{d}x = \int_{2}^{1/a} \frac{t}{(1+t^2)^{3/2}}\,\mathrm{d}t = \frac{1}{\sqrt{5}} - \frac{1}{\sqrt{1+a^{-2}}}.$$

As $a \to 0^+$, $a^{-2} \to \infty$ and the last term tends to 0. The left-hand side tends to the integral over $[0,\frac{1}{2}]$, since the integrand is continuous there. So that piece is $\frac{1}{\sqrt{5}}$ too, and

$$\int_{-2}^{2} \frac{1}{(1+x^2)^{3/2}}\,\mathrm{d}x = 2\Big(\frac{1}{\sqrt{5}} + \frac{1}{\sqrt{5}}\Big) = \frac{4}{\sqrt{5}}.$$

Part (iii)

(a) Write $J = \int_{1/2}^{2} \frac{1}{(1+x^2)^2}\,\mathrm{d}x$ and apply $x = 1/t$ again. Now $(1+x^2)^2 = (1+t^2)^2/t^4$, so

$$J = \int_{2}^{1/2} \frac{t^4}{(1+t^2)^2} \cdot \Big(-\frac{1}{t^2}\Big)\mathrm{d}t = \int_{1/2}^{2} \frac{t^2}{(1+t^2)^2}\,\mathrm{d}t,$$

which, after renaming the dummy variable t as x, is the first equality. Call this second integral K, so $J = K$. Adding the two integrands,

$$J + K = \int_{1/2}^{2} \frac{1+x^2}{(1+x^2)^2}\,\mathrm{d}x = \int_{1/2}^{2} \frac{1}{1+x^2}\,\mathrm{d}x,$$

and since $J = K$ this gives $J = \frac{1}{2}\int_{1/2}^{2} \frac{1}{1+x^2}\,\mathrm{d}x$, completing the chain of equalities.

Hence

$$J = \tfrac{1}{2}(\arctan 2 - \arctan \tfrac{1}{2}).$$

This can be tidied in either of two ways. Since $\arctan 2 + \arctan \frac{1}{2} = \frac{\pi}{2}$ (complementary angles in a right-angled triangle with legs 1 and 2), $J = \frac{\pi}{4} - \arctan \frac{1}{2}$. Or, by the subtraction formula for tan, $\tan(\arctan 2 - \arctan \frac{1}{2}) = \dfrac{2 - \frac{1}{2}}{1+1} = \frac{3}{4}$, and the difference lies in $(0, \frac{\pi}{2})$, so

$$\int_{1/2}^{2} \frac{1}{(1+x^2)^2}\,\mathrm{d}x = \tfrac{1}{2}\arctan\tfrac{3}{4} \approx 0.3218.$$

(b) Let L be the required integral. Before reaching for a hard antiderivative, try $x = 1/t$ once more. The integrand becomes

$$\frac{1-\frac{1}{t}}{\frac{1}{t}\sqrt{\frac{1+t^2}{t^2}}} = \frac{\frac{t-1}{t}}{\frac{\sqrt{1+t^2}}{t^2}} = \frac{t(t-1)}{\sqrt{1+t^2}},$$

using $t > 0$ again. Multiplying by $\mathrm{d}x = -\frac{1}{t^2}\,\mathrm{d}t$ and noting that the limits run from 2 to $\frac{1}{2}$,

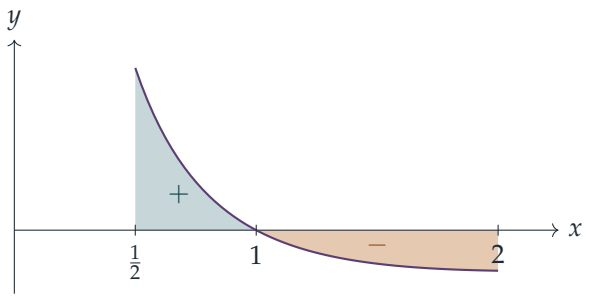
$$L = \int_2^{1/2} \frac{-(t-1)}{t\sqrt{1+t^2}}\,\mathrm{d}t = \int_{1/2}^{2} \frac{t-1}{t\sqrt{1+t^2}}\,\mathrm{d}t = -\int_{1/2}^{2} \frac{1-t}{t\sqrt{1+t^2}}\,\mathrm{d}t = -L.$$

? **Ask yourself** The substitution has turned L into $-L$. Is that a contradiction?

No: it just tells us the value. From $L = -L$ we get $2L = 0$, so

$$\int_{1/2}^{2} \frac{1-x}{x(1+x^2)^{1/2}}\,\mathrm{d}x = 0.$$

The picture agrees: the positive area over $[\frac{1}{2}, 1]$ exactly cancels the negative area over $[1, 2]$, because $x \mapsto 1/x$ swaps those two pieces and, by the working above, carries the area of one onto minus the area of the other.



Checks

- An antiderivative of $(1+x^2)^{-3/2}$ is $x(1+x^2)^{-1/2}$ (differentiate to confirm). It gives $\frac{2}{\sqrt{5}} - \frac{1}{\sqrt{5}} = \frac{1}{\sqrt{5}} \approx 0.4472$ for (ii)(a) and $\frac{4}{\sqrt{5}} \approx 1.7889$ for (ii)(b), matching our answers.
- In (iii)(a), $\frac{1}{2}\arctan\frac{3}{4} \approx 0.32175$ and $\frac{\pi}{4} - \arctan\frac{1}{2} \approx 0.32175$ agree. Numerical quadrature of all three integrals in the chain gives the same value.
- In (iii)(b) the integrand is positive for $x < 1$ and negative for $x > 1$, so a zero answer is plausible. Numerical quadrature gives 0 to within rounding error.

Common slips

- Losing the minus sign from $\mathrm{d}x = -t^{-2}\,\mathrm{d}t$, or forgetting that it is cancelled by swapping the limits.

- In (ii)(b), applying part (i) with $a = 0$, which would need the limit $t = 1/0$. Use a limit, or simply the antiderivative.
- Writing $\sqrt{(1+t^2)/t^2} = \sqrt{1+t^2}/t$ without noting that this needs $t > 0$.

Beyond STEP

The map $x \mapsto 1/x$ is a symmetry of the interval $[\frac{1}{2}, 2]$, and exploiting the symmetries of an integral is a recurring trick, as in $\int_0^{\pi/2} \ln \sin x \, \mathrm{d}x$ via $x \mapsto \frac{\pi}{2} - x$. Part (ii)(b) is a first taste of an improper integral: a substitution sent a finite endpoint to infinity, and we had to take a limit carefully. Abbott's *Understanding Analysis* treats limits of integrals rigorously, and Siklos's *Advanced Problems in Mathematics* has more STEP-style practice with integration.

WORKED SOLUTION

STEP II 2023, Question 2

Pure • Trigonometric Substitution

STEP II 2023, Question 2

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

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The key idea

Each right-hand side is a familiar multiple-angle formula in disguise: $\frac{2t}{1-t^2}$ is $\tan 2\alpha$ when $t = \tan\alpha$, $\frac{3t-t^3}{1-3t^2}$ is $\tan 3\alpha$, and $2t^2 - 1$ is $\cos 2\alpha$ when $t = \cos\alpha$. Substituting turns three coupled equations into a single trigonometric equation in one angle, which is easy to solve and, crucially, easy to *count*. The only care needed is to make sure the substitution captures every solution exactly once.

Solution

Part (i)

Every real number is $\tan\alpha$ for exactly one α in $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$, so we lose nothing by writing $x = \tan\alpha$ with α in that interval. For the first equation to make sense we need $x \neq \pm 1$, which says $\alpha \neq \pm\frac{1}{4}\pi$, so 2α is not an odd multiple of $\frac{1}{2}\pi$ and the double-angle formula applies:

$$y = \frac{2\tan\alpha}{1-\tan^2\alpha} = \tan 2\alpha.$$

Exactly the same step, applied to $y = \tan 2\alpha$ (with $y \neq \pm 1$ since z is defined), gives $z = \tan 4\alpha$, and then applied to z (with $z \neq \pm 1$) gives $x = \tan 8\alpha$. Since also $x = \tan\alpha$,

$$\tan\alpha = \tan 8\alpha,$$

as required. Two angles have the same tangent exactly when they differ by a multiple of π, so $8\alpha = \alpha + k\pi$ for some integer k, i.e.

$$\alpha = \frac{k\pi}{7}.$$

The condition $-\frac{1}{2}\pi < \alpha < \frac{1}{2}\pi$ means $-\frac{7}{2} < k < \frac{7}{2}$, so $k = -3, -2, \ldots, 3$.

Conversely, each of these seven values gives a genuine solution: $2\alpha = \frac{2k\pi}{7}$ and $4\alpha = \frac{4k\pi}{7}$ are never odd multiples of $\frac{1}{2}\pi$ (that would need $4k$ or $8k$ to equal 7 times an odd number,

so $7 \mid k$, forcing $k = 0$, which gives 0, not an odd multiple), so none of x, y, z equals ± 1, every denominator is non-zero, and the chain above runs backwards. Different k give different α in $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$, hence different x. So there are exactly seven solutions.

To quote them in the requested form, reduce 2α and 4α by multiples of π into $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$, using $\tan(\theta - \pi) = \tan\theta$. For example $\tan\frac{4\pi}{7} = \tan(-\frac{3\pi}{7})$ and $\tan\frac{8\pi}{7} = \tan\frac{\pi}{7}$. Writing $t_m = \tan\frac{m\pi}{7}$, the solutions (x, y, z) are:

k	(x, y, z)
0	$(0, 0, 0)$
1	$(t_1,\ t_2,\ t_{-3})$
2	$(t_2,\ t_{-3},\ t_1)$
3	$(t_3,\ t_{-1},\ t_{-2})$

k	(x, y, z)
-1	$(t_{-1},\ t_{-2},\ t_3)$
-2	$(t_{-2},\ t_3,\ t_{-1})$
-3	$(t_{-3},\ t_1,\ t_2)$

Notice that apart from $(0, 0, 0)$ they come as the cyclic rearrangements of (t_1, t_2, t_{-3}) and of (t_{-1}, t_{-2}, t_3), as they must, since the system is unchanged by $x \to y \to z \to x$.

Part (ii)

First the triple-angle formula: with $t = \tan\alpha$,

$$\tan 3\alpha = \frac{\tan 2\alpha + \tan\alpha}{1 - \tan 2\alpha \tan\alpha} = \frac{\frac{2t}{1-t^2} + t}{1 - \frac{2t^2}{1-t^2}} = \frac{2t + t - t^3}{1 - t^2 - 2t^2} = \frac{3t - t^3}{1 - 3t^2}.$$

As before, put $x = \tan\alpha$ with $-\frac{1}{2}\pi < \alpha < \frac{1}{2}\pi$. The equations require $1 - 3x^2$, $1 - 3y^2$, $1 - 3z^2$ to be non-zero; given that, the formula gives in turn $y = \tan 3\alpha$, $z = \tan 9\alpha$ and $x = \tan 27\alpha$. So $\tan\alpha = \tan 27\alpha$, giving $26\alpha = k\pi$, that is

$$\alpha = \frac{k\pi}{26}, \qquad -13 < k < 13.$$

That is 25 candidate values of α, all giving different values of x.

> **Ask yourself** Could some of these 25 candidates make a denominator vanish, so that they are not really solutions?

This is the point to check. A denominator $1 - 3\tan^2\beta$ vanishes when $\beta \equiv \pm\frac{1}{6}\pi \pmod{\pi}$, and the tangent itself is undefined when β is an odd multiple of $\frac{1}{2}\pi$. Here the angles involved are $\beta = 3^j \cdot \frac{k\pi}{26}$ for $j = 0, 1, 2$.

- $3^j k\pi/26 = \pm\frac{1}{6}\pi + m\pi$ would mean $3^{j+1}k = \pm 13 + 78m$. The left-hand side is a multiple of 3 but the right-hand side is not (as 78 is a multiple of 3 and 13 is not). Impossible.
- $3^j k\pi/26 = \frac{1}{2}(2m+1)\pi$ would mean $3^j k = 13(2m+1)$, so 13 divides k, forcing $k = 0$; but then the left-hand side is 0, which is not an odd multiple of 13. Impossible.

So every candidate is a genuine solution, and the system has exactly 25 real solutions. (The value $k = \pm 13$, which the interval excludes, corresponds to $\alpha = \pm\frac{1}{2}\pi$, where $\tan\alpha$ does not exist.)

Part (iii)(a)

Now $|x| \leq 1$, so we may write $x = \cos\alpha$ for exactly one $\alpha \in [0, \pi]$. The double-angle formula $\cos 2\alpha = 2\cos^2\alpha - 1$ gives $y = \cos 2\alpha$, then $z = \cos 4\alpha$ and $x = \cos 8\alpha$; these automatically satisfy $|y| \leq 1$ and $|z| \leq 1$. Hence the solutions with all three in $[-1, 1]$ correspond exactly to the $\alpha \in [0, \pi]$ with

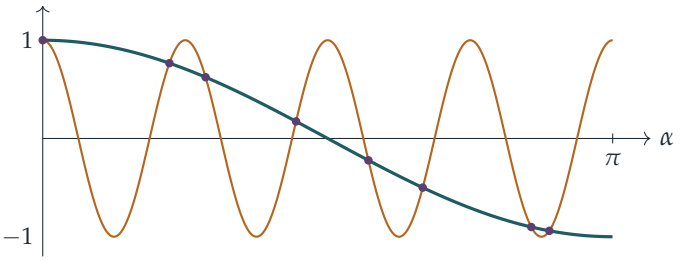
$$\cos 8\alpha = \cos\alpha.$$

Two angles have equal cosines exactly when they are equal or opposite up to a multiple of 2π, so

$$8\alpha = \alpha + 2m\pi \quad \text{or} \quad 8\alpha = -\alpha + 2n\pi, \qquad \text{i.e.} \qquad \alpha = \frac{2m\pi}{7} \quad \text{or} \quad \alpha = \frac{2n\pi}{9}.$$

In $[0, \pi]$ the first family gives $m = 0, 1, 2, 3$, that is $0, \frac{2\pi}{7}, \frac{4\pi}{7}, \frac{6\pi}{7}$, and the second gives $n = 0, 1, 2, 3, 4$, that is $0, \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{6\pi}{9}, \frac{8\pi}{9}$. The only overlap is $\alpha = 0$ (a common value would need $\frac{m}{7} = \frac{n}{9}$, i.e. $9m = 7n$, forcing $7 \mid m$). That leaves $4 + 5 - 1 = 8$ distinct values of α, and since cosine is one-to-one on $[0, \pi]$ they give 8 distinct values of x, each determining y and z. So there are exactly 8 solutions with $|x|, |y|, |z| \leq 1$.

The picture shows the eight crossings of $\cos 8\alpha$ (amber) with $\cos\alpha$ (teal) on $[0, \pi]$.



Part (iii)(b)

Now drop the restriction. Eliminating y and z: $y = 2x^2 - 1$ is a quadratic in x, so $z = 2y^2 - 1$ is a polynomial of degree 4 in x, and then $x = 2z^2 - 1$ becomes

$$x = 2\Big(2\big(2x^2 - 1\big)^2 - 1\Big)^2 - 1,$$

a polynomial equation of degree 8 in x (the leading term on the right is $2 \cdot (2 \cdot 2^2)^2 x^8 = 128x^8$, which nothing cancels). Expanded, it is

$$128x^8 - 256x^6 + 160x^4 - 32x^2 - x + 1 = 0.$$

Every solution (x, y, z) of the system has x a root of this octic, and x determines y and z. A polynomial of degree 8 has at most 8 real roots, and part (a) has already produced 8 distinct real roots, namely $\cos\alpha$ for the eight values of α found there. So these are *all* the roots, and every solution of the system is one of the eight from (a). In particular every solution has $|x| \leq 1$, $|y| \leq 1$, $|z| \leq 1$, which is what we had to show.

Ask yourself Why does (b) need the degree, rather than just the eight solutions from (a)?

Part (a) only found the solutions inside $[-1,1]$; on its own it says nothing about $x=5$, say. The degree bound is what rules out anything else: eight roots are all an octic can have, and (a) has used them up.

Checks

- In (i), $k=1$: $\tan\frac{\pi}{7}\approx 0.4816$, and $\frac{2(0.4816)}{1-0.4816^2}\approx 1.2540=\tan\frac{2\pi}{7}$. Two more steps return to 0.4816.
- A computer scan of the real line for roots of $f(f(f(x)))=x$ found exactly 7 roots for $f(t)=\frac{2t}{1-t^2}$ and exactly 25 for $f(t)=\frac{3t-t^3}{1-3t^2}$, the latter matching $\tan\frac{k\pi}{26}$, $|k|\le 12$.
- In (iii), $x=1$ gives $y=z=1$, the $\alpha=0$ solution, and indeed $128-256+160-32-1+1=0$. A scan of $[-50,50]$ for roots of the octic found exactly 8, all inside $[-1,1]$.

Common slips

- In (ii), allowing $k=\pm 13$ (or not restricting α at all), giving 27 solutions; $\alpha=\pm\frac{1}{2}\pi$ does not correspond to any real x.
- In (iii)(a), forgetting the second family $8\alpha=-\alpha+2n\pi$, or counting $\alpha=0$ twice.
- In (i), leaving answers such as $\tan\frac{4\pi}{7}$ unreduced, when the angle must lie in $(-\frac{1}{2}\pi,\frac{1}{2}\pi)$.

Beyond STEP

The polynomials T_n with $T_n(\cos\alpha)=\cos n\alpha$ are the *Chebyshev polynomials*; part (iii) is the equation $T_8(x)=x$, and the octic above is $T_8(x)-x$. Iterating a map like $x\mapsto 2x^2-1$ and asking for points that return after three steps is the starting point of the theory of dynamical systems: this map is "chaotic" on $[-1,1]$, and the angle substitution turns it into the simple doubling $\alpha\mapsto 2\alpha$. The same trick explains why the tangent maps in (i) and (ii) have so many periodic points. Siklos's *Advanced Problems in Mathematics* has more questions in this spirit.

WORKED SOLUTION

STEP II 2023, Question 3

Pure • Polynomials and Derivatives

STEP II 2023, Question 3

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

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The key idea

Differentiating q just shifts every term of the sum along by one, and the last derivative vanishes, so $q' = q - p$. That one relation drives everything. The three proofs that $q > 0$ then read it in three ways: as a fact about stationary points, as the derivative of $\mathrm{e}^{-x}q(x)$ (an integrating factor), and as the integral formula you get by integrating by parts again and again.

Solution

Throughout, write $p(x) = ax^n + \dots$ with $a \neq 0$. For $k \geq 1$, $p^{(k)}$ has degree less than n (or is zero), and $p^{(n+1)} \equiv 0$.

Part (i)

(a) A polynomial of odd degree tends to $+\infty$ at one end of the real line and to $-\infty$ at the other. Being continuous, it then takes the value 0 somewhere, by the intermediate value theorem. But $p(x) > 0$ for all x, so p has no real root, and n must be even.

Also $a > 0$: if $a < 0$ then $p(x) \to -\infty$ as $x \to \infty$, contradicting $p > 0$. In $q = p + p' + \cdots + p^{(n)}$ every term after the first has degree less than n, so q also has degree n with leading coefficient $a > 0$. Hence $q(x) \to +\infty$ as $x \to \infty$ (and, as n is even, also as $x \to -\infty$), so q certainly takes positive values. (If $n = 0$ then $q = p$ is a positive constant.)

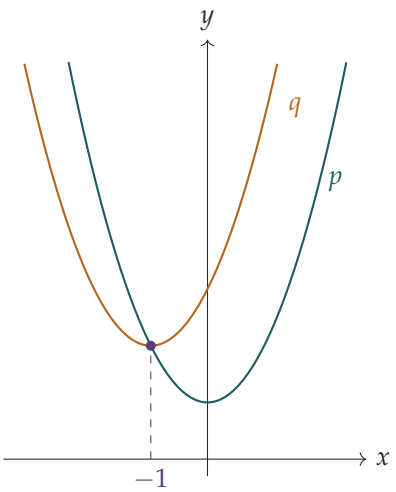
(b) Differentiating term by term,

$$q'(x) = \sum_{k=0}^{n} p^{(k+1)}(x) = \sum_{k=1}^{n+1} p^{(k)}(x) = \sum_{k=0}^{n} p^{(k)}(x) - p(x) + p^{(n+1)}(x) = q(x) - p(x),$$

since $p^{(n+1)} \equiv 0$ for a polynomial of degree n.

Part (ii)

(a) At a stationary point x_0 of $y = q(x)$ we have $q'(x_0) = 0$, so by (i)(b) $q(x_0) = p(x_0)$: the two curves meet there.



(The picture shows $p(x) = x^2 + 1$, for which $q(x) = x^2 + 2x + 3$. The minimum of q is at $x = -1$, exactly where the curves cross.)

Now for $q > 0$. If $n = 0$ then $q = p > 0$, so suppose $n \geq 2$. By (i)(a), $q(x) \to +\infty$ as $x \to \pm\infty$, so we can choose $R > 0$ with $q(x) > q(0)$ whenever $|x| > R$.

Ask yourself How do we know q has a lowest point at all, and that it is a stationary point?

On the closed interval $[-R, R]$ the continuous function q attains a minimum value at some x_0, and $q(x_0) \leq q(0)$. Outside the interval q is bigger than $q(0)$, so $q(x_0)$ is the minimum of q over the whole real line. Since x_0 is an interior point of $\mathbb{R}$ and q is differentiable, $q'(x_0) = 0$. By the first result, $q(x_0) = p(x_0) > 0$. So for every x,

$$q(x) \geq q(x_0) = p(x_0) > 0.$$

(b) Using (i)(b),

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{-x}q(x)\right) = \mathrm{e}^{-x}\left(q'(x) - q(x)\right) = -\mathrm{e}^{-x}p(x) < 0$$

for all x, since $p > 0$. So $\mathrm{e}^{-x}q(x)$ is (strictly) decreasing.

For $q > 0$: since q is a polynomial, $\mathrm{e}^{-X}q(X) \to 0$ as $X \to \infty$ (exponentials beat powers). A decreasing function is never below its limit at $+\infty$, so $\mathrm{e}^{-X}q(X) \geq 0$ for all X. Then, for any x, strict decrease gives

$$\mathrm{e}^{-x}q(x) > \mathrm{e}^{-(x+1)}q(x+1) \geq 0,$$

and multiplying by $e^x > 0$ gives $q(x) > 0$.

(c) All the integrals here converge, because a polynomial in t times e^{-t} is integrable on $[0, \infty)$. Integrate by parts with $u = p(x+t)$ and $dv = e^{-t}\,dt$ (treating x as a constant):

$$\begin{aligned}\int_0^\infty p(x+t)e^{-t}\,dt &= \Big[-p(x+t)e^{-t}\Big]_0^\infty + \int_0^\infty p^{(1)}(x+t)e^{-t}\,dt \\ &= p(x) + \int_0^\infty p^{(1)}(x+t)e^{-t}\,dt,\end{aligned}$$

because $p(x+t)e^{-t} \to 0$ as $t \to \infty$. This is the first result.

Nothing in that calculation used anything about p except that it is a polynomial, so the same holds for every derivative: writing

$$I_k(x) = \int_0^\infty p^{(k)}(x+t)e^{-t}\,dt, \qquad \text{we have} \qquad I_k(x) = p^{(k)}(x) + I_{k+1}(x)$$

for $k = 0, 1, \ldots, n$. Also $I_{n+1}(x) = 0$, since $p^{(n+1)} \equiv 0$. Applying the relation repeatedly,

$$I_0(x) = p(x) + I_1(x) = p(x) + p^{(1)}(x) + I_2(x) = \cdots = \sum_{k=0}^{n} p^{(k)}(x) + I_{n+1}(x) = q(x),$$

which is the second result.

Ask yourself Why does this formula make $q(x) > 0$ obvious?

For each fixed x, the integrand $p(x+t)e^{-t}$ is continuous and strictly positive for every $t \geq 0$, because $p > 0$ everywhere. The integral of a continuous, strictly positive function over an interval of positive length is strictly positive. So $q(x) = I_0(x) > 0$ for all x.

Checks

- With $p(x) = x^2 + 1$ we get $q(x) = x^2 + 2x + 3 = (x+1)^2 + 2$. Indeed $q' = 2x + 2 = q - p$, the minimum of q is 2 at $x = -1$, and $p(-1) = 2$ as (ii)(a) predicts.
- For the same p at $x = 0$: $\int_0^\infty (t^2+1)e^{-t}\,dt = 2! + 1 = 3 = q(0)$.
- By computer, for 300 random everywhere-positive polynomials of degree up to 6, we confirmed that $q' = q - p$ exactly, that $q > 0$ and $e^{-x}q(x)$ decreases on a grid over $[-20, 20]$, and that numerical integration reproduces $q(x)$ to about ten significant figures.

Common slips

- In (ii)(a), assuming a minimum exists without a reason. It needs $q \to +\infty$ at both ends, which uses n even and $a > 0$ from (i)(a).
- In (ii)(b), concluding only $e^{-x}q(x) \geq 0$ from the limit. The strict decrease is what upgrades this to $q(x) > 0$.

- Forgetting that the boundary term $p(x+t)\mathrm{e}^{-t}$ must vanish as $t \to \infty$ before discarding it.

Beyond STEP

The function e^{-x} in (ii)(b) is an integrating factor, and (ii)(c) solves $q' - q = -p$ explicitly. The formula $\int_0^\infty p(x+t)\mathrm{e}^{-t}\,\mathrm{d}t$ is a Laplace-transform-style integral, and the identity $q = (1 + D + D^2 + \cdots)p = (1 - D)^{-1}p$, where D means differentiation, is a first taste of operator methods for linear differential equations. Boyce and DiPrima, *Elementary Differential Equations*, covers integrating factors and Laplace transforms; Abbott, *Understanding Analysis*, makes the extreme-value argument of (ii)(a) rigorous.

WORKED SOLUTION

STEP II 2023, Question 4

Pure • Eliminating Surds from Polynomials

STEP II 2023, Question 4

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 6.3, *A Polynomial for* $\sqrt{2}+\sqrt{3}+\sqrt{5}$: https://beyondstep.keithmcnulty.org/ch06/field-extensions.html

The key idea

To clear a surd from an equation, put everything containing the surd on one side and everything else on the other, then square (or cube). Each part is the same move. Write the number as "something nice plus a surd", isolate that surd, and raise to a power until it disappears. When a surd survives after expanding, collect its terms on one side and repeat.

Solution

Part (i)

Expanding $(x-\sqrt{2})^2 = 3$ gives $x^2 - 2\sqrt{2}\,x + 2 = 3$, that is

$$x^2 - 1 = 2\sqrt{2}\,x.$$

The surd is now on its own, so we square both sides:

$$x^4 - 2x^2 + 1 = 8x^2 \quad \Longrightarrow \quad x^4 - 10x^2 + 1 = 0,$$

as required. (Squaring only goes one way, which is all we need: every x satisfying the first equation satisfies the second.)

For the deduction, take $x = \sqrt{2}+\sqrt{3}$. Then $x - \sqrt{2} = \sqrt{3}$, so $(x-\sqrt{2})^2 = 3$, and by what we have just shown $\mathrm{f}(\sqrt{2}+\sqrt{3}) = 0$.

Part (ii)

Let $x = \sqrt{2}+\sqrt{3}+\sqrt{5}$ and $y = x - \sqrt{5} = \sqrt{2}+\sqrt{3}$. By part (i), $y^4 - 10y^2 + 1 = 0$. We now write this in terms of x. First,

$$y^2 = (x-\sqrt{5})^2 = (x^2+5) - 2\sqrt{5}\,x,$$

and so

$$\begin{aligned} y^4 &= (x^2+5)^2 - 4\sqrt{5}\,x(x^2+5) + 20x^2 \\ &= x^4 + 30x^2 + 25 - 4\sqrt{5}\,x^3 - 20\sqrt{5}\,x. \end{aligned}$$

Therefore

$$\begin{aligned} 0 &= y^4 - 10y^2 + 1 \\ &= \left(x^4 + 30x^2 + 25 - 4\sqrt{5}\,x^3 - 20\sqrt{5}\,x\right) - 10\left(x^2 + 5 - 2\sqrt{5}\,x\right) + 1 \\ &= x^4 + 20x^2 - 24 - 4\sqrt{5}\,x^3. \end{aligned}$$

The $\sqrt{5}\,x$ terms cancel, which is lucky. Only one surd term remains, so we isolate it and square:

$$x^4 + 20x^2 - 24 = 4\sqrt{5}\,x^3 \quad\Longrightarrow\quad (x^4+20x^2-24)^2 = 80x^6.$$

Expanding the left-hand side,

$$\begin{aligned} (x^4+20x^2-24)^2 &= x^8 + 40x^6 + 400x^4 - 48x^4 - 960x^2 + 576 \\ &= x^8 + 40x^6 + 352x^4 - 960x^2 + 576. \end{aligned}$$

Subtracting $80x^6$, we can take

$$g(x) = x^8 - 40x^6 + 352x^4 - 960x^2 + 576,$$

which has degree 8 and integer coefficients, with $g(\sqrt{2}+\sqrt{3}+\sqrt{5}) = 0$.

Ask yourself Why do we eliminate $\sqrt{5}$ last, using part (i) for $\sqrt{2}+\sqrt{3}$, rather than starting from scratch?

Part (i) has already removed two of the three surds, so we only have to deal with one more. Removing a single surd by squaring doubles the degree, from 4 to 8, which is exactly the degree we were asked for. Starting from scratch would mean juggling all three surds at once.

Part (iii)

Let $x = a + \sqrt{2}$, so $a = x - \sqrt{2}$, and $a^3 - 3a + 1 = 0$. Expanding,

$$(x-\sqrt{2})^3 = x^3 - 3\sqrt{2}\,x^2 + 6x - 2\sqrt{2},$$

so

$$0 = (x-\sqrt{2})^3 - 3(x-\sqrt{2}) + 1 = x^3 + 3x + 1 - \sqrt{2}\,(3x^2 - 1).$$

We isolate the surd and square:

$$x^3 + 3x + 1 = \sqrt{2}\,(3x^2 - 1) \quad \Longrightarrow \quad (x^3 + 3x + 1)^2 = 2(3x^2 - 1)^2.$$

The two sides are

$$(x^3 + 3x + 1)^2 = x^6 + 6x^4 + 2x^3 + 9x^2 + 6x + 1,$$
$$2(3x^2 - 1)^2 = 18x^4 - 12x^2 + 2,$$

so we can take

$$\mathrm{h}(x) = x^6 - 12x^4 + 2x^3 + 21x^2 + 6x - 1.$$

We used only the fact that a satisfies $t^3 - 3t + 1 = 0$, so the same working applies to b and to c. Hence $\mathrm{h}(a+\sqrt{2}) = \mathrm{h}(b+\sqrt{2}) = \mathrm{h}(c+\sqrt{2}) = 0$.

Part (iv)

Squaring will not remove cube roots, so we need a different identity. Let $p = \sqrt[3]{2}$, $q = \sqrt[3]{3}$ and $x = p + q$. The binomial expansion gives

$$x^3 = p^3 + q^3 + 3pq(p+q) = 5 + 3pq\,x.$$

Ask yourself How do we get rid of pq, which is still irrational?

Here $pq = \sqrt[3]{6}$ is the only irrational quantity left, and it appears only as a factor. We isolate it and cube, using $(pq)^3 = 6$:

$$x^3 - 5 = 3pq\,x \quad \Longrightarrow \quad (x^3 - 5)^3 = 27 \cdot 6\,x^3 = 162x^3.$$

Since $(x^3-5)^3 = x^9 - 15x^6 + 75x^3 - 125$, we can take

$$\mathrm{k}(x) = x^9 - 15x^6 - 87x^3 - 125,$$

which has integer coefficients, and $\mathrm{k}(\sqrt[3]{2} + \sqrt[3]{3}) = 0$.

Checks

- In part (ii), $\sqrt{2}+\sqrt{3}+\sqrt{5} \approx 5.38233$. With $x^2 \approx 28.9695$, we get $x^8 \approx 704\,310$ and $40x^6 \approx 972\,485$. Then $352x^4 \approx 295\,410$ and $960x^2 \approx 27\,811$, and the terms of g cancel to within rounding error.
- Changing the signs of $\sqrt{2}$, $\sqrt{3}$ and $\sqrt{5}$ never changes $\mathrm{g}(x)$, so all eight numbers $\pm\sqrt{2} \pm \sqrt{3} \pm \sqrt{5}$ are roots. That accounts for the degree 8. Similarly, h has the six roots $a \pm \sqrt{2}$, $b \pm \sqrt{2}$, $c \pm \sqrt{2}$.
- The roots of $t^3 - 3t + 1 = 0$ are $2\cos 40^\circ$, $2\cos 160^\circ$ and $2\cos 280^\circ$. For example, $a \approx 1.53209$ gives $a + \sqrt{2} \approx 2.94630$, and $\mathrm{h}(2.94630)$ is zero to within rounding error.
- In part (iv), $\sqrt[3]{2} + \sqrt[3]{3} \approx 2.70217$, and $\mathrm{k}(2.70217)$ is zero to within rounding error. A computer check confirmed all four polynomials to about 10^{-10}.

Common slips

- Squaring before isolating the surd. If $\sqrt{5}$ terms are left on both sides, they will not cancel.
- In part (iv), trying to square, or cubing x without spotting that $x^3 = 5 + 3pqx$. The expansion $(p+q)^3 = p^3 + q^3 + 3pq(p+q)$ is the whole trick.
- Sign errors in the cubic expansion in part (iii), especially in the constant $-2\sqrt{2} + 3\sqrt{2} = \sqrt{2}$.

Beyond STEP

A number that is a root of a polynomial with integer coefficients is called *algebraic*. This question shows by hand that sums of algebraic numbers are algebraic. In general, the proof uses linear algebra: $\mathbb{Q}(\alpha, \beta)$ is a finite-dimensional vector space over $\mathbb{Q}$. So the powers $1, \alpha+\beta, (\alpha+\beta)^2, \ldots$ must eventually be linearly dependent. The degrees 4, 8, 6 and 9 that appear here are the dimensions of these *field extensions*. For more on this, see Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP II 2023, Question 5

Pure • Recurrence Relations and Convergence

STEP II 2023, Question 5

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 2.7, *Six Digits or Six Hundred*: https://beyondstep.keithmcnulty.org/ch02/newton-speed.html

The key idea

Both recurrences are machines for approximating $\sqrt{2}$, and the question is about how fast the error shrinks. The trick in each part is to measure the error in the right way: in (i) look at $x_n^2 - 2$, which gets multiplied by a factor of size at most $\frac{1}{4}$ (and changes sign) at every step; in (ii) look at $y_n - \sqrt{2}$, which gets *squared* (roughly) at every step. Multiplying by a constant gives steady progress; squaring doubles the number of correct digits each time, and that is why part (ii) reaches 10^{-600} in the same ten steps.

Solution

Part (i)

(a) We can rewrite the recurrence as

$$x_{n+1} = \frac{(x_n + 1) + 1}{x_n + 1} = 1 + \frac{1}{x_n + 1}.$$

We have $x_0 = 1 \geq 1$. If $x_n \geq 1$, then $x_n + 1 > 0$, so $\frac{1}{x_n+1} > 0$ and $x_{n+1} > 1$. By induction, $x_n \geq 1$ for all n.

(b) Put everything over a common denominator:

$$\begin{aligned} x_{n+1}^2 - 2 &= \frac{(x_n+2)^2 - 2(x_n+1)^2}{(x_n+1)^2} \\ &= \frac{x_n^2 + 4x_n + 4 - 2x_n^2 - 4x_n - 2}{(x_n+1)^2} \\ &= \frac{-(x_n^2-2)}{(x_n+1)^2}. \end{aligned}$$

The denominator is positive, so $x_{n+1}^2 - 2$ and $x_n^2 - 2$ have opposite signs. (Neither is ever zero: every x_n is rational, by induction, and $\sqrt{2}$ is irrational.)

Taking moduli, and using $x_n + 1 \geq 2$ from (a), so that $(x_n+1)^2 \geq 4$,

$$|x_{n+1}^2 - 2| = \frac{|x_n^2 - 2|}{(x_n+1)^2} \leq \tfrac{1}{4}|x_n^2 - 2|,$$

as required.

(c) Since $x_0^2 - 2 = -1$, applying (b) ten times gives

$$|x_{10}^2 - 2| \leq \left(\tfrac{1}{4}\right)^{10} |x_0^2 - 2| = \frac{1}{4^{10}} = \frac{1}{1\,048\,576} < 10^{-6}.$$

For the sign: $x_0^2 - 2 < 0$ and the sign flips at every step, so $x_n^2 - 2$ is negative for every even n. In particular $x_{10}^2 - 2 < 0$, and so

$$-10^{-6} < x_{10}^2 - 2 < 0, \qquad \text{that is,} \qquad 2 - 10^{-6} \leq x_{10}^2 \leq 2.$$

Part (ii)

(a) First, every y_n is positive: $y_0 = 1 > 0$, and if $y_n > 0$ then $y_{n+1} = \frac{y_n^2+2}{2y_n} > 0$. So the recurrence never divides by zero. Now

$$y_{n+1} - \sqrt{2} = \frac{y_n^2 + 2 - 2\sqrt{2}\,y_n}{2y_n} = \frac{(y_n - \sqrt{2})^2}{2y_n},$$

since $(y_n - \sqrt{2})^2 = y_n^2 - 2\sqrt{2}\,y_n + 2$. This is the required identity.

For the deduction: $y_0 = 1$. For $n \geq 1$, the identity (applied with $n-1$) shows that $y_n - \sqrt{2}$ is a square divided by the positive number $2y_{n-1}$, so $y_n - \sqrt{2} \geq 0$, and hence $y_n \geq \sqrt{2} > 1$. So $y_n \geq 1$ for all $n \geq 0$.

(b) Write $e_n = y_n - \sqrt{2}$ and $r = \dfrac{\sqrt{2}-1}{2}$. From (a), $e_n \geq 0$ for $n \geq 1$, and since $y_n \geq 1$,

$$e_{n+1} = \frac{e_n^2}{2y_n} \leq \frac{e_n^2}{2}.$$

We prove $e_n \leq 2r^{2^n}$ for $n \geq 1$ by induction.

Base case $n = 1$. $y_1 = \frac{3}{2}$, so $e_1 = \frac{3}{2} - \sqrt{2}$, while

$$2r^2 = 2 \cdot \frac{(\sqrt{2}-1)^2}{4} = \frac{3 - 2\sqrt{2}}{2} = \tfrac{3}{2} - \sqrt{2}.$$

So the bound holds with equality.

Inductive step. Suppose $0 \le e_n \le 2r^{2^n}$ for some $n \ge 1$.

? **Ask yourself** Why are we allowed to square both sides of $e_n \le 2r^{2^n}$?

Squaring preserves an inequality between *non-negative* numbers, and $e_n \ge 0$ is exactly what part (a) gave us. (This is why the bound starts at $n = 1$, not $n = 0$: $e_0 = 1 - \sqrt{2}$ is negative.) So

$$e_{n+1} \le \frac{e_n^2}{2} \le \frac{\left(2r^{2^n}\right)^2}{2} = \frac{4r^{2^{n+1}}}{2} = 2r^{2^{n+1}},$$

which completes the induction.

(c) The lower bound $y_{10} \ge \sqrt{2}$ was shown in (a). For the upper bound, $\sqrt{2} - 1 < \frac{1}{2}$ gives $0 < r < \frac{1}{4}$, so by (b)

$$y_{10} - \sqrt{2} \le 2r^{2^{10}} = 2r^{1024} < 2\left(\tfrac{1}{4}\right)^{1024} = 2 \cdot 2^{-2048} = 2^{-2047}.$$

To compare this with a power of 10, use $2^{10} = 1024 > 10^3$:

$$2^{2047} > 2^{2040} = \left(2^{10}\right)^{204} > \left(10^3\right)^{204} = 10^{612} > 10^{600}.$$

Hence $y_{10} - \sqrt{2} < 2^{-2047} < 10^{-600}$, and altogether

$$\sqrt{2} \le y_{10} \le \sqrt{2} + 10^{-600}.$$

? **Ask yourself** Where does the extraordinary speed in (ii) come from?

In (i) the error is multiplied by at most $\frac{1}{4}$ each time, so ten steps gain only about six decimal places. In (ii) the error is (roughly) squared each time, so the exponent 2^n doubles: ten steps take us from an error of about $\frac{1}{10}$ to one below 10^{-600}.

Checks

- The first few terms: $x_1 = \frac{3}{2}$, $x_2 = \frac{7}{5}$, $x_3 = \frac{17}{12}$, and $x_n^2 - 2$ goes -1, $\frac{1}{4}$, $-\frac{1}{25}$, $\frac{1}{144}$: alternating, and shrinking by at least a factor of 4 each time.
- Computing exactly with fractions, $x_{10}^2 \approx 2 - 3.0 \times 10^{-8}$, comfortably inside the window $[2 - 10^{-6}, 2]$.
- $y_2 = \frac{17}{12}$, so $y_2 - \sqrt{2} \approx 0.00245$, against the bound $2r^4 \approx 0.00368$. A computer check with exact fractions gives $y_{10} - \sqrt{2} \approx 10^{-783}$, well below the guaranteed $2^{-2047} \approx 10^{-616}$.

Common slips

- Forgetting the sign in (i)(c): the modulus bound alone gives $2 - 10^{-6} \leq x_{10}^2 \leq 2 + 10^{-6}$; the upper bound 2 needs the alternating signs.
- Squaring $e_n \leq 2r^{2^n}$ without first knowing $e_n \geq 0$.
- Stopping at 2^{-2047} without actually comparing it to 10^{-600}; the inequality $2^{10} > 10^3$ is the cleanest way.

Beyond STEP

The recurrence in (ii) is Newton's method applied to $f(y) = y^2 - 2$, and the identity in (ii)(a) is the statement that Newton's method converges *quadratically* near a simple root, whereas (i) is an example of *linear* convergence with rate at most $\frac{1}{4}$. The numbers x_n are also the convergents of the continued fraction for $\sqrt{2}$. Orders of convergence and fixed-point iteration are standard topics in a first numerical analysis course, and the analysis behind them is developed carefully in Bartle and Sherbert, *Introduction to Real Analysis*.

WORKED SOLUTION

STEP II 2023, Question 6

Pure • Fibonacci Matrix Identities

STEP II 2023, Question 6

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Once the Fibonacci numbers are packaged as the entries of the powers of a single 2×2 matrix, every identity about them becomes an identity about powers of $\mathbf{Q}$, where ordinary algebra (index laws, determinants, the binomial theorem) does the work. Each part follows the same recipe: find two ways of writing the same matrix, then read off one entry, almost always the top-right one, which is F_n itself.

Solution

The matrix formula

For $n = 1$ the left-hand side is $\begin{pmatrix} F_2 & F_1 \\ F_1 & F_0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \mathbf{Q}$, since $F_2 = F_1 + F_0 = 1$. So the result holds for $n = 1$.

Suppose it holds for some $n \geq 1$. Then

$$\mathbf{Q}^{n+1} = \mathbf{Q}^n\mathbf{Q} = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} F_{n+1}+F_n & F_{n+1} \\ F_n+F_{n-1} & F_n \end{pmatrix} = \begin{pmatrix} F_{n+2} & F_{n+1} \\ F_{n+1} & F_n \end{pmatrix},$$

using the recurrence twice. This is the formula with n replaced by $n+1$, so by induction it holds for all positive integers n.

Part (i)

Determinants are multiplicative, so $\det(\mathbf{Q}^n) = (\det \mathbf{Q})^n = (0-1)^n = (-1)^n$. Computing the same determinant from the entries,

$$\det(\mathbf{Q}^n) = F_{n+1}F_{n-1} - F_n^2,$$

and therefore $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$, as required. (This is Cassini's identity.)

Part (ii)

By the index law, $\mathbf{Q}^{m+n} = \mathbf{Q}^m\mathbf{Q}^n$:

$$\begin{pmatrix} F_{m+n+1} & F_{m+n} \\ F_{m+n} & F_{m+n-1} \end{pmatrix} = \begin{pmatrix} F_{m+1} & F_m \\ F_m & F_{m-1} \end{pmatrix}\begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}.$$

The top-right entry of the product is the first row of $\mathbf{Q}^m$ times the second column of $\mathbf{Q}^n$, namely $F_{m+1}F_n + F_mF_{n-1}$. Comparing top-right entries gives

$$F_{m+n} = F_{m+1}F_n + F_mF_{n-1}.$$

Part (iii)

Directly,

$$\mathbf{Q}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \mathbf{I} + \mathbf{Q}.$$

(This is the matrix version of $x^2 = 1 + x$, the equation satisfied by the golden ratio.)

In the parts below, note that the top-right entry of $\mathbf{Q}^k$ is F_k for every $k \geq 1$, and also for $k = 0$, since $\mathbf{Q}^0 = \mathbf{I}$ has top-right entry $0 = F_0$. Taking the top-right entry of a sum of matrices is linear, so the top-right entry of $\sum c_k\mathbf{Q}^k$ is $\sum c_kF_k$.

(a) Using $\mathbf{Q}^2 = \mathbf{I} + \mathbf{Q}$ and the binomial theorem with $\mathbf{A} = \mathbf{Q}$,

$$\mathbf{Q}^{2n} = (\mathbf{Q}^2)^n = (\mathbf{I} + \mathbf{Q})^n = \sum_{k=0}^{n}\binom{n}{k}\mathbf{Q}^k.$$

The top-right entry of the left side is F_{2n}, and that of the right side is $\sum_k \binom{n}{k}F_k$. Hence $F_{2n} = \sum_{k=0}^{n}\binom{n}{k}F_k$.

(b) First find $\mathbf{Q}^3$ in the same style:

$$\mathbf{Q}^3 = \mathbf{Q}\,\mathbf{Q}^2 = \mathbf{Q}(\mathbf{I} + \mathbf{Q}) = \mathbf{Q} + \mathbf{Q}^2 = \mathbf{Q} + \mathbf{I} + \mathbf{Q} = \mathbf{I} + 2\mathbf{Q}.$$

Now apply the binomial theorem with $\mathbf{A} = 2\mathbf{Q}$:

$$\mathbf{Q}^{3n} = (\mathbf{I} + 2\mathbf{Q})^n = \sum_{k=0}^{n}\binom{n}{k}2^k\mathbf{Q}^k,$$

and comparing top-right entries gives $F_{3n} = \sum_{k=0}^{n}\binom{n}{k}2^kF_k$.

For the second identity, write $\mathbf{Q}^{3n} = \mathbf{Q}^n\mathbf{Q}^{2n}$ and use the expansion from (a):

$$\mathbf{Q}^{3n} = \mathbf{Q}^n\sum_{k=0}^{n}\binom{n}{k}\mathbf{Q}^k = \sum_{k=0}^{n}\binom{n}{k}\mathbf{Q}^{n+k}.$$

The top-right entry of $\mathbf{Q}^{n+k}$ is F_{n+k}, so $F_{3n} = \sum_{k=0}^{n}\binom{n}{k}F_{n+k}$.

(c)

Ask yourself Which matrix should we raise to the nth power to produce the sign $(-1)^{n+k}$?

The sign $(-1)^{n+k} = (-1)^{n-k}$ is what appears when we expand $(\mathbf{Q}-\mathbf{I})^n$. To fit the form of the binomial theorem we are allowed to use, write $\mathbf{Q}-\mathbf{I} = -\big(\mathbf{I}+(-\mathbf{Q})\big)$, so

$$(\mathbf{Q}-\mathbf{I})^n = (-1)^n\big(\mathbf{I}+(-\mathbf{Q})\big)^n = (-1)^n\sum_{k=0}^{n}\binom{n}{k}(-1)^k\mathbf{Q}^k = \sum_{k=0}^{n}(-1)^{n+k}\binom{n}{k}\mathbf{Q}^k.$$

Multiplying by $\mathbf{Q}^n$,

$$\sum_{k=0}^{n}(-1)^{n+k}\binom{n}{k}\mathbf{Q}^{n+k} = \mathbf{Q}^n(\mathbf{Q}-\mathbf{I})^n = \big(\mathbf{Q}(\mathbf{Q}-\mathbf{I})\big)^n,$$

where the last step is allowed because $\mathbf{Q}$ and $\mathbf{Q}-\mathbf{I}$ commute (both are polynomials in $\mathbf{Q}$), so the factors can be regrouped. But by (iii), $\mathbf{Q}(\mathbf{Q}-\mathbf{I}) = \mathbf{Q}^2-\mathbf{Q} = \mathbf{I}$, so the whole sum equals $\mathbf{I}^n = \mathbf{I}$.

The top-right entry of $\mathbf{I}$ is 0, while that of the left side is $\sum_k(-1)^{n+k}\binom{n}{k}F_{n+k}$. Therefore

$$\sum_{k=0}^{n}(-1)^{n+k}\binom{n}{k}F_{n+k} = 0.$$

As a bonus, the top-left entries give $\sum_k(-1)^{n+k}\binom{n}{k}F_{n+k+1} = 1$.

Checks

- (i) with $n=5$: $F_6F_4 - F_5^2 = 8\cdot 3 - 25 = -1 = (-1)^5$.
- (ii) with $m=3$, $n=4$: $F_4F_4 + F_3F_3 = 9+4 = 13 = F_7$.
- (a) with $n=4$: $0+4\cdot 1+6\cdot 1+4\cdot 2+1\cdot 3 = 21 = F_8$.
- (b) with $n=3$: $3\cdot 2\cdot 1+3\cdot 4\cdot 1+8\cdot 2 = 34 = F_9$, and also $F_3+3F_4+3F_5+F_6 = 2+9+15+8 = 34$.
- (c) with $n=3$: $-F_3+3F_4-3F_5+F_6 = -2+9-15+8 = 0$.

All five identities were also confirmed by computer for every n up to 59 (and every m,n up to 49 in (ii)).

Common slips

- In (ii), comparing the wrong entry, or multiplying a row by a row, gives a plausible-looking but false formula. The top-right entry of $\mathbf{Q}^m\mathbf{Q}^n$ uses the first *row* of $\mathbf{Q}^m$ and the second *column* of $\mathbf{Q}^n$.
- Using the binomial theorem for $(\mathbf{A}+\mathbf{B})^n$ with matrices that do not commute. Here every matrix is a polynomial in $\mathbf{Q}$, so everything commutes, but it is worth saying so.

- Forgetting the $k = 0$ term: it contributes $\mathbf{I}$, whose top-right entry is $F_0 = 0$.

Beyond STEP

The relation $\mathbf{Q}^2 = \mathbf{I} + \mathbf{Q}$ is the Cayley–Hamilton theorem for $\mathbf{Q}$: every square matrix satisfies its own characteristic equation, here $\lambda^2 - \lambda - 1 = 0$. Diagonalising $\mathbf{Q}$, whose eigenvalues are $\frac{1}{2}(1 \pm \sqrt{5})$, gives Binet's closed formula for F_n, and the matrix powers also give a fast way to compute F_n by repeated squaring. Strang's *Introduction to Linear Algebra* treats Fibonacci numbers in exactly this way when it introduces eigenvalues.

WORKED SOLUTION

STEP II 2023, Question 7

Pure • Sums of Two Squares

STEP II 2023, Question 7

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 1.6, *Sums of Two Squares*: https://beyondstep.keithmcnulty.org/ch01/two-squares.html

The key idea

A sum of two squares $x^2 + y^2$ is $|x + y\mathrm{i}|^2$. Part (i) then says that the product of two sums of two squares is again a sum of two squares, and multiplying the complex numbers tells you which squares. Every later part works the same way: write each factor as a sum of two squares (often by spotting a nearby perfect square) and multiply. The last part runs this backwards, dividing instead of multiplying.

Solution

Part (i)

We have $zw = (a + b\mathrm{i})(c + d\mathrm{i}) = (ac - bd) + (ad + bc)\mathrm{i}$, so

$$\begin{aligned} |zw|^2 &= (ac - bd)^2 + (ad + bc)^2 \\ &= a^2c^2 - 2abcd + b^2d^2 + a^2d^2 + 2abcd + b^2c^2 \\ &= a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2 \\ &= (a^2 + b^2)(c^2 + d^2) = |z|^2|w|^2. \end{aligned}$$

Both $|zw|$ and $|z||w|$ are non-negative, so taking square roots gives $|zw| = |z||w|$, as required.

In what follows we use the squared form: if z and w have integer real and imaginary parts, then

$$|z|^2\,|w|^2 = |zw|^2 = (\mathrm{Re}\,zw)^2 + (\mathrm{Im}\,zw)^2.$$

Part (ii)

Here $|2+\mathrm{i}|^2 = 4+1 = 5$ and $|10+11\mathrm{i}|^2 = 100+121 = 221$. The product is

$$(2+\mathrm{i})(10+11\mathrm{i}) = 20+22\mathrm{i}+10\mathrm{i}-11 = 9+32\mathrm{i},$$

so by part (i), $9^2+32^2 = 5\times 221$. (Indeed $81+1024 = 1105$.) So $h=9$, $k=32$ works.

Using $10-11\mathrm{i}$ instead (which has the same modulus) gives $(2+\mathrm{i})(10-11\mathrm{i}) = 31-12\mathrm{i}$, so $h=31$, $k=12$ also works.

Part (iii)

The number ends in 5, so $8045 = 5\times 1609$, and $5 = |2+\mathrm{i}|^2$ as before. We now need 1609 as a sum of two squares. It is just above $40^2 = 1600$, and $1609-1600 = 9 = 3^2$, so $1609 = |40+3\mathrm{i}|^2$. Then

$$(2+\mathrm{i})(40+3\mathrm{i}) = 80+6\mathrm{i}+40\mathrm{i}-3 = 77+46\mathrm{i},$$

and part (i) gives

$$77^2+46^2 = 5\times 1609 = 8045.$$

So $m=77$, $n=46$. (Using $3+40\mathrm{i}$ instead gives $-34+83\mathrm{i}$, so $34^2+83^2 = 8045$ too.)

Part (iv)

Here the extra factor is $36 = 6^2 = |6|^2$, the modulus squared of the real number 6. Multiplying $102+201\mathrm{i}$ by 6 simply scales both parts:

$$(6\times 102)^2+(6\times 201)^2 = 36\,(102^2+201^2) = 36\times 50805.$$

So $p=612$, $q=1206$.

Part (v)

The idea is to find two different ways of writing 25 as $|\alpha|^2$ with Gaussian integers, and multiply each by one fixed representation of 1002082.

First, 1002082 is just above $1001^2 = 1002001$, and

$$1002082-1002001 = 81 = 9^2,$$

so $1002082 = 1001^2+9^2 = |1001+9\mathrm{i}|^2$.

Next, $25 = |5|^2 = |3+4\mathrm{i}|^2 = |3-4\mathrm{i}|^2$. Each of these, multiplied by $1001+9\mathrm{i}$, gives a pair:

$$\begin{aligned}
5(1001+9\mathrm{i}) &= 5005+45\mathrm{i},\\
(3+4\mathrm{i})(1001+9\mathrm{i}) &= 3003+27\mathrm{i}+4004\mathrm{i}-36 = 2967+4031\mathrm{i},\\
(3-4\mathrm{i})(1001+9\mathrm{i}) &= 3003+27\mathrm{i}-4004\mathrm{i}+36 = 3039-3977\mathrm{i}.
\end{aligned}$$

Each has modulus squared 25×1002082, by part (i). Taking absolute values of the parts and putting the smaller one first, three distinct pairs with $r < s$ are

$$(r, s) = (45, 5005), \qquad (2967, 4031), \qquad (3039, 3977).$$

Ask yourself Why not use $4 + 3\mathrm{i}$ as well, to get a fourth pair?

Because $4 + 3\mathrm{i} = \mathrm{i}(3 - 4\mathrm{i})$, and multiplying by i just swaps the real and imaginary parts (up to sign). It reproduces $(3039, 3977)$ with the order reversed, so it gives nothing new. The genuinely different choices are 5, $3 + 4\mathrm{i}$ and $3 - 4\mathrm{i}$.

Part (vi)

Now we run the method backwards. Spot a nearby square again: $1002037 - 1002001 = 36 = 6^2$, so

$$1002037 = 1001^2 + 6^2 = |1001 + 6\mathrm{i}|^2,$$

and also $109 = 10^2 + 3^2 = |10 + 3\mathrm{i}|^2$. If we can write $1001 + 6\mathrm{i} = (10 + 3\mathrm{i})(t + u\mathrm{i})$ with t, u integers, then part (i) gives

$$1002037 = 109\,(t^2 + u^2),$$

and dividing by 109 (using $109 \times 9193 = 1002037$) gives $t^2 + u^2 = 9193$.

So we divide, multiplying top and bottom by the conjugate $10 - 3\mathrm{i}$:

$$\frac{1001 + 6\mathrm{i}}{10 + 3\mathrm{i}} = \frac{(1001 + 6\mathrm{i})(10 - 3\mathrm{i})}{109} = \frac{10010 - 3003\mathrm{i} + 60\mathrm{i} + 18}{109}$$
$$= \frac{10028 - 2943\mathrm{i}}{109} = 92 - 27\mathrm{i},$$

since $109 \times 92 = 10028$ and $109 \times 27 = 2943$. The division comes out exactly, so $1001 + 6\mathrm{i} = (10 + 3\mathrm{i})(92 - 27\mathrm{i})$, and therefore

$$92^2 + 27^2 = 9193.$$

So $t = 92$, $u = 27$. As a check, $8464 + 729 = 9193$.

Ask yourself What if the division hadn't come out exactly?

Then we would try the other conjugate. Dividing by $10 - 3\mathrm{i}$ instead gives $(9992 + 3063\mathrm{i})/109$, which is not a Gaussian integer (9992 is not a multiple of 109). Since 109 is prime, one of the two choices must work; this is the Gaussian-integer fact mentioned in Beyond STEP below. Dividing $1001 - 6\mathrm{i}$ gives nothing new, just the conjugates of these.

Checks

Each answer can be checked by direct arithmetic: $9^2 + 32^2 = 1105 = 5 \times 221$; $77^2 + 46^2 = 5929 + 2116 = 8045$; $612^2 + 1206^2 = 1828980 = 36 \times 50805$; for (v), each of $45^2 + 5005^2$,

$2967^2 + 4031^2$ and $3039^2 + 3977^2$ equals 25052050; and $92^2 + 27^2 = 9193$. A computer search confirms these, and finds that 25×1002082 actually has six representations with $0 < r < s$ (the other three are $(721, 4953)$, $(2079, 4553)$ and $(2395, 4395)$). They come from a second representation of 1002082 itself, so any three distinct pairs are acceptable. Similarly $9193 = 48^2 + 83^2$ as well.

Common slips

- In (v), counting $(3 + 4\mathrm{i})$ and $(4 + 3\mathrm{i})$ as giving different pairs: they give the same pair in the opposite order.
- In (vi), dividing by the wrong conjugate and getting a non-integer, then giving up. Try the other one.
- Sign errors in $\mathrm{i}^2 = -1$ when expanding products: always check the final sum of squares directly.

Beyond STEP

The numbers $a + b\mathrm{i}$ with a, b integers form the *Gaussian integers*, a ring in which, like the ordinary integers, factorisation into primes is unique. An ordinary prime $p \equiv 1 \pmod 4$ splits as $(x + y\mathrm{i})(x - y\mathrm{i})$, which is Fermat's two-squares theorem. Counting the ways of sharing these factors out explains exactly how many representations a number has, and why 1002082 has more than the one we used. Davenport's *The Higher Arithmetic* treats sums of two squares beautifully; Allenby's *Rings, Fields and Groups* covers the Gaussian integers as a ring.

WORKED SOLUTION

STEP II 2023, Question 8

Pure • Vectors in Three Dimensions

STEP II 2023, Question 8

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Name the three pairs of opposite edges, and part (i) becomes three linear equations. After that, everything follows from turning each length condition into a dot product, using $|\mathbf{u}-\mathbf{v}|^2 = |\mathbf{u}|^2 + |\mathbf{v}|^2 - 2\,\mathbf{u}\cdot\mathbf{v}$. Once all three dot products $\mathbf{a}\cdot\mathbf{b}$, $\mathbf{b}\cdot\mathbf{c}$, $\mathbf{c}\cdot\mathbf{a}$ are written in terms of $|\mathbf{a}|^2$, $|\mathbf{b}|^2$ and $|\mathbf{c}|^2$, parts (iii) and (iv) are just expanding brackets.

Solution

Part (i)

Label the tetrahedron $OABC$. Its six edges form three pairs of edges that don't share a vertex:

$$OA \text{ and } BC, \qquad OB \text{ and } CA, \qquad OC \text{ and } AB.$$

Write $OA = p$, $BC = p'$, $OB = q$, $CA = q'$, $OC = r$, $AB = r'$. The four faces have perimeters

$$OAB : p+q+r', \qquad OAC : p+r+q',$$
$$OBC : q+r+p', \qquad ABC : p'+q'+r'.$$

If the tetrahedron is isosceles, then $p = p'$, $q = q'$, $r = r'$, and each of the four perimeters equals $p+q+r$.

Conversely, suppose all four perimeters are equal. Put $X = p-p'$, $Y = q-q'$, $Z = r-r'$. Comparing each of the first three faces with ABC:

$$(p+q+r') - (p'+q'+r') = X+Y = 0,$$
$$(p+r+q') - (p'+q'+r') = X+Z = 0,$$
$$(q+r+p') - (p'+q'+r') = Y+Z = 0.$$

Adding all three gives $2(X+Y+Z) = 0$. Subtracting each equation in turn from $X+Y+Z = 0$ gives $Z = 0$, $Y = 0$ and $X = 0$. So $p = p'$, $q = q'$ and $r = r'$: the tetrahedron is isosceles. That proves both directions.

Part (ii)

Since $\overrightarrow{BC} = \mathbf{c} - \mathbf{b}$, the condition $OA = BC$ says

$$|\mathbf{a}|^2 = |\mathbf{c} - \mathbf{b}|^2 = |\mathbf{b}|^2 + |\mathbf{c}|^2 - 2\,\mathbf{b} \cdot \mathbf{c},$$

which rearranges to $2\,\mathbf{b} \cdot \mathbf{c} = |\mathbf{b}|^2 + |\mathbf{c}|^2 - |\mathbf{a}|^2$, as required.

The same argument with the other two pairs of opposite edges ($OB = CA$ and $OC = AB$) gives

$$2\,\mathbf{a} \cdot \mathbf{c} = |\mathbf{a}|^2 + |\mathbf{c}|^2 - |\mathbf{b}|^2, \qquad 2\,\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|^2 + |\mathbf{b}|^2 - |\mathbf{c}|^2.$$

Adding these two, the $|\mathbf{b}|^2$ and $|\mathbf{c}|^2$ terms cancel:

$$2\,\mathbf{a} \cdot \mathbf{b} + 2\,\mathbf{a} \cdot \mathbf{c} = 2|\mathbf{a}|^2, \quad \text{so} \quad \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = |\mathbf{a}|^2.$$

The three dot-product formulas are symmetric in $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, so in the same way

$$\mathbf{b} \cdot (\mathbf{c} + \mathbf{a}) = |\mathbf{b}|^2, \qquad \mathbf{c} \cdot (\mathbf{a} + \mathbf{b}) = |\mathbf{c}|^2.$$

Part (iii)

Let $S = |\mathbf{a}|^2 + |\mathbf{b}|^2 + |\mathbf{c}|^2$. Adding the three formulas for $2\,\mathbf{b} \cdot \mathbf{c}$, $2\,\mathbf{c} \cdot \mathbf{a}$ and $2\,\mathbf{a} \cdot \mathbf{b}$ from part (ii), each squared length appears twice with a plus sign and once with a minus sign, so

$$2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = S.$$

Distance from O.

$$OG^2 = \tfrac{1}{16}|\mathbf{a} + \mathbf{b} + \mathbf{c}|^2 = \tfrac{1}{16}\big(S + 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a})\big) = \tfrac{1}{16}(2S) = \tfrac{1}{8}S.$$

Distance from A. Here $\overrightarrow{AG} = \tfrac{1}{4}(\mathbf{b} + \mathbf{c} - 3\mathbf{a})$, and

$$\begin{aligned} |\mathbf{b} + \mathbf{c} - 3\mathbf{a}|^2 &= |\mathbf{b} + \mathbf{c}|^2 - 6\,\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) + 9|\mathbf{a}|^2 \\ &= |\mathbf{b}|^2 + |\mathbf{c}|^2 + 2\,\mathbf{b} \cdot \mathbf{c} - 6|\mathbf{a}|^2 + 9|\mathbf{a}|^2 \qquad \text{using } \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = |\mathbf{a}|^2 \\ &= |\mathbf{b}|^2 + |\mathbf{c}|^2 + \big(|\mathbf{b}|^2 + |\mathbf{c}|^2 - |\mathbf{a}|^2\big) + 3|\mathbf{a}|^2 \\ &= 2S. \end{aligned}$$

So $AG^2 = \tfrac{1}{16}(2S) = \tfrac{1}{8}S$ too.

Distances from B and C. Every fact we used about $\mathbf{a}$ holds equally for $\mathbf{b}$ and for $\mathbf{c}$ (the end of part (ii)), and S is symmetric. So the identical calculation gives $BG^2 = CG^2 = \tfrac{1}{8}S$.

Hence G is at distance $\sqrt{S/8}$ from all four vertices.

Part (iv)

Expand the suggested vector:

$$\begin{aligned} |\mathbf{a} - \mathbf{b} - \mathbf{c}|^2 &= S - 2\,\mathbf{a} \cdot \mathbf{b} - 2\,\mathbf{a} \cdot \mathbf{c} + 2\,\mathbf{b} \cdot \mathbf{c} \\ &= S - 2|\mathbf{a}|^2 + 2\,\mathbf{b} \cdot \mathbf{c} \qquad \text{by part (ii)} \\ &= \big(|\mathbf{b}|^2 + |\mathbf{c}|^2 - |\mathbf{a}|^2\big) + 2\,\mathbf{b} \cdot \mathbf{c} = 4\,\mathbf{b} \cdot \mathbf{c}. \end{aligned}$$

So

$$\mathbf{b}\cdot\mathbf{c} = \tfrac{1}{4}|\mathbf{a}-\mathbf{b}-\mathbf{c}|^2 \geq 0,$$

and the angle BOC is not obtuse.

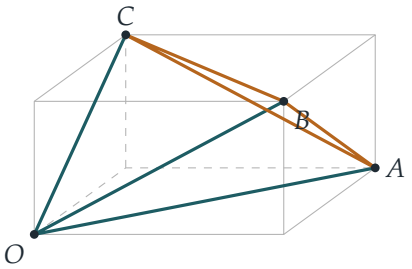
Ask yourself That is one angle out of twelve. How do we get all the others without repeating the work?

The same calculation with $\mathbf{b}-\mathbf{c}-\mathbf{a}$ and $\mathbf{c}-\mathbf{a}-\mathbf{b}$ gives $\mathbf{c}\cdot\mathbf{a}\geq 0$ and $\mathbf{a}\cdot\mathbf{b}\geq 0$, so no angle at O is obtuse. Being isosceles is a property of the tetrahedron, not of how we label it, so any vertex could have been called O. The same argument therefore works at A, B and C. (Alternatively, every face has sides p, q, r, so all four faces are congruent triangles. Their angles are exactly the three angles BOC, COA, AOB at O.) Hence none of the angles between edges meeting at a vertex is obtuse.

Right angles? Equality $\mathbf{b}\cdot\mathbf{c}=0$ happens only if $|\mathbf{a}-\mathbf{b}-\mathbf{c}|=0$, that is $\mathbf{a}=\mathbf{b}+\mathbf{c}$. But then A lies in the plane through O, B, C (in fact $OBAC$ is a rectangle), and $OABC$ is flat, not a tetrahedron. The same holds at every angle, so **no angle can be a right angle**: in a genuine isosceles tetrahedron every face angle is acute.

Checks

- A concrete example: take a cuboid with side lengths x,y,z and pick four of its vertices, no two joined by an edge of the cuboid, e.g. $O=(0,0,0)$, $A=(x,y,0)$, $B=(x,0,z)$, $C=(0,y,z)$. Opposite edges are diagonals of opposite faces, so they're equal. With $x=1$, $y=2$, $z=3$ we get $\mathbf{b}\cdot\mathbf{c}=9$ and $\mathbf{a}-\mathbf{b}-\mathbf{c}=(0,0,-6)$, and indeed $\frac{1}{4}\cdot 36=9$. Also $G=(\frac{1}{2},1,\frac{3}{2})$ is the centre of the cuboid, at distance $\sqrt{14}/2$ from every vertex, which matches $\sqrt{S/8}$ with $S=5+10+13=28$.
- The flat case: $\mathbf{b}=(3,0,0)$, $\mathbf{c}=(0,4,0)$, $\mathbf{a}=(3,4,0)$ gives a rectangle whose opposite "edges" (sides and diagonals) are equal and which has a right angle. That is exactly the degenerate case we excluded.
- These identities were also confirmed by computer on thousands of random cuboid-based examples.



Common slips

- In part (i), proving only one direction. "If and only if" needs both, and the converse is the part with content.
- In part (iv), forgetting to rule out the right angle properly. $\mathbf{b} \cdot \mathbf{c} = 0$ forces $\mathbf{a} = \mathbf{b} + \mathbf{c}$, which makes the four points coplanar.
- Checking only the angles at O and stopping. Say why the same holds at the other three vertices.

Beyond STEP

The cuboid picture above is the whole story: every isosceles tetrahedron (also called a *disphenoid*) is obtained by choosing alternate vertices of a cuboid. Then part (iii) says the centroid is the cuboid's centre, and part (iv) says the cuboid's sides have positive length. Converting length conditions into dot products, and so into a *Gram matrix* of inner products, is a standard tool in first-year linear algebra and geometry. Strang's *Introduction to Linear Algebra* is a good place to see it developed.

WORKED SOLUTION

STEP II 2023, Question 9

Mechanics • Connected Particles on Slopes

STEP II 2023, Question 9

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

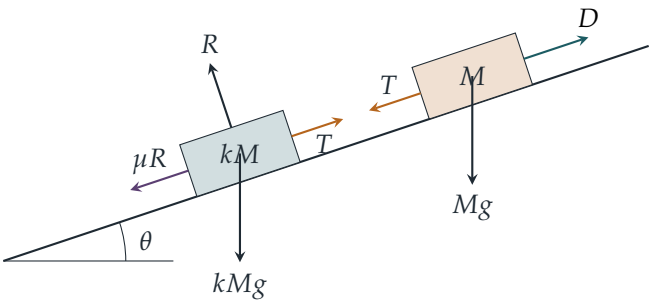
The key idea

Write down Newton's second law separately for the truck and for the trailer in each of the three situations. Going down the slope is just going up a slope of angle $-\alpha$, so one set of equations covers all three cases. The truck's equation alone gives the relation in (i) almost for free. For (ii), the whole problem comes down to comparing $1 - \cos\alpha$ with $\sin\alpha$.

Solution

We model the truck and trailer as particles joined by a light rigid bar along the line of motion. Both move in the direction of the truck, so the trailer's friction acts against the motion with magnitude μR, where R is the normal reaction on the trailer. The tension T pulls the trailer forwards and the truck backwards. (If T came out negative, the bar would be in thrust; the algebra is unchanged.)

Treat the slope as having angle θ, measured so that $\theta = \alpha$ is uphill, $\theta = 0$ is level and $\theta = -\alpha$ is downhill.



With acceleration a up the slope, resolving along the slope for each body, and perpen-

dicular to it for the trailer ($R = kMg\cos\theta$), gives

$$\text{truck:} \qquad D - T - Mg\sin\theta = Ma, \tag{1}$$

$$\text{trailer:} \qquad T - kMg\sin\theta - \mu kMg\cos\theta = kMa. \tag{2}$$

Adding (1) and (2) eliminates T:

$$a = \frac{D}{(1+k)M} - g\sin\theta - \frac{\mu k g\cos\theta}{1+k}.$$

Substituting into (1), $T = D - Mg\sin\theta - Ma$, so

$$T = D - \frac{D}{1+k} + \frac{\mu kMg\cos\theta}{1+k} = \frac{kD + \mu kMg\cos\theta}{1+k}.$$

The $\sin\theta$ terms cancel, so the tension depends on θ only through $\cos\theta$.

Part (i)

Since $\cos(-\alpha) = \cos\alpha$, the cases $\theta = \alpha$ and $\theta = -\alpha$ give the same tension:

$$T_1 = T_3 = \frac{kD + \mu kMg\cos\alpha}{1+k}, \qquad T_2 = \frac{kD + \mu kMg}{1+k}.$$

For the second result we only need the truck equation (1) in the three cases:

$$\begin{aligned} Ma_1 &= D - T_1 - Mg\sin\alpha, \\ Ma_2 &= D - T_2, \\ Ma_3 &= D - T_3 + Mg\sin\alpha. \end{aligned}$$

Adding the first and third equations and using $T_3 = T_1$ gives $M(a_1 + a_3) = 2D - 2T_1$. Subtracting twice the middle equation,

$$M(a_1 + a_3 - 2a_2) = (2D - 2T_1) - (2D - 2T_2) = 2(T_2 - T_1),$$

as required.

Part (ii)(a)

First inequality. By (i), $a_2 < \frac{1}{2}(a_1 + a_3)$ is equivalent to $T_2 > T_1$ (since $M > 0$). From the formulas above,

$$T_2 - T_1 = \frac{\mu kMg\,(1 - \cos\alpha)}{1+k} > 0,$$

because $0 < \alpha < \frac{1}{2}\pi$ gives $\cos\alpha < 1$, and $\mu > 0$ since friction acts. So $a_2 < \frac{1}{2}(a_1 + a_3)$.

Second inequality. $\frac{1}{2}(a_1 + a_3) < a_3$ is equivalent to $a_1 < a_3$. From the formula for a, the D and friction terms are the same for $\theta = \pm\alpha$, so

$$a_3 - a_1 = 2g\sin\alpha > 0.$$

(Equivalently, subtract the truck equations: $M(a_3 - a_1) = 2Mg\sin\alpha$, using $T_1 = T_3$.) Hence

$$a_2 < \tfrac{1}{2}(a_1 + a_3) < a_3.$$

Part (ii)(b)

Using the formula for a with $\theta = 0$ and $\theta = \alpha$,

$$a_2 - a_1 = g\sin\alpha - \frac{\mu k g\,(1-\cos\alpha)}{1+k} = g\left[\sin\alpha - \frac{\mu k}{1+k}(1-\cos\alpha)\right].$$

Ask yourself Going uphill costs $g\sin\alpha$ of acceleration, but the reduced normal reaction saves some friction. Which effect wins?

The friction saving is the smaller one. First, $\dfrac{k}{1+k} < 1$ and $\mu < 1$, so $0 \le \dfrac{\mu k}{1+k} < 1$. Second, using the half-angle formulae,

$$1-\cos\alpha = 2\sin^2\tfrac{1}{2}\alpha, \qquad \sin\alpha = 2\sin\tfrac{1}{2}\alpha\cos\tfrac{1}{2}\alpha,$$

so $\dfrac{1-\cos\alpha}{\sin\alpha} = \tan\tfrac{1}{2}\alpha < 1$, since $0 < \tfrac{1}{2}\alpha < \tfrac{1}{4}\pi$. Therefore

$$\frac{\mu k}{1+k}(1-\cos\alpha) < 1-\cos\alpha < \sin\alpha,$$

where the first inequality is strict because $1-\cos\alpha > 0$. So $a_2 - a_1 > 0$, that is, $a_1 < a_2$. This is where $\mu < 1$ is used. With a very heavy trailer, a large μ and a steep slope, the friction saving can exceed $g\sin\alpha$.

Checks

- Take $M = 1000\,\text{kg}$, $k = 0.5$, $\mu = 0.4$, $\alpha = 30^\circ$, $D = 5000\,\text{N}$ and $g = 9.8\,\text{m s}^{-2}$. Then $a_1 \approx -2.698$, $a_2 \approx 2.027$ and $a_3 \approx 7.102\,\text{m s}^{-2}$, with $T_1 = T_3 \approx 2798\,\text{N}$ and $T_2 \approx 2973\,\text{N}$. Indeed $\frac{1}{2}(a_1 + a_3) \approx 2.202$ lies strictly between a_2 and a_3, and $a_1 < a_2$. A computer check over many random parameter sets with $\mu < 1$ confirmed all the identities and inequalities.
- If $\mu = 0$, the tension is $kD/(1+k)$ on every slope and a_2 is exactly the mean of a_1 and a_3. That is why (ii)(a) needs friction to be present.
- With $\mu = 3$, $k = 100$ and $\alpha = 1.5$ radians, $a_1 > a_2$, so the condition $\mu < 1$ in (b) really matters.

Common slips

- Using $\mu k M g$ for the friction on the slopes. The normal reaction there is $kMg\cos\alpha$, and this difference is the whole content of part (ii).
- Getting the sign of the weight component wrong when going downhill. Writing everything in terms of θ and then setting $\theta = -\alpha$ avoids this.
- In (ii)(b), arguing only that $\mu k/(1+k) < 1$ without comparing $1-\cos\alpha$ with $\sin\alpha$. Both facts are needed.

Beyond STEP

Splitting a system into bodies and applying Newton's laws to each, with internal forces appearing in equal and opposite pairs, is the basic method of rigid-body and multi-particle mechanics. Adding the equations gives the motion of the centre of mass, and the internal force drops out. The inequality $1 - \cos\alpha < \sin\alpha$ is also a small example of comparing functions using half-angle identities, a technique that is useful throughout analysis. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas carefully.

WORKED SOLUTION

STEP II 2023, Question 10

Mechanics • Projectile Envelope

STEP II 2023, Question 10

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

A trajectory and the curve E *touch* exactly when the equation for their common points has a repeated root, so part (i) is a discriminant calculation. Asking for this to happen for every angle forces E to be the *envelope* of all the trajectories: the boundary of where particles of speed u can reach. The rest of the question looks at that boundary from three angles: which points can be hit, where the particles are at a fixed time, and a second particle sliding along the envelope itself.

Solution

Throughout, the particles are modelled as points moving freely under gravity, with no air resistance, and $c = \cos\alpha$, $s = \sin\alpha$.

Part (i)

With t the time since projection, P_α stays in the plane $y = 0$ with

$$x = uct, \qquad z = ust - \tfrac{1}{2}gt^2.$$

Eliminating $t = x/(uc)$ gives the trajectory

$$z = x\tan\alpha - \frac{gx^2}{2u^2c^2}.$$

Common points with E satisfy $A - Bx^2 = x\tan\alpha - \dfrac{gx^2}{2u^2c^2}$, that is

$$\left(\frac{g}{2u^2c^2} - B\right)x^2 - x\tan\alpha + A = 0. \qquad (*)$$

If the coefficient of x^2 were zero, $(*)$ would be linear with $\tan\alpha \neq 0$, giving a single crossing rather than a touching, so $(*)$ is a genuine quadratic. The curves touch exactly

once when $(*)$ has a repeated root, i.e. when its discriminant vanishes:

$$\tan^2\alpha = 4A\left(\frac{g}{2u^2c^2} - B\right).$$

Multiplying by u^2c^2:

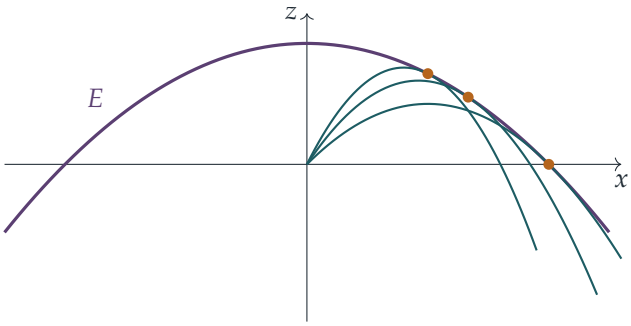
$$\begin{aligned} u^2s^2 &= 2gA - 4ABu^2c^2 \\ u^2 - u^2c^2 &= 2gA - 4ABu^2c^2 \qquad (\text{using } s^2 = 1 - c^2) \\ u^2 - 2gA &= u^2(1 - 4AB)\cos^2\alpha, \end{aligned}$$

as required.

If this holds for every α in $(0, \frac{1}{2}\pi)$, then the left-hand side is a constant while $\cos^2\alpha$ takes every value in $(0, 1)$. A constant equal to $k\cos^2\alpha$ for all these values forces $k = 0$ and then the constant is 0. So $u^2 = 2gA$ and $4AB = 1$:

$$A = \frac{u^2}{2g}, \qquad B = \frac{g}{2u^2}.$$

Conversely these values make the identity true for every α. The touching point is the repeated root $x = \dfrac{\tan\alpha}{2(g/(2u^2c^2) - B)} = \dfrac{u^2}{g\tan\alpha}$.



The figure (drawn with $u^2/g = 2$) shows three trajectories each touching E once, at the amber points.

Part (ii)

Every particle moves in the vertical plane containing its initial velocity, and everything is symmetric about the z-axis. So take a point at horizontal distance r from the z-axis and height z, and work in the vertical plane through it, with $\theta \in (-\frac{1}{2}\pi, \frac{1}{2}\pi)$ the angle of projection above the horizontal towards the point.

If $r = 0$, only particles fired straight up or down can arrive; the one fired upwards reaches every height up to $u^2/(2g)$ and the one fired downwards every negative height, so the reachable points on the axis are those with $z \le u^2/(2g)$.

If $r > 0$, exactly as in part (i) the particle passes through the point if

$$z = r\tan\theta - \frac{gr^2}{2u^2}(1 + \tan^2\theta),$$

using $1/\cos^2\theta = 1 + \tan^2\theta$. Writing $T = \tan\theta$, which can take any real value, this is the quadratic

$$\frac{gr^2}{2u^2}T^2 - rT + \left(z + \frac{gr^2}{2u^2}\right) = 0,$$

which has a real root precisely when

$$r^2 \geq \frac{2gr^2}{u^2}\left(z + \frac{gr^2}{2u^2}\right) \iff z \leq \frac{u^2}{2g} - \frac{gr^2}{2u^2},$$

dividing by $2gr^2/u^2 > 0$.

So, with $r^2 = x^2 + y^2$, the points that can be hit are exactly those on or below the paraboloid

$$z = \frac{u^2}{2g} - \frac{g}{2u^2}(x^2 + y^2),$$

which is the surface obtained by rotating E (with the values of part (i)) about the z-axis. Points strictly below it are reached by two directions in the plane, points on it by exactly one (the repeated root, which matches the touching in part (i)), and points above it by none.

Part (iii)

Let the particle have initial velocity $u\mathbf{n}$, where $\mathbf{n}$ is a unit vector. After time t its position is

$$\mathbf{r} = ut\,\mathbf{n} - \tfrac{1}{2}gt^2\,\mathbf{k},$$

where $\mathbf{k}$ is the upward unit vector. Hence

$$\left|\mathbf{r} - (-\tfrac{1}{2}gt^2\,\mathbf{k})\right| = |ut\,\mathbf{n}| = ut.$$

So at time t every particle lies on the sphere with centre $(0, 0, -\frac{1}{2}gt^2)$ and radius ut. Physically: gravity accelerates every particle equally, so relative to a point falling freely from the origin they all fly outwards in straight lines at speed u.

Part (iv)

With τ the time since Q was projected, Q has $y = 0$ and

$$x = u\tau, \qquad z = A - \tfrac{1}{2}g\tau^2.$$

Substituting $\tau = x/u$:

$$z = A - \frac{g}{2u^2}x^2 = A - Bx^2,$$

using $B = g/(2u^2)$ from part (i). So Q is always on E.

Part (v)

Suppose Q is projected at time T after the explosion, and that Q and P_α are at the same point at time t ($\geq T$) after the explosion. Then $\tau = t - T$ and both y-coordinates are 0, so we need the x- and z-coordinates to agree:

$$uct = u(t - T),$$
$$ust - \tfrac{1}{2}gt^2 = A - \tfrac{1}{2}g(t - T)^2.$$

The first gives $t - T = ct$, so $T = t(1 - c)$. Putting $t - T = ct$ into the second and using $1 - c^2 = s^2$:

$$ust - \tfrac{1}{2}gs^2t^2 = A = \frac{u^2}{2g}.$$

Ask yourself How do we solve this for t without a messy formula?

Multiply by $2g$ and collect everything on one side: $g^2s^2t^2 - 2gus\,t + u^2 = 0$, which is the perfect square

$$(gst - u)^2 = 0, \qquad \text{so} \qquad t = \frac{u}{g\sin\alpha}.$$

A repeated root is just what we should expect: the collision has to happen where the trajectory of P_α touches E, since that is the only point the two paths share. Indeed P_α is then at $x = uct = u^2/(g\tan\alpha)$, the touching point found in part (i). Therefore

$$T = t(1 - \cos\alpha) = \frac{u(1 - \cos\alpha)}{g\sin\alpha},$$

as required. Since $0 < 1 - \cos\alpha < 1$, this gives $0 < T < t$, so Q is indeed launched after the explosion and before the collision.

Checks

- In part (i), $\alpha = \frac{1}{4}\pi$ gives the touching point $x = u^2/g$, $z = 0$: the maximum range on level ground, which the boundary of the reachable region should indeed pass through.
- In part (ii), on the axis $r = 0$ the bound is $z \leq u^2/(2g)$, the greatest height of a particle fired straight up.
- In part (v), with $u = 7$, $g = 9.8$ and $\alpha = 0.5$, the formula gives $T \approx 0.182$; stepping both motions forward in time confirms that the particles meet at $t = u/(g\sin\alpha) \approx 1.49$, and that launching Q even 0.05 later makes them miss.
- As $\alpha \to \frac{1}{2}\pi$, $T \to u/g$ and $t \to u/g$: P_α goes almost straight up to the top of E at $(0, 0, A)$, arriving at time u/g, just as Q sets off from there.

Common slips

- Setting the discriminant to zero without first rewriting the common-points equation as a single quadratic in x, and so dropping the $-B$ term from the x^2 coefficient.
- In part (ii), describing only the boundary surface: the question asks for the whole set, which is the solid region on and below the paraboloid.
- In part (v), using the same time variable for both particles: Q has only been moving for $t - T$.

Beyond STEP

The curve E is the *envelope* of the family of trajectories: a curve tangent to every member of the family. In general the envelope of a family $F(x, z, \alpha) = 0$ is found by eliminating α between $F = 0$ and $\partial F/\partial \alpha = 0$, which is the calculus version of the repeated-root condition used here. The paraboloid in part (ii) is often called the *paraboloid of safety*, and part (iii) is an example of changing to a freely falling frame, where gravity disappears. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both ideas.

WORKED SOLUTION

STEP II 2023, Question 11

Probability and Statistics • Mixture Distributions

STEP II 2023, Question 11

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 13.5, *Asking Awkward Questions*: https://beyondstep.keithmcnulty.org/ch13/awkward-questions.html

The key idea

Y is a *mixture*: first pick a distribution at random, then draw from it. Its mean is the weighted average of the two means, but its variance has an extra term, because the spread between the two means adds variability of its own. Parts (ii) and (iii) both toss a coin before using a die, yet they behave completely differently. In (ii) a fresh coin is tossed for every throw, so the mixing averages out. In (iii) a single coin decides everything, and that extra term ends up dominating.

Solution

Part (i)

Choosing which variable to use is done independently of the values of X_1 and X_2, so

$$\mathrm{P}(Y = x_i) = p\,a_i + q\,b_i \qquad (i = 1, \ldots, n).$$

Hence

$$\mathrm{E}(Y) = \sum_i x_i(pa_i + qb_i) = p\sum_i x_i a_i + q\sum_i x_i b_i = p\mu_1 + q\mu_2.$$

In the same way, $\mathrm{E}(Y^2) = p\,\mathrm{E}(X_1^2) + q\,\mathrm{E}(X_2^2) = p(\sigma_1^2 + \mu_1^2) + q(\sigma_2^2 + \mu_2^2)$. So

$$\begin{aligned}\mathrm{Var}(Y) &= p\sigma_1^2 + q\sigma_2^2 + p\mu_1^2 + q\mu_2^2 - (p\mu_1 + q\mu_2)^2 \\ &= p\sigma_1^2 + q\sigma_2^2 + p(1-p)\mu_1^2 + q(1-q)\mu_2^2 - 2pq\mu_1\mu_2 \\ &= p\sigma_1^2 + q\sigma_2^2 + pq\mu_1^2 + pq\mu_2^2 - 2pq\mu_1\mu_2 \\ &= p\sigma_1^2 + q\sigma_2^2 + pq(\mu_1 - \mu_2)^2,\end{aligned}$$

using $1 - p = q$ and $1 - q = p$. This is the required result.

Part (ii)

Let H be the event that the coin shows heads, and S the event that the die shows a six. Coin and die are independent, so

$$\mathrm{P}(B=1) = \mathrm{P}(H)\mathrm{P}(S) + \mathrm{P}(H^c)\mathrm{P}(S^c) = \tfrac{1}{2}\cdot\tfrac{1}{6} + \tfrac{1}{2}\cdot\tfrac{5}{6} = \tfrac{1}{2}.$$

So each B is 0 or 1 with probability $\frac{1}{2}$ each. Z_1 is the number of 1s among n independent copies of B, each a success with the same probability $\frac{1}{2}$. That is exactly the binomial set-up, so $Z_1 \sim \mathrm{B}(n, \frac{1}{2})$, with mean $\frac{n}{2}$ and variance $\frac{n}{4}$.

For large n we approximate Z_1 by $\mathrm{N}(\frac{n}{2}, \frac{n}{4})$, whose standard deviation is $\frac{\sqrt{n}}{2}$. Being within 10% of the mean means $|Z_1 - \frac{n}{2}| \le \frac{n}{20}$. Standardising, with $Z \sim \mathrm{N}(0,1)$,

$$\mathrm{P}\big(|Z_1 - \tfrac{n}{2}| \le \tfrac{n}{20}\big) \approx \mathrm{P}\Big(|Z| \le \frac{n/20}{\sqrt{n}/2}\Big) = \mathrm{P}\Big(|Z| \le \frac{\sqrt{n}}{10}\Big) = 2\Phi\Big(\frac{\sqrt{n}}{10}\Big) - 1.$$

As $n \to \infty$, $\sqrt{n}/10 \to \infty$, so $\Phi(\sqrt{n}/10) \to 1$ and the probability tends to $2 - 1 = 1$. (A continuity correction changes the bound by only $\frac{1}{\sqrt{n}}$ standard deviations and does not affect the limit.)

Part (iii)

Let X_1 be the number of sixes and X_2 the number of non-sixes among the n dice. Then $X_1 \sim \mathrm{B}(n, \frac{1}{6})$ and $X_2 \sim \mathrm{B}(n, \frac{5}{6})$. Now Z_2 takes the value of X_1 with probability $p = \frac{1}{2}$ and that of X_2 with probability $q = \frac{1}{2}$, and the coin is independent of the dice. This is exactly the situation of part (i), with

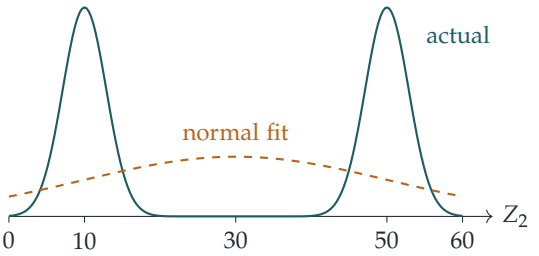
$$\mu_1 = \tfrac{n}{6}, \quad \mu_2 = \tfrac{5n}{6}, \quad \sigma_1^2 = \sigma_2^2 = n\cdot\tfrac{1}{6}\cdot\tfrac{5}{6} = \tfrac{5n}{36}.$$

So

$$\mathrm{E}(Z_2) = \tfrac{1}{2}\cdot\tfrac{n}{6} + \tfrac{1}{2}\cdot\tfrac{5n}{6} = \tfrac{n}{2},$$
$$\mathrm{Var}(Z_2) = \tfrac{1}{2}\cdot\tfrac{5n}{36} + \tfrac{1}{2}\cdot\tfrac{5n}{36} + \tfrac{1}{4}\Big(\tfrac{4n}{6}\Big)^2 = \frac{5n}{36} + \frac{n^2}{9}.$$

Ask yourself What does the distribution of Z_2 actually look like for large n?

With probability $\frac{1}{2}$, Z_2 is close to $\frac{n}{6}$, and with probability $\frac{1}{2}$ it is close to $\frac{5n}{6}$. Each of these humps has standard deviation $\frac{\sqrt{5n}}{6}$, which is tiny compared with their separation $\frac{2n}{3}$. So the distribution is *bimodal*: two narrow peaks, with almost no probability near the mean $\frac{n}{2}$. A normal distribution has a single peak at its mean, so it is exactly wrong here: it puts the most probability where Z_2 almost never lands. There is a second problem too. The standard deviation is about $\frac{n}{3}$, so $\mathrm{N}(\frac{n}{2}, \frac{n^2}{9})$ gives probability about $\mathrm{P}(Z < -1.5) \approx 0.067$ to negative values, and the same to values above n, though $0 \le Z_2 \le n$ always. The sketch shows the two pictures for $n = 60$.



Now for the final part. Z_2 is within 10% of its mean when $0.45n \le Z_2 \le 0.55n$. Conditioning on the coin,

$$\mathrm{P}(0.45n \le Z_2 \le 0.55n) = \tfrac{1}{2}\,\mathrm{P}(0.45n \le X_1 \le 0.55n) + \tfrac{1}{2}\,\mathrm{P}(0.45n \le X_2 \le 0.55n).$$

For the first term, $\mathrm{P}(0.45n \le X_1 \le 0.55n) \le \mathrm{P}(X_1 \ge 0.45n)$, and $0.45n - \frac{n}{6} = \frac{17n}{60}$. Using the normal approximation $X_1 \approx \mathrm{N}(\frac{n}{6}, \frac{5n}{36})$, whose standard deviation is $\frac{\sqrt{5n}}{6}$,

$$\mathrm{P}(X_1 \ge 0.45n) \approx \mathrm{P}\Big(Z \ge \frac{17n/60}{\sqrt{5n}/6}\Big) = 1 - \Phi\Big(\frac{17\sqrt{n}}{10\sqrt{5}}\Big) \longrightarrow 1 - 1 = 0.$$

The second term is the mirror image: $X_2 = n - X_1$, so $0.45n \le X_2 \le 0.55n$ exactly when $0.45n \le X_1 \le 0.55n$, and the second term equals the first. Hence the probability that Z_2 is within 10% of its mean is at most about $1 - \Phi\big(\frac{17\sqrt{n}}{10\sqrt{5}}\big)$, which tends to 0 as $n \to \infty$.

This step does not really depend on the approximation. Chebyshev's inequality gives exactly

$$\mathrm{P}\big(|X_1 - \tfrac{n}{6}| \ge \tfrac{17n}{60}\big) \le \frac{5n/36}{(17n/60)^2} = \frac{500}{289\,n} \longrightarrow 0.$$

Checks

- In (i), if $\mu_1 = \mu_2$ the extra term vanishes, and if $\sigma_1 = \sigma_2 = 0$ then Y takes the value μ_1 or μ_2 only, with variance $pq(\mu_1 - \mu_2)^2$, as for a scaled Bernoulli variable.
- In (ii), the exact binomial probability for $n = 400$ is 0.9598, against the normal value $2\Phi(2) - 1 = 0.9545$. For $n = 1000$ the two values are 0.9986 and 0.9984.
- In (iii), a computer simulation of Z_2 with $n = 30$ gave mean 15.00 and variance 104.2, against the formula's 15 and 104.17. The exact probability of being within 10% of the mean is about 3×10^{-11} when $n = 100$.

Common slips

- Giving $\mathrm{Var}(Y) = p\sigma_1^2 + q\sigma_2^2$, forgetting that the choice between the two means is itself a source of variation.
- Treating Z_2 in (iii) like Z_1 in (ii). In (ii) the coin is re-tossed for every die, but in (iii) one toss governs all n dice, so the outcomes are not independent.

- Explaining the poor fit only by "the variance is large". The real problem is the shape: two separate peaks, with very little probability near the mean.

Beyond STEP

Part (i) is the simplest case of the *law of total variance*, $\mathrm{Var}(Y) = \mathrm{E}\big(\mathrm{Var}(Y \mid C)\big) + \mathrm{Var}\big(\mathrm{E}(Y \mid C)\big)$, where C is the coin. Part (ii) is the weak law of large numbers, which a first course proves rigorously using Chebyshev's inequality. Part (iii) shows why that law needs independence: one shared coin is enough to destroy it. Mixture distributions like Z_2 are a staple of statistical modelling. Blitzstein and Hwang, *Introduction to Probability*, covers all three ideas with many examples.

WORKED SOLUTION

STEP II 2023, Question 12

Probability and Statistics • Distribution of a Maximum

STEP II 2023, Question 12

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

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The key idea

The maximum of several numbers is at most t exactly when *every* one of them is at most t, so the cumulative distribution function of Y is a power of the cumulative distribution function of a single X_i. Everything else follows from working with cumulative distribution functions rather than densities: the median comes from solving $\mathrm{P}(Y \leq m) = \frac{1}{2}$ directly, and the mean comes out neatly from integrating by parts so that the distribution function, not the density, appears in the integral.

Solution

Part (i)

Write $F(t) = \mathrm{P}(X_1 \leq t)$. For $0 \leq t \leq \pi$,

$$F(t) = \int_0^t \tfrac{1}{2} \sin x \, \mathrm{d}x = \tfrac{1}{2}(1 - \cos t).$$

The event $\{Y \leq t\}$ happens exactly when $X_1 \leq t$, $X_2 \leq t$, …, $X_n \leq t$ all happen: if the largest is at most t then so are all the others, and conversely. The X_i are independent and identically distributed, so the probability of this intersection is the product of the individual probabilities:

$$\mathrm{P}(Y \leq t) = \mathrm{P}(X_1 \leq t)\,\mathrm{P}(X_2 \leq t) \cdots \mathrm{P}(X_n \leq t) = \big[\mathrm{P}(X_1 \leq t)\big]^n.$$

Hence, writing G for the cumulative distribution function of Y,

$$G(t) = \frac{(1 - \cos t)^n}{2^n} \qquad (0 \leq t \leq \pi),$$

with $G(t) = 0$ for $t < 0$ and $G(t) = 1$ for $t > \pi$. Differentiating by the chain rule, the probability density function of Y is

$$g(t) = \frac{n\,(1 - \cos t)^{n-1} \sin t}{2^n} \quad \text{for } 0 \leq t \leq \pi,$$

and zero otherwise.

Part (ii)

The median $m = m(n)$ satisfies $G(m) = \frac{1}{2}$:

$$\Big(\frac{1-\cos m}{2}\Big)^n = \frac{1}{2} \iff \frac{1-\cos m}{2} = 2^{-1/n}$$
$$\iff \cos m = 1 - 2^{1-1/n},$$

taking the positive real nth root since $(1-\cos m)/2 \geq 0$. As m must lie in $[0, \pi]$, where cosine is one-to-one,

$$m(n) = \arccos\big(1 - 2^{1-1/n}\big).$$

As n increases, $1/n$ decreases, so $2^{1-1/n}$ increases (towards 2) and $1 - 2^{1-1/n}$ decreases (towards -1). Since arccos is a decreasing function, $m(n)$ *increases* with n, and $m(n) \to \arccos(-1) = \pi$ as $n \to \infty$. (For $n = 1$ we get $m(1) = \arccos 0 = \pi/2$, as the symmetry of $\frac{1}{2}\sin x$ about $\pi/2$ demands.) This makes sense: the maximum of more and more values is pushed towards the top of the range.

Part (iii)

By definition $\mu(n) = \int_0^\pi t\, g(t)\, \mathrm{d}t$. Since $g = G'$, integrate by parts with $u = t$ and $\mathrm{d}v = g(t)\,\mathrm{d}t$, $v = G(t)$:

$$\mu(n) = \Big[\, t\, G(t)\Big]_0^\pi - \int_0^\pi G(t)\, \mathrm{d}t$$
$$= \pi \cdot 1 - 0 - \frac{1}{2^n}\int_0^\pi (1-\cos x)^n\, \mathrm{d}x,$$

using $G(\pi) = \big(\frac{1+1}{2}\big)^n = 1$. This is the required formula.

> **?** **Ask yourself** Why integrate by parts rather than trying to evaluate $\int t\,(1-\cos t)^{n-1}\sin t\,\mathrm{d}t$ directly?

The target formula contains $(1-\cos x)^n$, which is $2^n G(x)$, the *distribution* function rather than the density. Integrating by parts is precisely the move that trades $t\,g(t)$ for $G(t)$, and it removes the awkward factor of t at the same time.

(a) Put $h(x) = \frac{1}{2}(1-\cos x)$, so that $0 \leq h(x) \leq 1$ on $[0, \pi]$, with $0 < h(x) < 1$ for $0 < x < \pi$. The formula reads $\mu(n) = \pi - \int_0^\pi h(x)^n\, \mathrm{d}x$, so

$$\mu(n+1) - \mu(n) = \int_0^\pi \big(h(x)^n - h(x)^{n+1}\big)\, \mathrm{d}x = \int_0^\pi h(x)^n\big(1-h(x)\big)\, \mathrm{d}x.$$

On $0 < x < \pi$ the integrand is continuous and strictly positive, because $h(x) > 0$ and $1 - h(x) > 0$ there. So the integral is strictly positive and $\mu(n+1) > \mu(n)$ for every $n \geq 1$: $\mu(n)$ increases with n.

(b) With $n = 2$, expand and use $\cos^2 x = \frac{1}{2}(1+\cos 2x)$:

$$\int_0^\pi (1-\cos x)^2\, \mathrm{d}x = \int_0^\pi \Big(1 - 2\cos x + \tfrac{1}{2} + \tfrac{1}{2}\cos 2x\Big)\mathrm{d}x = \tfrac{3}{2}\pi,$$

since $\cos x$ and $\cos 2x$ both integrate to zero over $[0, \pi]$. Hence

$$\mu(2) = \pi - \tfrac{1}{4} \cdot \tfrac{3}{2}\pi = \frac{5\pi}{8}.$$

From part (ii), $m(2) = \arccos\left(1 - \sqrt{2}\right)$. Both $5\pi/8$ and $m(2)$ lie in $[0, \pi]$, where cosine is strictly decreasing, so

$$\mu(2) < m(2) \iff \cos \tfrac{5\pi}{8} > \cos m(2) = 1 - \sqrt{2}.$$

Ask yourself How do we get an exact value for $\cos \frac{5\pi}{8}$?

Write $\cos \frac{5\pi}{8} = \cos\left(\frac{\pi}{2} + \frac{\pi}{8}\right) = -\sin \frac{\pi}{8}$, and use the half-angle formula $\sin^2 \frac{\pi}{8} = \frac{1}{2}\left(1 - \cos \frac{\pi}{4}\right) = \frac{2-\sqrt{2}}{4}$. As $\sin \frac{\pi}{8} > 0$,

$$\cos \tfrac{5\pi}{8} = -\tfrac{1}{2}\sqrt{2 - \sqrt{2}}.$$

So we need $-\frac{1}{2}\sqrt{2-\sqrt{2}} > 1 - \sqrt{2}$, that is, $\sqrt{2} - 1 > \frac{1}{2}\sqrt{2-\sqrt{2}}$. Both sides are positive, so this is equivalent to the inequality between their squares, multiplied by 4:

$$\begin{aligned} 4\left(3 - 2\sqrt{2}\right) > 2 - \sqrt{2} &\iff 10 > 7\sqrt{2} \\ &\iff 100 > 98, \end{aligned}$$

where the last step squares two positive numbers. This is true, and every step was an equivalence, so $\cos \frac{5\pi}{8} > 1 - \sqrt{2}$ and therefore $\mu(2) < m(2)$.

Checks

- With $n = 1$, $Y = X_1$ is symmetric about $\pi/2$, and indeed $\mu(1) = \pi - \frac{1}{2}\int_0^\pi (1 - \cos x)\,\mathrm{d}x = \pi/2 = m(1)$.
- Numerically, $\mu(2) = 5\pi/8 \approx 1.9635$ and $m(2) = \arccos(1 - \sqrt{2}) \approx 1.9979$, so the mean sits just below the median. A computer simulation of 10^5 maxima for each n from 1 to 7 agreed with both formulas to within 0.01 (for instance $\mu(3) \approx 2.1598$, $m(3) \approx 2.1986$), and both sequences increased.
- The density of Y for $n = 2$ is $\frac{1}{2}(1 - \cos t)\sin t$, which is skewed to the left (a long tail towards 0), and a left-skewed distribution typically has its mean below its median.

Common slips

- Forgetting to justify $\mathrm{P}(Y \le t) = [F(t)]^n$: both the "all are at most t" equivalence and independence are needed.
- Integrating by parts but forgetting the boundary term $\left[tG(t)\right]_0^\pi = \pi$, which is where the leading π comes from.
- Comparing angles via cosines without noting that cosine is *decreasing* on $[0, \pi]$, and so getting the inequality the wrong way round.

Beyond STEP

The maximum of a sample is an example of an *order statistic*, and the trick in part (i) gives the distribution of any sample maximum. The identity $\mathrm{E}[Y] = \pi - \int_0^\pi G(t)\,\mathrm{d}t$ is a case of the general tail-integral formula $\mathrm{E}[Y] = \int_0^\infty \mathrm{P}(Y > t)\,\mathrm{d}t$ for non-negative random variables, which you will meet early in a university probability course. Both ideas are developed in Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP III 2023, Question 1

Pure • Tangents to Conics

STEP III 2023, Question 1

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

A line is tangent to a circle exactly when its perpendicular distance from the centre equals the radius. So we write down the chord PQ in terms of p and q, compute that distance, and watch the tangency condition collapse to the given relation. After that, everything is about a single symmetric equation in two parameters: the final part asks whether the relation still holds for the *third* side, which is a job for sums and products of roots.

Solution

Part (i)

The gradient of PQ is

$$\frac{aq^2 - ap^2}{2aq - 2ap} = \frac{p+q}{2},$$

dividing by $q - p \neq 0$ because the points are distinct. The line through P is therefore $y - ap^2 = \frac{1}{2}(p+q)(x - 2ap)$, which simplifies to

$$(p+q)x - 2y - 2apq = 0.$$

The distance from the centre $C(0, 3a)$ to this line is

$$\frac{|-6a - 2apq|}{\sqrt{(p+q)^2 + 4}} = \frac{2a|pq+3|}{\sqrt{(p+q)^2+4}}.$$

The line is tangent to the circle exactly when this equals $2a$, that is (squaring, since both sides are positive, and cancelling $4a^2 \neq 0$) when

$$(pq+3)^2 = (p+q)^2 + 4 \iff p^2q^2 + 6pq + 5 = (p+q)^2.$$

This is $(*)$, so when $(*)$ holds the line PQ is a tangent to the circle. (Every step was reversible, so the converse holds too, and we shall need that in part (iii).)

Part (ii)

Expanding $(*)$ and collecting powers of q:

$$(1-p^2)q^2 - 4pq + (p^2-5) = 0.$$

Since $p^2 \neq 1$, the leading coefficient is nonzero, so this is a genuine quadratic in q. Its discriminant is

$$\begin{aligned} 16p^2 - 4(1-p^2)(p^2-5) &= 16p^2 - 4(-p^4+6p^2-5) \\ &= 4(p^4 - 2p^2 + 5) \\ &= 4\big((p^2-1)^2 + 4\big) > 0, \end{aligned}$$

so there are two distinct real roots q_1, q_2. Neither equals p: putting $q = p$ in the left-hand side gives $-p^4 - 2p^2 - 5 < 0$. So Q_1 and Q_2 really are points distinct from P.

From the quadratic,

$$q_1 + q_2 = \frac{4p}{1-p^2}, \qquad q_1 q_2 = \frac{p^2-5}{1-p^2}.$$

Part (iii)

Take the triangle PQ_1Q_2, where $Q_i = (2aq_i, aq_i^2)$ and q_1, q_2 are the roots from part (ii). Its vertices all lie on $x^2 = 4ay$, and they are three distinct points (as shown in (ii)); three distinct points of a parabola are never collinear, since a line meets a parabola at most twice. So this is a genuine triangle.

By part (i), the sides PQ_1 and PQ_2 are tangent to the circle, since (p, q_1) and (p, q_2) satisfy $(*)$.

> **Ask yourself** How can we show that the third side Q_1Q_2 is tangent when we don't want to solve for q_1 and q_2 explicitly?

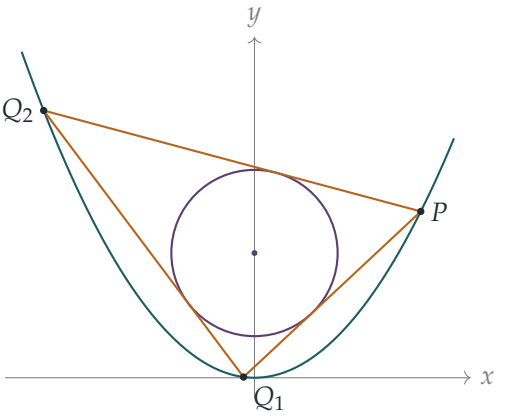
By the reversible argument of part (i) (with p, q replaced by q_1, q_2), Q_1Q_2 is tangent precisely when

$$(q_1+q_2)^2 = q_1^2 q_2^2 + 6q_1q_2 + 5,$$

and both sides involve only $s = q_1 + q_2$ and $t = q_1 q_2$, which we know. Multiply through by $(1-p^2)^2$. The left-hand side becomes $16p^2$. The right-hand side becomes

$$\begin{aligned} &(p^2-5)^2 + 6(p^2-5)(1-p^2) + 5(1-p^2)^2 \\ &\quad = (p^4 - 10p^2 + 25) + (-6p^4 + 36p^2 - 30) + (5p^4 - 10p^2 + 5) \\ &\quad = 16p^2. \end{aligned}$$

The two sides agree, so Q_1Q_2 is also tangent to the circle, and PQ_1Q_2 is the required triangle.



The figure shows the case $a = 1$, $p = 2$, where $q_{1,2} = \frac{1}{3}(-4 \pm \sqrt{13})$.

Checks

- Take $a = 1$, $p = 2$: the quadratic is $3q^2 + 8q + 1 = 0$, with $q_1 + q_2 = -\frac{8}{3} = \frac{4p}{1-p^2}$ and $q_1 q_2 = \frac{1}{3} = \frac{p^2-5}{1-p^2}$. Then $(q_1 + q_2)^2 = \frac{64}{9}$ and $q_1^2 q_2^2 + 6q_1 q_2 + 5 = \frac{1}{9} + 2 + 5 = \frac{64}{9}$, as part (iii) requires.
- Numerically, with $q_1 \approx -0.1315$, $q_2 \approx -2.5352$, all three side lines are at distance 2.000 from $(0, 3)$; a computer check over thousands of random a and p agreed to about 10^{-13}.
- The excluded case $p = \pm 1$ is instructive: for $p = 1$ the quadratic degenerates to $-4q - 4 = 0$, giving only $q = -1$. The "missing" tangent from $P(2a, a)$ is the vertical line $x = 2a$, which touches the circle but meets the parabola only at P.

Common slips

- Forgetting the modulus in the distance formula, or squaring without noting that both sides are positive.
- Using only the "if" direction of part (i): part (iii) needs the converse to turn tangency of Q_1Q_2 into an algebraic condition, so say that the steps are reversible.
- Not checking that $q_1, q_2 \neq p$, so that the triangle is not degenerate.

Beyond STEP

This is an instance of *Poncelet's porism*: if a triangle can be inscribed in one conic and circumscribed about another, then such a triangle can be started from *any* point of the first conic. Here the parabola and the circle are such a pair, and part (iii) proves closure

directly with Vieta's formulas. The general proof uses projective geometry and, most elegantly, elliptic curves. Siklos's *Advanced Problems in Mathematics* has more questions in this style of parametric conic geometry.

WORKED SOLUTION

STEP III 2023, Question 2

Pure • Polar Curve Areas

STEP III 2023, Question 2

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Both curves are drawn by a radius sweeping from $\theta = 0$ to $\theta = \pi$, so the whole question is about which curve is further out at each angle. Comparing $k(1 + \sin\theta)$ with $k + \cos\theta$ shows that they swap over exactly once, at $\theta = \alpha$. Regions A and B together are then the overlap of the two curves, and every area comes from the polar formula $\frac{1}{2}\int r^2 \, \mathrm{d}\theta$ with $\sin^2$ and $\cos^2$ handled by double angles.

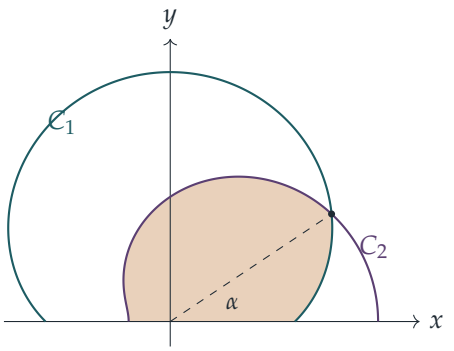
Solution

Part (i)

Some values help with the sketch:

	$\theta = 0$	$\theta = \frac{\pi}{2}$	$\theta = \pi$
C_1: $r = k(1 + \sin\theta)$	k	$2k$	k
C_2: $r = k + \cos\theta$	$k + 1$	k	$k - 1$

Both radii are positive throughout (as $k > 1$). C_1 is the upper half of a cardioid, symmetric about the y-axis and reaching $(0, 2k)$. C_2 is a slightly squashed semicircle of "radius about k", pushed 1 unit to the right at $\theta = 0$ and 1 unit in at $\theta = \pi$. The sketch below has $k = 1.5$; the shaded region is where the curves overlap.



At an intersection the radii agree, so

$$k(1+\sin\alpha) = k+\cos\alpha \iff k\sin\alpha = \cos\alpha.$$

Here $\cos\alpha \neq 0$ (if it were, we would need $k\sin\alpha = 0$, but $\sin\alpha = \pm 1$ then), so dividing gives $\tan\alpha = \dfrac{1}{k}$, as required. Since $0 < \frac{1}{k} < 1$ there is exactly one such angle in $[0,\pi]$, with $0 < \alpha < \frac{\pi}{4}$.

More usefully, $k+\cos\theta - k(1+\sin\theta) = \cos\theta - k\sin\theta$, which is positive for $0 \leq \theta < \alpha$ and negative for $\alpha < \theta \leq \pi$ (using the values of $\sin\alpha$, $\cos\alpha$ below, it equals $\sqrt{1+k^2}\,\sin(\alpha-\theta)$, and $\alpha-\theta$ lies in $(0,\alpha]$ or in $[\alpha-\pi, 0)$ respectively). So C_2 is outside C_1 before α, and inside it afterwards.

We will also use, from $\tan\alpha = \frac{1}{k}$,

$$\sin\alpha = \frac{1}{\sqrt{1+k^2}}, \qquad \cos\alpha = \frac{k}{\sqrt{1+k^2}}.$$

Part (ii)

Region A is swept out by the radius up to C_1 for $0 \leq \theta \leq \alpha$, so its area is

$$\begin{aligned}
\tfrac{1}{2}\int_0^\alpha k^2(1+\sin\theta)^2\,\mathrm{d}\theta &= \frac{k^2}{2}\int_0^\alpha \left(1+2\sin\theta+\tfrac{1}{2}-\tfrac{1}{2}\cos 2\theta\right)\mathrm{d}\theta \\
&= \frac{k^2}{2}\left[\tfrac{3}{2}\theta - 2\cos\theta - \tfrac{1}{4}\sin 2\theta\right]_0^\alpha \\
&= \frac{k^2}{2}\left(\tfrac{3}{2}\alpha - \tfrac{1}{2}\sin\alpha\cos\alpha - 2\cos\alpha + 2\right) \\
&= \frac{k^2}{4}(3\alpha - \sin\alpha\cos\alpha) + k^2(1-\cos\alpha),
\end{aligned}$$

using $\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$ and $\sin 2\alpha = 2\sin\alpha\cos\alpha$. This is the required expression.

Part (iii)

Similarly, using $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$, the area of B is

$$\begin{aligned}\tfrac{1}{2}\int_\alpha^\pi (k+\cos\theta)^2\,\mathrm{d}\theta &= \tfrac{1}{2}\int_\alpha^\pi \left(k^2+\tfrac{1}{2}+2k\cos\theta+\tfrac{1}{2}\cos 2\theta\right)\mathrm{d}\theta \\ &= \tfrac{1}{2}\left[\left(k^2+\tfrac{1}{2}\right)\theta + 2k\sin\theta + \tfrac{1}{4}\sin 2\theta\right]_\alpha^\pi \\ &= \frac{(2k^2+1)(\pi-\alpha)}{4} - k\sin\alpha - \tfrac{1}{4}\sin\alpha\cos\alpha.\end{aligned}$$

If we want to remove the trigonometric functions of α, the values from part (i) give $k\sin\alpha = \dfrac{k}{\sqrt{1+k^2}}$ and $\sin\alpha\cos\alpha = \dfrac{k}{1+k^2}$, so

$$\text{area of B} = \frac{(2k^2+1)(\pi-\alpha)}{4} - \frac{k}{\sqrt{1+k^2}} - \frac{k}{4(1+k^2)}.$$

Part (iv)

Ask yourself What shape is the region made of A and B together?

By part (i), for $\theta < \alpha$ the boundary of A is the inner curve C_1, and for $\theta > \alpha$ the boundary of B is the inner curve C_2. So R is the area of the overlap of the two regions (shaded in the sketch). That already suggests the answers: for large k, C_2 lies almost entirely inside C_1, so the overlap should be almost all of T.

The full areas are, by the same integrals taken over $[0,\pi]$ (the $\sin\theta$ and $\sin 2\theta$ terms give 0 at both ends; $-2\cos\theta$ gives 4):

$$T = \tfrac{1}{2}\int_0^\pi (k+\cos\theta)^2\,\mathrm{d}\theta = \frac{\pi(2k^2+1)}{4}, \qquad S = \tfrac{1}{2}\int_0^\pi k^2(1+\sin\theta)^2\,\mathrm{d}\theta = \frac{k^2(3\pi+8)}{4}.$$

As $k\to\infty$, $\alpha = \arctan\frac{1}{k} \to 0$. Rather than expand A, we bound it: on $[0,\alpha]$ we have $r = k(1+\sin\theta) \le 2k$, so A lies inside a sector of radius $2k$ and angle α, and since $\alpha < \tan\alpha = \frac{1}{k}$,

$$0 < \text{area of A} \le \tfrac{1}{2}(2k)^2\alpha < 2k.$$

Hence $\dfrac{\mathrm{A}}{T} < \dfrac{8k}{\pi(2k^2+1)} \to 0$. For B, dividing part (iii) by T,

$$\frac{\mathrm{B}}{T} = \frac{\pi-\alpha}{\pi} - \frac{4\left(k\sin\alpha + \frac{1}{4}\sin\alpha\cos\alpha\right)}{\pi(2k^2+1)} \to 1 - 0 = 1,$$

because $\alpha\to 0$ and $0 < k\sin\alpha + \frac{1}{4}\sin\alpha\cos\alpha < k+1$, which is small compared with $2k^2+1$. Therefore

$$\frac{R}{T} = \frac{\mathrm{A}}{T} + \frac{\mathrm{B}}{T} \to 1.$$

Finally

$$\frac{R}{S} = \frac{R}{T}\cdot\frac{T}{S} = \frac{R}{T}\cdot\frac{\pi(2k^2+1)}{k^2(3\pi+8)} \to 1\cdot\frac{2\pi}{3\pi+8} = \frac{2\pi}{3\pi+8}.$$

Checks

- At $\theta = 0$, C_2 ($r = k+1$) is outside C_1 ($r = k$), and at $\theta = \frac{\pi}{2}$ it is inside ($k < 2k$), so the curves must cross between, consistent with $0 < \alpha < \frac{\pi}{4}$.
- With $k = 2$: $\alpha = \arctan\frac{1}{2} \approx 0.4636$, A ≈ 1.4132, B ≈ 5.0309, $T = \frac{9}{4}\pi \approx 7.0686$, so $R/T \approx 0.912$. Direct numerical integration of $\frac{1}{2}r^2$ agrees with both formulas.
- For $k = 50$, $R/T \approx 0.99987$ and $R/S \approx 0.36061$, close to $\frac{2\pi}{3\pi+8} \approx 0.36059$.
- A sanity check on the limit: for huge k, C_2 is nearly a semicircle of radius k (area $\frac{1}{2}\pi k^2$), and C_1 is k times a fixed half-cardioid of area $\frac{3\pi}{4} + 2$; the ratio is $\frac{2\pi}{3\pi+8}$.

Common slips

- Forgetting the factor $\frac{1}{2}$ in the polar area formula, or squaring $1 + \sin\theta$ as $1 + \sin^2\theta$.
- In part (iv), arguing that $k^2(1 - \cos\alpha)$ "tends to 0 because $\alpha \to 0$": it is multiplied by k^2, and in fact it tends to $\frac{1}{2}$. Only its size compared with T matters.

Beyond STEP

The area of the overlap of two regions given in polar form is a first taste of the area integrals $\iint r\,\mathrm{d}r\,\mathrm{d}\theta$ met in first-year multivariable calculus, where the hard part is always describing the region, exactly as here. The limit arguments in part (iv) (bounding a term rather than computing it exactly) are the style of real analysis; Abbott's *Understanding Analysis* is a good place to see that style developed carefully.

WORKED SOLUTION

STEP III 2023, Question 3

Pure • Complex Roots of Cubics

STEP III 2023, Question 3

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Measured from a, the three points are $sb\mathrm{i}$, $-sb\mathrm{i}$ and b. Multiplying by i turns a vector through a right angle, so a is the midpoint of the base and b points along the axis of symmetry. With that picture, part (ii) is a matter of Vieta's formulas: the roots sum to zero, which pins b down in terms of a, and then a cancels from p^3/q^2. Parts (iii) and (iv) turn the result into a one-variable function of $x = s^2 > 0$, whose graph shows how small it can be.

Solution

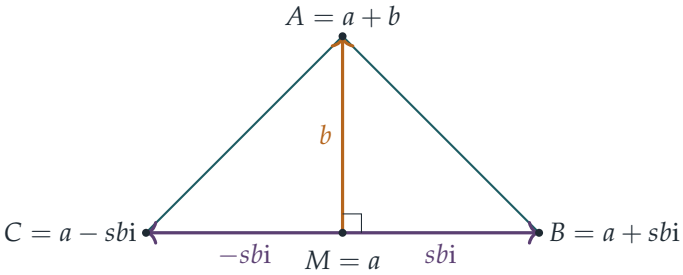
Part (i)

Distances from $a + b$ to the other two points are

$$\big|(a + sb\mathrm{i}) - (a + b)\big| = |b|\,|s\mathrm{i} - 1| = |b|\sqrt{s^2 + 1},$$
$$\big|(a - sb\mathrm{i}) - (a + b)\big| = |b|\,|-s\mathrm{i} - 1| = |b|\sqrt{s^2 + 1}.$$

These are equal. We still need the three points to form a genuine triangle. Relative to a they are $sb\mathrm{i}$, $-sb\mathrm{i}$ and b. The first two are distinct (as $s > 0$ and $b \neq 0$) and lie on the line through a in the direction of $b\mathrm{i}$. Since $b\mathrm{i}$ is b rotated through a right angle, the third point $a + b$ lies off that line, because $b \neq 0$. So the points form an isosceles triangle with apex $a + b$, base from $a - sb\mathrm{i}$ to $a + sb\mathrm{i}$, and base midpoint a.

Recovering a, b, s. Let the triangle have apex A (the vertex where the two equal sides meet; in an equilateral triangle any vertex will do) and other vertices B, C. Let M be the midpoint of BC.



- Take $a = M = \frac{1}{2}(B + C)$.
- Take $b = A - M$. This is non-zero, as A is not on BC.
- Since $AB = AC$, A lies on the perpendicular bisector of BC, so $MB \perp MA$. Hence $B - M$ is b turned through a right angle and rescaled: $B - M = \lambda b\mathrm{i}$ for some real $\lambda \neq 0$. If $\lambda < 0$, swap the labels B and C, which changes the sign of λ. Then $s = \lambda > 0$ is
$$s = \frac{MB}{MA} = \frac{\frac{1}{2}BC}{\text{height of the triangle}}.$$

Then $a + b = A$, $a + sb\mathrm{i} = B$ and $a - sb\mathrm{i} = M - (B - M) = C$, as required.

Part (ii)

By part (i), we can label the roots $a + sb\mathrm{i}$, $a - sb\mathrm{i}$, $a + b$ with $b \neq 0$, $s > 0$. The cubic has no z^2 term, so the roots sum to zero:

$$3a + b = 0, \qquad \text{so } b = -3a,$$

and $a \neq 0$ because $b \neq 0$. The roots are therefore

$$a(1 - 3s\mathrm{i}), \qquad a(1 + 3s\mathrm{i}), \qquad -2a.$$

The first two have product $a^2(1 + 9s^2)$ and sum $2a$. So, by Vieta's formulas,

$$p = a^2(1 + 9s^2) + (-2a)(2a) = a^2(9s^2 - 3) = 3a^2(3s^2 - 1),$$
$$q = -(\text{product of roots}) = -(-2a)\,a^2(1 + 9s^2) = 2a^3(9s^2 + 1).$$

Here $q \neq 0$ since $a \neq 0$, so we may divide:

$$\frac{p^3}{q^2} = \frac{27a^6(3s^2 - 1)^3}{4a^6(9s^2 + 1)^2} = \frac{27(3s^2 - 1)^3}{4(9s^2 + 1)^2},$$

with s a non-zero (indeed positive) real number, as required.

Part (iii)

Let $y = \dfrac{(3x-1)^3}{(9x+1)^2}$, defined for $x \neq -\frac{1}{9}$. By the quotient rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{9(3x-1)^2(9x+1)^2 - (3x-1)^3 \cdot 18(9x+1)}{(9x+1)^4}$$
$$= \frac{9(3x-1)^2\big[(9x+1) - 2(3x-1)\big]}{(9x+1)^3} = \frac{27(3x-1)^2(x+1)}{(9x+1)^3}.$$

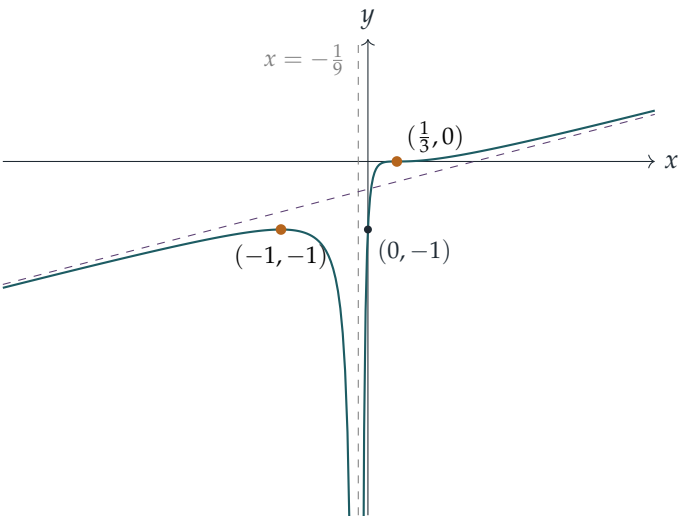
So the stationary points are at $x = -1$ and $x = \frac{1}{3}$.

- At $x = -1$: $y = (-4)^3/(-8)^2 = -1$. For $x < -1$, the factors $x+1$ and $(9x+1)^3$ are both negative, so $y' > 0$. For $-1 < x < -\frac{1}{9}$, only $(9x+1)^3$ is negative, so $y' < 0$. So $(-1,-1)$ is a **local maximum**.
- At $x = \frac{1}{3}$: $y = 0$. On both sides (for $x > -\frac{1}{9}$) every factor except $(3x-1)^2$ is positive, and $(3x-1)^2 > 0$ for $x \neq \frac{1}{3}$. So $y' > 0$ on either side, and $(\frac{1}{3}, 0)$ is a **stationary point of inflection**.

In particular, y is strictly increasing on $x > -\frac{1}{9}$, since $y' > 0$ there except at the single point $x = \frac{1}{3}$.

Other features:

- Vertical asymptote $x = -\frac{1}{9}$. There the numerator tends to $(-\frac{4}{3})^3 < 0$ and the denominator to 0^+, so $y \to -\infty$ from both sides.
- Dividing $27x^3 - 27x^2 + 9x - 1$ by $81x^2 + 18x + 1$ gives quotient $\frac{1}{3}x - \frac{11}{27}$, so $y = \frac{1}{3}x - \frac{11}{27}$ is an oblique asymptote as $x \to \pm\infty$.
- The curve meets the axes at $(0,-1)$ and $(\frac{1}{3}, 0)$.



The dashed line is the oblique asymptote $y = \frac{1}{3}x - \frac{11}{27}$. The left branch never rises above -1. The right branch increases from $-\infty$ through every real value.

Part (iv)

By part (ii), with the real number $s \neq 0$ found there and $x = s^2$,

$$\frac{p^3}{q^2} = \frac{27}{4} \cdot \frac{(3x-1)^3}{(9x+1)^2}.$$

Since s is real, the right-hand side is real, and therefore so is p^3/q^2.

> **Ask yourself** The graph goes down to $-\infty$, so why can't p^3/q^2 be very negative?

Because $x = s^2$ is not just any real number: s is real and non-zero, so $x > 0$. The values of x we care about therefore lie on the right branch, where (from part (iii)) y is strictly increasing. Hence, for $x > 0$,

$$\frac{(3x-1)^3}{(9x+1)^2} > \frac{(0-1)^3}{(0+1)^2} = -1,$$

and so

$$\frac{p^3}{q^2} > -\frac{27}{4}.$$

The inequality is strict because $x = 0$ (that is, $s = 0$) is excluded. It would make the triangle collapse to a line segment.

Checks

- Equilateral triangle: the roots of $z^3 - 1 = 0$ form an equilateral triangle. Here $p = 0$, and the formula gives $p = 0$ exactly when $s^2 = \frac{1}{3}$, i.e. when the half-base is $\frac{1}{\sqrt{3}}$ times the height, which is the equilateral shape.
- A concrete case: $a = 1$, $s = 1$ gives roots $1 \pm 3\mathrm{i}$ and -2, so $z^3 + 6z + 20 = 0$. Then $p^3/q^2 = 216/400 = \frac{27}{50}$, and the formula gives $\frac{27 \cdot 8}{4 \cdot 100} = \frac{27}{50}$.
- As $s \to 0$ the triangle flattens towards a double root at a, and $p^3/q^2 \to -\frac{27}{4}$: $4p^3 + 27q^2 = 0$ is exactly the condition for a repeated root. For $s = 10^{-4}$ the value is about -6.749998.
- Random isosceles triangles with centroid at the origin were checked by computer: p^3/q^2 was always real and greater than -6.75.

Common slips

- Treating a and b as real numbers in part (i). They are complex, so the triangle can sit at any angle.

- Getting the sign of q wrong: for a monic cubic, q is *minus* the product of the roots.
- In part (iv), using the whole graph and forgetting that $x = s^2 > 0$, which is what keeps us on the right branch.

Beyond STEP

The quantity $-(4p^3 + 27q^2)$ is the discriminant of the cubic, and p^3/q^2 is a shape invariant: it is unchanged when the roots are rotated or scaled about their centroid (replacing z by kz multiplies p by k^2 and q by k^3), so it records only the *shape* of the triangle of roots. Invariants like this are the starting point of classifying polynomials up to change of variable. Needham's *Visual Complex Analysis* is a beautiful guide to thinking geometrically with complex numbers in the way this question asks.

WORKED SOLUTION

STEP III 2023, Question 4

Pure • Chebyshev Polynomial Coefficients

STEP III 2023, Question 4

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Writing $\cos\big((2n+1)\theta\big)$ as the real part of $(\cos\theta + \mathrm{i}\sin\theta)^{2n+1}$ turns p into an explicit sum, and each coefficient of p becomes a sum of binomial coefficients. Sums of the form $\sum\binom{N}{2r}$ or $\sum r\binom{N}{2r}$ that run over even indices only are found by putting $t=1$ and $t=-1$ into $(1+t)^N$ (or its derivative) and adding or subtracting, because the odd-index terms then cancel. For the final part we compare the top few coefficients on both sides of the factorisation.

Solution

Throughout, write $c=\cos\theta$ and $s=\sin\theta$.

Part (i)

By de Moivre's theorem and the binomial theorem,

$$\cos\big((2n+1)\theta\big)+\mathrm{i}\sin\big((2n+1)\theta\big)=(c+\mathrm{i}s)^{2n+1}=\sum_{k=0}^{2n+1}\binom{2n+1}{k}c^{2n+1-k}(\mathrm{i}s)^k.$$

Since c and s are real, the real terms are exactly those with k even, say $k=2r$ with $0\le r\le n$, and then $(\mathrm{i}s)^{2r}=(-1)^r s^{2r}=(-s^2)^r=(c^2-1)^r$. Taking real parts,

$$\cos\big((2n+1)\theta\big)=\sum_{r=0}^{n}\binom{2n+1}{2r}\cos^{2n+1-2r}\theta\,(\cos^2\theta-1)^r,$$

as required.

Part (ii)

Part (i) shows that

$$\mathrm{p}(x)=1+\sum_{r=0}^{n}\binom{2n+1}{2r}x^{2n+1-2r}(x^2-1)^r$$

for every $x = \cos\theta \in [-1, 1]$. Both sides are polynomials that agree at infinitely many points, so they are the same polynomial.

The rth term has degree $(2n+1-2r)+2r = 2n+1$ with leading coefficient $\binom{2n+1}{2r}$, and the constant 1 does not affect x^{2n+1}. So the coefficient of x^{2n+1} in p is

$$S_0 = \sum_{r=0}^{n} \binom{2n+1}{2r},$$

the sum of the even-index binomial coefficients. Putting $t = 1$ and $t = -1$ into $(1+t)^{2n+1} = \sum_k \binom{2n+1}{k} t^k$ gives

$$2^{2n+1} = \sum_k \binom{2n+1}{k}, \qquad 0 = \sum_k (-1)^k \binom{2n+1}{k}.$$

Adding, the odd-k terms cancel and the even-k terms double: $2S_0 = 2^{2n+1}$. Hence the coefficient of x^{2n+1} in $\mathrm{p}(x)$ is $S_0 = 2^{2n}$.

Part (iii)

In the rth term, $x^{2n+1-2r}(x^2-1)^r$, we need the x^{2r-2} term of $(x^2-1)^r$ to reach x^{2n-1}. That term is $\binom{r}{1}(x^2)^{r-1}(-1) = -rx^{2r-2}$ (and it is absent when $r = 0$, consistent with $-r = 0$). Since $n \geq 1$ we have $2n-1 \geq 1$, so the constant 1 in p plays no part. The coefficient of x^{2n-1} is therefore

$$-\sum_{r=0}^{n} r \binom{2n+1}{2r}.$$

Ask yourself How can we pick out the even-index terms when each one is weighted by r?

Differentiating brings down the index as a factor. From $(1+t)^{2n+1} = \sum_k \binom{2n+1}{k} t^k$,

$$(2n+1)(1+t)^{2n} = \sum_{k=1}^{2n+1} k \binom{2n+1}{k} t^{k-1}.$$

At $t = 1$ this gives $\sum_k k\binom{2n+1}{k} = (2n+1)2^{2n}$. At $t = -1$ the left-hand side is 0 (since $2n \geq 1$), so

$$0 = \sum_k (-1)^{k-1} k \binom{2n+1}{k} = \sum_{k \text{ odd}} k \binom{2n+1}{k} - \sum_{k \text{ even}} k \binom{2n+1}{k}.$$

So the even-k and odd-k parts are equal, and each is half the total:

$$\sum_{k \text{ even}} k \binom{2n+1}{k} = (2n+1)2^{2n-1}.$$

With $k = 2r$, this says $\sum_r 2r\binom{2n+1}{2r} = (2n+1)2^{2n-1}$, that is, $\sum_r r\binom{2n+1}{2r} = (2n+1)2^{2n-2}$. Hence the coefficient of x^{2n-1} in $\mathrm{p}(x)$ is $-(2n+1)2^{2n-2}$, as required.

Part (iv)

The leading coefficient. By (ii), p has degree $2n+1$ with leading coefficient 2^{2n}. If q has degree m and leading coefficient a, then $(x+1)[\mathrm{q}(x)]^2$ has degree $2m+1$ and leading coefficient a^2. So $m=n$ and $a^2=2^{2n}$, and since $a>0$, the coefficient of x^n in q is 2^n.

One more fact about p. Every term $x^{2n+1-2r}(x^2-1)^r$ contains only odd powers of x, because $x^{2n+1-2r}$ is an odd power and $(x^2-1)^r$ has only even powers. The only even power in p is the constant 1. As $2n\geq 2$, the coefficient of x^{2n} in p is 0.

Comparing coefficients. Now take $n\geq 2$ and write

$$\mathrm{q}(x)=2^n x^n+bx^{n-1}+cx^{n-2}+\cdots,$$

so that

$$[\mathrm{q}(x)]^2=2^{2n}x^{2n}+2^{n+1}b\,x^{2n-1}+\left(b^2+2^{n+1}c\right)x^{2n-2}+\cdots.$$

Multiplying by $x+1$, the coefficients of x^{2n} and x^{2n-1} in $(x+1)[\mathrm{q}(x)]^2$ are

$$2^{n+1}b+2^{2n}\qquad\text{and}\qquad\left(b^2+2^{n+1}c\right)+2^{n+1}b.$$

The coefficient of x^{2n} in p is 0, so $b=-2^{2n}/2^{n+1}=-2^{n-1}$. Then $b^2=2^{2n-2}$ and $2^{n+1}b=-2^{2n}$, and matching the coefficient of x^{2n-1} with part (iii) gives

$$2^{2n-2}+2^{n+1}c-2^{2n}=-(2n+1)2^{2n-2}.$$

Rearranging,

$$2^{n+1}c=2^{2n-2}\big(-(2n+1)-1+4\big)=2^{2n-2}(2-2n)=2^{2n-1}(1-n),$$

so the coefficient of x^{n-2} in q is

$$c=2^{n-2}(1-n),$$

as required.

Checks

- For $n=1$: $\cos 3\theta=4\cos^3\theta-3\cos\theta$, so $\mathrm{p}(x)=4x^3-3x+1=(x+1)(2x-1)^2$. The leading coefficient is 2^2, the x-coefficient is $-3=-3\cdot 2^0$, and q has leading coefficient 2^1.
- For $n=2$: $\mathrm{p}(x)=16x^5-20x^3+5x+1=(x+1)(4x^2-2x-1)^2$. The x^3-coefficient is $-20=-5\cdot 2^2$, and the constant term of q is $-1=2^0(1-2)$.
- For $n=3$ we found $\mathrm{q}(x)=8x^3-4x^2-4x+1$, with coefficient of x equal to $-4=2^1(1-3)$. A computer check confirmed every coefficient formula for $n=1,\ldots,12$ by expanding $\cos\big((2n+1)\theta\big)+1$ exactly and taking the square root of $\mathrm{p}(x)/(x+1)$.

Common slips

- Forgetting the x^{2n} coefficient of p. It is 0, and that is exactly what fixes the x^{n-1} coefficient b of q, which you need before you can find c.
- Writing $(\mathrm{i}s)^{2r} = s^{2r}$ and losing the sign $(-1)^r$. That sign is what produces $\cos^2\theta - 1$ rather than $1 - \cos^2\theta$.
- Putting only $t = 1$ into the derivative in (iii), which gives the sum over all k, not just the even ones.

Beyond STEP

The polynomial T_N with $T_N(\cos\theta) = \cos N\theta$ is the Chebyshev polynomial of degree N, and parts (ii) and (iii) find its top two coefficients. The factorisation in (iv) reflects the fact that $T_{2n+1}(x) + 1$ has a simple root at $x = -1$ and a double root at each other point where $\cos\big((2n+1)\theta\big) = -1$. Chebyshev polynomials are central in approximation theory: among monic polynomials of degree N, $2^{1-N}T_N$ is the one with the smallest maximum size on $[-1, 1]$. Siklos, *Advanced Problems in Mathematics*, has more questions in this vein.

WORKED SOLUTION

STEP III 2023, Question 5

Pure • Number Theory

STEP III 2023, Question 5

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Each part turns an equation in integers into a *product* equal to something whose factors we know completely: 49, then pq, then $3p^2q$. Once one side is a product of two integers, the size of each factor (which one is bigger, which is smaller than p) cuts the list of possibilities down to one or two cases. In part (iii) the factor we control is $p+q-n$, and the inequalities we are asked to prove are exactly what stop it sharing a factor with p or q.

Solution

Part (i)

Multiplying through by $7xy$ (both non-zero) gives

$$7y+14x=2xy \quad \Longleftrightarrow \quad 2xy-14x-7y=0.$$

Since $(2x-7)(y-7)=2xy-14x-7y+49$, adding 49 to both sides gives $(2x-7)(y-7)=49$, as required.

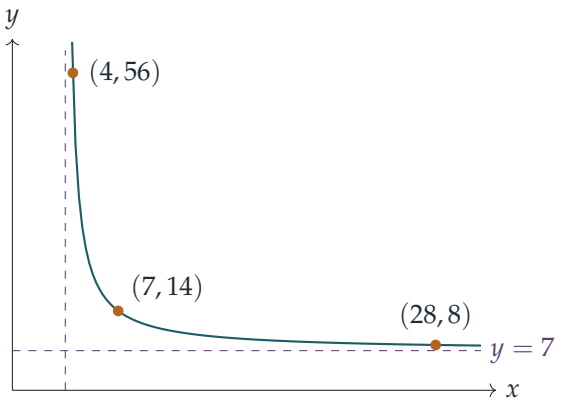
Now let x and y be positive integers. Then $2x-7$ and $y-7$ are integers whose product is 49, so they are a factor pair of 49:

$$(2x-7,\ y-7)\in\{(1,49),\,(7,7),\,(49,1),\,(-1,-49),\,(-7,-7),\,(-49,-1)\}.$$

Positivity rules out the negative pairs: $x\geq 1$ forces $2x-7\geq -5$, and $y\geq 1$ forces $y-7\geq -6$, so neither factor can be -7 or -49, and every negative pair contains one of these. The positive pairs give

$$(x,y)=(4,56),\quad (7,14),\quad (28,8),$$

and each does satisfy the original equation: for example $\frac{1}{4}+\frac{2}{56}=\frac{7}{28}+\frac{1}{28}=\frac{2}{7}$.



The curve is a rectangular hyperbola with asymptotes $x = \frac{7}{2}$ and $y = 7$; the three solutions are the lattice points on its positive branch.

Part (ii)

Expanding $(p+q)^2 = p^2 + 2pq + q^2 = n^2 + pq$, so

$$(p+q+n)(p+q-n) = (p+q)^2 - n^2 = pq.$$

The right-hand side is positive and $p+q+n > 0$, so $p+q-n$ is a *positive* integer. The positive divisors of pq are $1, p, q, pq$ (just $1, p, p^2$ if $p = q$), and $p+q+n$ is one of them. But $n > 0$ means $p+q+n > p$ and $p+q+n > q$, so $p+q+n$ is neither p nor q; the only divisor left is pq. Hence $p+q+n = pq$ and the co-factor is

$$p+q-n = 1, \qquad \text{that is,} \qquad p+q = n+1.$$

Substituting $n = p+q-1$ into $p+q+n = pq$ gives $2p+2q-1 = pq$, which rearranges to

$$pq - 2p - 2q + 4 = 3 \quad \Longleftrightarrow \quad (p-2)(q-2) = 3.$$

Primes satisfy $p-2 \geq 0$, so both factors are non-negative, and the product is 3, so $\{p-2, q-2\} = \{1, 3\}$. Therefore

$$(p,q) = (3,5) \quad \text{or} \quad (5,3),$$

and indeed $9 + 15 + 25 = 49 = 7^2$, with $n = 7 = 3 + 5 - 1$.

Part (iii)

Here n is the positive real cube root of $p^3 + q^3 + 3pq^2$.

The inequalities. Since $p, q > 0$,

$$n^3 = q^3 + \left(p^3 + 3pq^2\right) > q^3, \qquad n^3 = p^3 + \left(q^3 + 3pq^2\right) > p^3,$$

and cubing is strictly increasing, so $n > q$ and $n > p$. These rearrange to $p + q - n < p$ and $p + q - n < q$ respectively. We shall also need the lower bound:

$$(p+q)^3 = p^3 + 3p^2q + 3pq^2 + q^3 = n^3 + 3p^2q > n^3,$$

so $p + q > n$, i.e. $p + q - n > 0$.

No prime solutions. Suppose p and q are primes and n is an integer. Put $d = p + q - n$, an integer with

$$0 < d < p, \qquad 0 < d < q.$$

From the expansion above, $(p+q)^3 - n^3 = 3p^2q$, and factorising a difference of cubes,

$$d\left((p+q)^2 + (p+q)n + n^2\right) = 3p^2q,$$

so d divides $3p^2q$.

Ask yourself Why can d not share a factor with p or q?

Because p is prime, the only way for d to share a factor with p is for p to divide d, which would need $d \geq p$. But $d < p$. In the same way $d < q$ means q does not divide d. So d is coprime to p^2q, and $d \mid 3p^2q$ forces $d \mid 3$. This is exactly what the inequalities were for. Hence $d = 1$ or $d = 3$. Write $s = p + q$.

Case $d = 1$. Then $n = s - 1$ and

$$3p^2q = s^3 - (s-1)^3 = 3s^2 - 3s + 1,$$

which is impossible: the left-hand side is a multiple of 3 and the right-hand side leaves remainder 1 on division by 3.

Case $d = 3$. Now $3 = d < p$ and $3 < q$, so neither p nor q is 3. Here $n = s - 3$ and

$$3p^2q = s^3 - (s-3)^3 = 9s^2 - 27s + 27, \qquad \text{so} \qquad p^2q = 3(s^2 - 3s + 3).$$

Thus $3 \mid p^2q$, and since 3 is prime it divides p or q, making one of them equal to 3. That contradicts $p, q > 3$.

Both cases are impossible, so there are no primes p and q for which $p^3 + q^3 + 3pq^2$ is the cube of an integer. ■

Checks

A brute-force search over all positive integers $x, y < 2000$ finds exactly the three pairs in (i). Over all primes $p, q < 1500$, $p^2 + pq + q^2$ is a perfect square only for $\{p, q\} = \{3, 5\}$ (giving 49), and $p^3 + q^3 + 3pq^2$ is never a perfect cube. Random positive reals confirm $0 < p + q - n < \min(p, q)$ in (iii). A quick hand check of (ii): $(3 + 5 + 7)(3 + 5 - 7) = 15 \times 1 = 15 = pq$.

Common slips

- Forgetting the negative factor pairs of 49 in (i). They give no solutions here, but they have to be ruled out, not ignored.
- In (ii), assuming without justification that $p+q-n$ is the smaller factor and positive: both follow from $n>0$ and $pq>0$, and they need saying.
- In (iii), stopping at $d \mid 3$. The two cases $d=1$ and $d=3$ still have to be eliminated, and the second needs the fact that $d<p,q$ once more.

Beyond STEP

Every part uses the same trick: rewrite a Diophantine equation as a product of integers equal to a number whose factorisation you know, then use size and coprimality to pin the factors down. Part (i) belongs to the family of Egyptian-fraction equations $\frac{1}{x}+\frac{1}{y}=\frac{1}{n}$, and part (ii) is about integer triangles with a 120° angle, since $p^2+pq+q^2=n^2$ is the cosine rule for that angle. Davenport's *The Higher Arithmetic* is a lovely introduction to this kind of elementary number theory.

WORKED SOLUTION

STEP III 2023, Question 6

Pure • Hyperbolic Functions and Monotonicity

STEP III 2023, Question 6

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Both functions are a difference of $\tan^{-1} x$ and a multiple of $\tanh x$, so their derivatives are a difference of $\dfrac{1}{1+x^2}$ and a multiple of $\operatorname{sech}^2 x$. Everything therefore comes down to comparing $\cosh^2 x$ with $1+x^2$: the inequality in (i) settles the sign of f', and showing that the ratio $\cosh^2 x/(1+x^2)$ is increasing tells us exactly how often it can equal $\frac{1}{2}\pi$, which is where g' vanishes.

Solution

Part (i)

Using $\cosh^2 x = \frac{1}{2}(1+\cosh 2x)$ and $\cosh 2x = \frac{1}{2}(\mathrm{e}^{2x}+\mathrm{e}^{-2x})$, the Maclaurin series for e^x gives (the odd powers cancel)

$$\cosh 2x = \sum_{n=0}^{\infty} \frac{(2x)^{2n}}{(2n)!} = 1 + 2x^2 + \frac{2}{3}x^4 + \cdots.$$

Every term is non-negative, so $\cosh 2x \geq 1+2x^2$, and hence

$$\cosh^2 x = \tfrac{1}{2}(1+\cosh 2x) \geq \tfrac{1}{2}(2+2x^2) = 1+x^2,$$

with equality only when $x=0$ (for $x \neq 0$ the x^4 term is strictly positive).

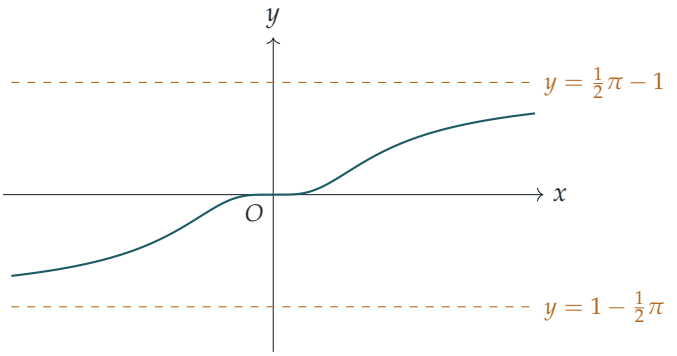
Now differentiate:

$$\mathrm{f}'(x) = \frac{1}{1+x^2} - \operatorname{sech}^2 x = \frac{1}{1+x^2} - \frac{1}{\cosh^2 x}.$$

Since $\cosh^2 x \geq 1+x^2 > 0$, the reciprocals satisfy $\dfrac{1}{\cosh^2 x} \leq \dfrac{1}{1+x^2}$, so $\mathrm{f}'(x) \geq 0$ for all x, with equality only at $x=0$. A function whose derivative is positive except at a single point is (strictly) increasing, as required.

Sketch. Some features to put on the graph:

- f is odd (both $\tan^{-1}$ and tanh are odd), and $f(0) = 0$.
- $f'(0) = 0$ but f is increasing, so the origin is a stationary point of inflection. In fact $\tan^{-1} x = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \cdots$ and $\tanh x = x - \frac{1}{3}x^3 + \frac{2}{15}x^5 - \cdots$, so $f(x) \approx \frac{1}{15}x^5$ near 0: the graph is very flat there.
- As $x \to \infty$, $\tan^{-1} x \to \frac{1}{2}\pi$ and $\tanh x \to 1$, so $f(x) \to \frac{1}{2}\pi - 1 \approx 0.571$ from below; by oddness $f(x) \to 1 - \frac{1}{2}\pi$ as $x \to -\infty$.



Part (ii)(a)

Here

$$g'(x) = \frac{1}{1+x^2} - \frac{\pi}{2\cosh^2 x},$$

which is continuous and even. At $x = 0$, $g'(0) = 1 - \frac{1}{2}\pi < 0$. At $x = 2$,

$$g'(2) = \tfrac{1}{5} - \frac{\pi}{2\cosh^2 2},$$

and $\cosh 2 > \frac{1}{2}e^2 > 3.6$, so $\cosh^2 2 > 12.9$ and $\dfrac{\pi}{2\cosh^2 2} < \dfrac{1.6}{12.9} < \frac{1}{5}$; thus $g'(2) > 0$. By the intermediate value theorem g' has a zero in $(0, 2)$, and since g' is even it has another in $(-2, 0)$. So g has at least two stationary points.

Part (ii)(b)

Let $h(x) = (1 + x^2)\sinh x - x\cosh x$. Then

$$\begin{aligned} h'(x) &= 2x\sinh x + (1+x^2)\cosh x - \cosh x - x\sinh x \\ &= x\sinh x + x^2\cosh x. \end{aligned}$$

For $x \geq 0$ both $x\sinh x$ and $x^2\cosh x$ are non-negative (and positive for $x > 0$), so h is increasing on $[0, \infty)$. Since $h(0) = 0$, we get $h(x) \geq 0$ for $x \geq 0$, and in fact $h(x) > 0$ for $x > 0$.

Part (ii)(c)

By the quotient rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\cosh^2 x}{1+x^2}\right) = \frac{2\cosh x \sinh x\,(1+x^2) - 2x\cosh^2 x}{(1+x^2)^2} = \frac{2\cosh x\; h(x)}{(1+x^2)^2}.$$

For $x \geq 0$, $\cosh x > 0$ and $h(x) \geq 0$ by (b), so the derivative is non-negative; it is zero only at $x = 0$. Hence $\cosh^2 x/(1+x^2)$ is (strictly) increasing for $x \geq 0$.

Part (ii)(d)

Ask yourself How can a statement about $\cosh^2 x/(1+x^2)$ count the zeros of g'?

Multiplying $\mathrm{g}'(x) = 0$ through by the positive quantity $(1+x^2)\cosh^2 x$ shows that

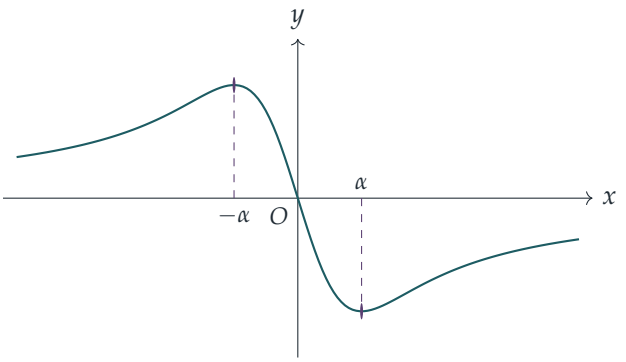
$$\mathrm{g}'(x) = 0 \iff \frac{\cosh^2 x}{1+x^2} = \frac{\pi}{2}.$$

Write $R(x) = \cosh^2 x/(1+x^2)$. On $[0,\infty)$, R is continuous and strictly increasing by (c), with $R(0) = 1 < \frac{1}{2}\pi$, so it takes the value $\frac{1}{2}\pi$ at most once there; by (a) it does take it, at some α with $0 < \alpha < 2$. So g' has exactly one zero in $[0,\infty)$. Since R is even, the zeros for $x < 0$ are the negatives of those for $x > 0$, giving exactly one more, at $-\alpha$. Hence g has exactly two stationary points, at $x = \pm\alpha$ (numerically $\alpha \approx 1.406$).

Part (ii)(e)

Collect the facts:

- g is odd, $\mathrm{g}(0) = 0$ and $\mathrm{g}'(0) = 1 - \frac{1}{2}\pi < 0$.
- From (d), $\mathrm{g}' < 0$ on $(-\alpha,\alpha)$ (where $R < \frac{1}{2}\pi$) and $\mathrm{g}' > 0$ for $|x| > \alpha$. So there is a local maximum at $x = -\alpha$ and a local minimum at $x = \alpha$, with $\mathrm{g}(\alpha) = -\mathrm{g}(-\alpha) < 0$ (about -0.44).
- As $x \to \infty$, $\mathrm{g}(x) \to \frac{1}{2}\pi - \frac{1}{2}\pi \cdot 1 = 0$. Because g is increasing on (α,∞) towards the limit 0, it stays negative there: the graph approaches the x-axis from below. By oddness, $\mathrm{g}(x) \to 0$ from above as $x \to -\infty$.
- So the only zero of g is at the origin: $\mathrm{g} > 0$ for $x < 0$ and $\mathrm{g} < 0$ for $x > 0$.



Checks

- At $x = 1$: $\cosh^2 1 \approx 2.381 \geq 2$, consistent with (i).
- $f(30) \approx 0.537$, approaching $\frac{1}{2}\pi - 1 \approx 0.571$ from below, and $f(0.1) \approx 6.6 \times 10^{-7}$, close to $\frac{1}{15}(0.1)^5$.
- Solving $\cosh^2 x = \frac{1}{2}\pi(1 + x^2)$ numerically gives $\alpha \approx 1.406$, inside $(0, 2)$ as (a) predicts, with $g(\alpha) \approx -0.440$. A fine scan of g' on $[-20, 20]$ finds exactly two sign changes, and $g(20) \approx -0.050$ is small and negative, as the sketch shows.

Common slips

- Forgetting that $\cosh^2 x \geq 1 + x^2$ must be turned round to $\operatorname{sech}^2 x \leq \dfrac{1}{1 + x^2}$; this is valid because both sides are positive.
- In (d), proving only that there is one positive root and forgetting to use evenness (or to note that $x = 0$ is not a root) to count the negative one.
- Sketching g with an asymptote other than $y = 0$, or crossing the axis again for large x; the monotonicity on (α, ∞) rules this out.

Beyond STEP

The method in (ii) (turn "how many solutions of $g' = 0$?" into "how often does a monotonic function hit a value?") is the basic way analysts count roots: continuity plus the intermediate value theorem gives existence, and strict monotonicity gives uniqueness. These ideas are made rigorous in a first analysis course; Abbott's *Understanding Analysis* is an excellent place to see them done carefully.

WORKED SOLUTION

STEP III 2023, Question 7

Pure • Integral Inequalities

STEP III 2023, Question 7

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

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The key idea

The integral of a square of a continuous function is never negative, and it is zero only when the function is identically zero. Each part hands us an equation involving $\int(\text{something})^2$ that, after a little rearranging, becomes "a perfect-square integral equals zero". Part (i) supplies the tool for (ii), integration by parts does the same job in (iii), and in (iv) we have to spot the square ourselves by completing it.

Solution

Part (i)

Substitute $x = u^2$ in the left-hand integral, so $\mathrm{d}x = 2u\,\mathrm{d}u$. As x runs from 0 to 1, so does $u \geq 0$, and $\sqrt{x} = u$. Hence

$$\int_0^1 \mathrm{f}(\sqrt{x})\,\mathrm{d}x = \int_0^1 \mathrm{f}(u)\,2u\,\mathrm{d}u = 2\int_0^1 x\,\mathrm{f}(x)\,\mathrm{d}x,$$

renaming the dummy variable at the end. This is the required result.

Part (ii)

By part (i) with $\mathrm{f} = \mathrm{g}$, the given condition becomes

$$\int_0^1 \big(\mathrm{g}(x)\big)^2\,\mathrm{d}x = 2\int_0^1 x\,\mathrm{g}(x)\,\mathrm{d}x - \frac{1}{3}.$$

Now expand the target integral and use $\int_0^1 x^2\,\mathrm{d}x = \frac{1}{3}$:

$$\begin{aligned}\int_0^1 \big(\mathrm{g}(x) - x\big)^2\,\mathrm{d}x &= \int_0^1 \big(\mathrm{g}(x)\big)^2\,\mathrm{d}x - 2\int_0^1 x\,\mathrm{g}(x)\,\mathrm{d}x + \int_0^1 x^2\,\mathrm{d}x \\ &= -\frac{1}{3} + \frac{1}{3} = 0,\end{aligned}$$

as required.

Ask yourself Why does a zero integral force $g(x) = x$ everywhere, and not just "mostly"?

Let $\phi(x) = \big(g(x) - x\big)^2$. It is continuous (because g is) and $\phi(x) \geq 0$. Suppose $\phi(c) > 0$ for some $c \in [0,1]$. By continuity there is an interval $I \subseteq [0,1]$ of some positive length δ, containing c, on which $\phi(x) > \frac{1}{2}\phi(c)$. Since $\phi \geq 0$ everywhere else,

$$\int_0^1 \phi(x)\,\mathrm{d}x \;\geq\; \int_I \phi(x)\,\mathrm{d}x \;\geq\; \tfrac{1}{2}\phi(c)\,\delta > 0,$$

a contradiction. So $\phi(x) = 0$, that is $g(x) = x$, for every x in $0 \leq x \leq 1$. (Continuity really matters: a function that differs from x at a single point would still give a zero integral.)

Part (iii)

The term $2h(1) - 2\int_0^1 h(x)\,\mathrm{d}x$ should remind us of integration by parts. Integrating $1 \cdot h(x)$ by parts, with x as the antiderivative of 1,

$$\int_0^1 h(x)\,\mathrm{d}x = \Big[x\,h(x)\Big]_0^1 - \int_0^1 x\,h'(x)\,\mathrm{d}x = h(1) - \int_0^1 x\,h'(x)\,\mathrm{d}x.$$

So $2h(1) - 2\int_0^1 h(x)\,\mathrm{d}x = 2\int_0^1 x\,h'(x)\,\mathrm{d}x$, and the condition reads

$$\int_0^1 \big(h'(x)\big)^2\,\mathrm{d}x = 2\int_0^1 x\,h'(x)\,\mathrm{d}x - \frac{1}{3}.$$

This is exactly the condition of part (ii) (in the form we reached via part (i)) with h' in place of g. Repeating that argument, which needs h' to be continuous as the question intends,

$$\int_0^1 \big(h'(x) - x\big)^2\,\mathrm{d}x = 0 \quad\Longrightarrow\quad h'(x) = x \text{ for } 0 \leq x \leq 1.$$

Integrating, $h(x) = \frac{1}{2}x^2 + C$, and $h(0) = 0$ gives $C = 0$. Hence

$$h(x) = \tfrac{1}{2}x^2.$$

Part (iv)

Note first that $a \neq 0$, since the condition contains $1/a$. Comparing with the earlier parts, we look for a square of the form $e^{ax}\big(k(x) - p(x)\big)^2$ whose cross term is $-2k(x)$. Expanding gives cross term $-2e^{ax}p(x)k(x)$, so we need $p(x) = e^{-ax}$:

$$e^{ax}\big(k(x) - e^{-ax}\big)^2 = e^{ax}\big(k(x)\big)^2 - 2k(x) + e^{-ax}.$$

Integrate over $[0,1]$, use the given condition for the first two terms, and $\int_0^1 \mathrm{e}^{-ax}\,\mathrm{d}x = \dfrac{1-\mathrm{e}^{-a}}{a}$ for the last:

$$\begin{aligned}\int_0^1 \mathrm{e}^{ax}\big(\mathrm{k}(x)-\mathrm{e}^{-ax}\big)^2\,\mathrm{d}x &= \frac{\mathrm{e}^{-a}}{a}-\frac{1}{a^2}-\frac{1}{4}+\frac{1-\mathrm{e}^{-a}}{a}\\ &= \frac{1}{a}-\frac{1}{a^2}-\frac{1}{4} = -\Big(\frac{1}{a}-\frac{1}{2}\Big)^2 = -\frac{(a-2)^2}{4a^2}.\end{aligned}$$

The left-hand side is the integral of a non-negative continuous function, so it is ≥ 0; the right-hand side is ≤ 0. Both must therefore be zero, which forces $(a-2)^2=0$, so $a=2$.

With $a=2$ the integral of the continuous non-negative function $\mathrm{e}^{2x}\big(\mathrm{k}(x)-\mathrm{e}^{-2x}\big)^2$ is zero, so by the argument of part (ii) it vanishes everywhere on $[0,1]$. Since $\mathrm{e}^{2x}>0$, this gives

$$\mathrm{k}(x)=\mathrm{e}^{-2x}, \qquad 0\leq x\leq 1.$$

Checks

- Part (i) with $\mathrm{f}(x)=x$: the left side is $\int_0^1 \sqrt{x}\,\mathrm{d}x = \frac{2}{3}$ and the right side is $2\int_0^1 x^2\,\mathrm{d}x = \frac{2}{3}$.
- Part (iii): with $\mathrm{h}=\frac{1}{2}x^2$, $\int_0^1 x^2\,\mathrm{d}x=\frac{1}{3}$ and $2\cdot\frac{1}{2}-2\cdot\frac{1}{6}-\frac{1}{3}=\frac{1}{3}$. They agree.
- Part (iv): with $a=2$ and $\mathrm{k}=\mathrm{e}^{-2x}$ the left side is $\int_0^1 \mathrm{e}^{-2x}\,\mathrm{d}x=\frac{1}{2}(1-\mathrm{e}^{-2})\approx 0.43233$, and the right side is $(1-\mathrm{e}^{-2})+\frac{1}{2}\mathrm{e}^{-2}-\frac{1}{4}-\frac{1}{4}=\frac{1}{2}-\frac{1}{2}\mathrm{e}^{-2}$, the same. A computer check of the identity "left minus right $=\int \mathrm{e}^{ax}(\mathrm{k}-\mathrm{e}^{-ax})^2+(a-2)^2/(4a^2)$" for random a and random polynomial k also agreed.

Common slips

- Concluding $\mathrm{g}(x)=x$ from a zero integral without using continuity and non-negativity: a zero integral alone says nothing.
- In part (iv), forgetting to state that $a\neq 0$, or trying to guess k before completing the square.

Beyond STEP

The quantity $\int_0^1 \mathrm{f}(x)\mathrm{g}(x)\,\mathrm{d}x$ behaves like a dot product of the "vectors" f and g, and $\int_0^1(\mathrm{g}-x)^2\,\mathrm{d}x$ is the squared distance between g and x. Each part says "this distance is zero", and the weight e^{ax} in (iv) gives a weighted inner product. This viewpoint (inner product spaces, and the fact that a continuous function with zero norm is zero) is central to first-year linear algebra and analysis; see Strang, *Introduction to Linear Algebra*, and Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2023, Question 8

Pure • Piecewise Differential Equations

STEP III 2023, Question 8

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

The modulus makes the equation non-linear, but only mildly: on any stretch where y' keeps one sign, $|y'|$ is just y' or $-y'$ and the equation is an ordinary constant-coefficient one. So we follow the solution until y' hits zero, switch to the other linear equation, and match y and y' at the join. Part (iv) is the labour-saving trick: reflecting a solution, $x \mapsto c - x$, flips the sign of the first-derivative term, which is exactly the switch between the two linear equations.

Solution

Part (i)

With $\mathrm{f}(x) = x\mathrm{e}^{-x}$,

$$\mathrm{f}'(x) = (1-x)\mathrm{e}^{-x}, \qquad \mathrm{f}''(x) = -\mathrm{e}^{-x} - (1-x)\mathrm{e}^{-x} = (x-2)\mathrm{e}^{-x}.$$

Hence

$$\mathrm{f}'' + 2\mathrm{f}' + \mathrm{f} = \big[(x-2) + 2(1-x) + x\big]\mathrm{e}^{-x} = 0$$

for every real x. Also $\mathrm{f}(0) = 0$ and $\mathrm{f}'(0) = 1 \cdot \mathrm{e}^{0} = 1$, as required.

Since $\mathrm{e}^{-x} > 0$, the sign of $\mathrm{f}'(x)$ is the sign of $1 - x$, which is ≥ 0 for $x \leq 1$. So $\mathrm{f}'(x) \geq 0$ for $x \leq 1$.

Part (ii)

On $x \leq 1$ we have $\mathrm{f}' \geq 0$, so $|\mathrm{f}'| = \mathrm{f}'$ and the equation of (ii) is the same as the equation of (i). Thus f satisfies the new equation and its initial conditions on $x \leq 1$, and

$$\mathrm{g}_1(x) = x\mathrm{e}^{-x}.$$

At the join, $\mathrm{g}_1(1) = \mathrm{e}^{-1}$ and $\mathrm{g}_1'(1) = 0$. Continuous differentiability means g_2 must start with the same values: $\mathrm{g}_2(1) = \mathrm{e}^{-1}$, $\mathrm{g}_2'(1) = 0$.

Ask yourself Which sign does y' take just after $x = 1$?

At $x = 1$ the equation gives $y'' = -2|y'| - y = -\mathrm{e}^{-1} < 0$, so y' is decreasing through 0 and is negative just beyond $x = 1$. There $|y'| = -y'$, and the equation becomes

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2\frac{\mathrm{d}y}{\mathrm{d}x} + y = 0.$$

The auxiliary equation $m^2 - 2m + 1 = 0$ has the repeated root $m = 1$, so $\mathrm{g}_2(x) = (A + Bx)\mathrm{e}^x$ and $\mathrm{g}_2'(x) = (A + B + Bx)\mathrm{e}^x$. The conditions at $x = 1$ give

$$(A + B)\mathrm{e} = \mathrm{e}^{-1}, \qquad (A + 2B)\mathrm{e} = 0,$$

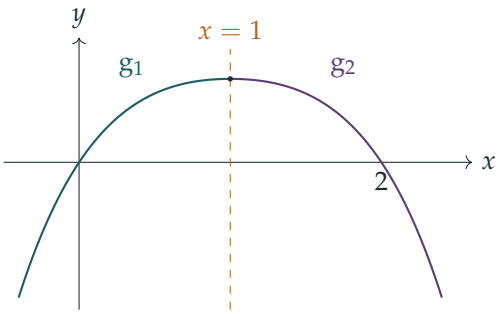
so $B = -\mathrm{e}^{-2}$ and $A = 2\mathrm{e}^{-2}$. Therefore

$$\mathrm{g}_2(x) = (2 - x)\mathrm{e}^{x-2}.$$

We must check this is consistent with the assumption $y' \le 0$: $\mathrm{g}_2'(x) = (1 - x)\mathrm{e}^{x-2} \le 0$ for all $x \ge 1$. So $|\mathrm{g}_2'| = -\mathrm{g}_2'$ throughout $x \ge 1$, and g_2 really does satisfy the modulus equation on the whole of $x \ge 1$.

Part (iii)

Replacing x by $2 - x$ in g_1 gives $(2 - x)\mathrm{e}^{-(2-x)} = (2 - x)\mathrm{e}^{x-2} = \mathrm{g}_2(x)$. So $\mathrm{g}_2(x) = \mathrm{g}_1(2 - x)$: the curve $y = \mathrm{g}_2(x)$ is the reflection of $y = \mathrm{g}_1(x)$ in the line $x = 1$.



Part (iv)

Let $Y(x) = \mathrm{k}(c - x)$ and write $u = c - x$. By the chain rule, with $\mathrm{d}u/\mathrm{d}x = -1$,

$$Y'(x) = -\mathrm{k}'(u), \qquad Y''(x) = \mathrm{k}''(u).$$

Hence

$$Y'' - pY' + qY = \mathrm{k}''(u) + p\,\mathrm{k}'(u) + q\,\mathrm{k}(u),$$

and the right-hand side is zero whenever $r \le u \le s$, because k solves the original equation there. Now $r \le c - x \le s$ is equivalent to

$$c - s \le x \le c - r,$$

so $y = \mathrm{k}(c - x)$ satisfies $y'' - py' + qy = 0$ on the interval $[c - s, c - r]$.

This explains (iii): with $p = 2$, $q = 1$, $\mathrm{k} = \mathrm{g}_1$ on $(-\infty, 1]$ and $c = 2$, the reflected function $\mathrm{g}_1(2 - x)$ solves $y'' - 2y' + y = 0$ on $[1, \infty)$.

Part (v)

With $\mathrm{h}(x) = \mathrm{e}^{-x}\sin x$,

$$\mathrm{h}'(x) = \mathrm{e}^{-x}(\cos x - \sin x) = \sqrt{2}\,\mathrm{e}^{-x}\cos\left(x + \tfrac{1}{4}\pi\right),$$

so $\mathrm{h}'(\frac{1}{4}\pi) = \sqrt{2}\,\mathrm{e}^{-\pi/4}\cos\frac{1}{2}\pi = 0$. (The second form also shows $\mathrm{h}' \geq 0$ on $[-\frac{3}{4}\pi, \frac{1}{4}\pi]$, as given, and that $\mathrm{h}'(-\frac{3}{4}\pi) = 0$, which we shall need.)

The plan mirrors part (ii). On an interval where $y' \leq 0$ the equation is $y'' - 2y' + 2y = 0$; where $y' \geq 0$ it is $y'' + 2y' + 2y = 0$, which h satisfies.

(a) At $x = \frac{1}{4}\pi$ we have $y = \mathrm{h}(\frac{1}{4}\pi) = \frac{1}{\sqrt{2}}\mathrm{e}^{-\pi/4} > 0$ and $y' = 0$, so $y'' = -2y < 0$ and y' turns negative. We therefore want a solution of $y'' - 2y' + 2y = 0$, and part (iv) with $p = q = 2$ hands us one: $y = \mathrm{h}(c - x)$ works for $c - \frac{1}{4}\pi \leq x \leq c + \frac{3}{4}\pi$. Choosing $c = \frac{1}{2}\pi$ makes this interval exactly $[\frac{1}{4}\pi, \frac{5}{4}\pi]$, and

$$y = \mathrm{h}\left(\tfrac{1}{2}\pi - x\right) = \mathrm{e}^{x-\pi/2}\sin\left(\tfrac{1}{2}\pi - x\right) = \mathrm{e}^{x-\pi/2}\cos x.$$

? **Ask yourself** How do we know this is the right solution, and not just a solution?

It matches at the join: $\mathrm{h}(\frac{1}{2}\pi - \frac{1}{4}\pi) = \mathrm{h}(\frac{1}{4}\pi)$, and its derivative $-\mathrm{h}'(\frac{1}{2}\pi - x)$ equals $-\mathrm{h}'(\frac{1}{4}\pi) = 0$ at $x = \frac{1}{4}\pi$. So y and y' are continuous there. It is also self-consistent: for $x \in [\frac{1}{4}\pi, \frac{5}{4}\pi]$ we have $\frac{1}{2}\pi - x \in [-\frac{3}{4}\pi, \frac{1}{4}\pi]$, where $\mathrm{h}' \geq 0$, so $y' = -\mathrm{h}'(\frac{1}{2}\pi - x) \leq 0$ and $|y'| = -y'$ throughout. Since a linear equation with given y and y' at a point has only one solution, this is the continuation. So on $\frac{1}{4}\pi \leq x \leq \frac{5}{4}\pi$,

$$y = \mathrm{e}^{x-\pi/2}\cos x.$$

(b) At $x = \frac{5}{4}\pi$: $y = \mathrm{e}^{3\pi/4}\cos\frac{5}{4}\pi = -\frac{1}{\sqrt{2}}\mathrm{e}^{3\pi/4} < 0$, and $y' = -\mathrm{h}'(-\frac{3}{4}\pi) = 0$. Now $y'' = -2y > 0$, so y' turns positive and the equation reverts to $y'' + 2y' + 2y = 0$.

This equation has constant coefficients, so a translate of a solution is a solution. Since $\sin(x - 2\pi) = \sin x$, the function $\mathrm{h}(x - 2\pi) = \mathrm{e}^{2\pi - x}\sin x$ solves it, and on $[\frac{5}{4}\pi, \frac{9}{4}\pi]$ its argument $x - 2\pi$ runs over $[-\frac{3}{4}\pi, \frac{1}{4}\pi]$, where $\mathrm{h}' \geq 0$, so it is consistent with $y' \geq 0$. Check the join at $x = \frac{5}{4}\pi$:

$$\mathrm{e}^{2\pi - 5\pi/4}\sin\tfrac{5}{4}\pi = -\tfrac{1}{\sqrt{2}}\mathrm{e}^{3\pi/4}, \qquad \mathrm{h}'(\tfrac{5}{4}\pi - 2\pi) = \mathrm{h}'(-\tfrac{3}{4}\pi) = 0,$$

both matching. So on $\frac{5}{4}\pi \leq x \leq \frac{9}{4}\pi$,

$$y = \mathrm{e}^{2\pi - x}\sin x.$$

At $x = \frac{9}{4}\pi$ this is $e^{-\pi/4}\sin\frac{1}{4}\pi = h(\frac{1}{4}\pi)$ with zero derivative: the solution is back where it was at $\frac{1}{4}\pi$, so for $x \geq \frac{1}{4}\pi$ it is periodic with period 2π.

Checks

- The pieces join smoothly by construction, but it is worth a direct look: at $x = \frac{5}{4}\pi$ both (a) and (b) give $y = -\frac{1}{\sqrt{2}}e^{3\pi/4} \approx -7.4605$, with zero slope.
- Stepping the modulus equations forward numerically from $y(0) = 0$, $y'(0) = 1$ reproduces g_1, g_2 on $[0, 4]$ and the three pieces of (v) on $[0, \frac{9}{4}\pi]$ to within 10^{-9}.
- Differentiate (a) directly: $y' = e^{x-\pi/2}(\cos x - \sin x)$, $y'' = -2e^{x-\pi/2}\sin x$, and $y'' - 2y' + 2y = 0$ as required.

Common slips

- Keeping $|y'| = y'$ after the turning point: f itself is not the solution beyond $x = 1$.
- Forgetting to check that the new piece keeps y' of the assumed sign over the whole interval.
- In (iv), giving the interval as $[r, s]$; the reflection maps it to $[c - s, c - r]$.

Beyond STEP

Equations like these, whose right-hand side is smooth except across a switching surface (here $y' = 0$), are called piecewise-smooth or switching systems; the solution is built by gluing linear pieces, and existence and uniqueness still follow from the Picard–Lindelöf theorem because $|y'|$ is Lipschitz. The 2π-periodicity in (v) is something a linear damped equation could never produce: with $E = \frac{1}{2}y'^2 + y^2$ we get $dE/dx = -2y'|y'|$, so the "damping" term removes energy while y rises and puts it back while y falls. Boyce and DiPrima, *Elementary Differential Equations*, covers the linear theory and uniqueness.

WORKED SOLUTION

STEP III 2023, Question 9

Mechanics • Two-Particle String Motion

STEP III 2023, Question 9

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

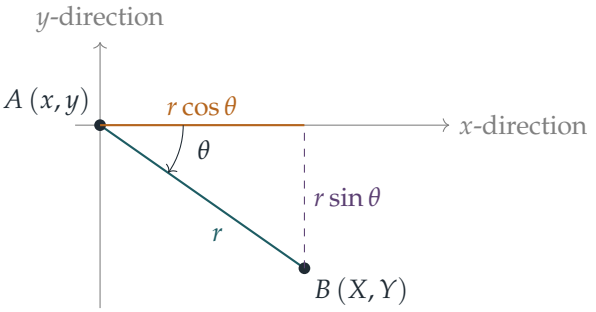
Describe the system by the position of A and the direction θ of the string. The tension acts only along the string, so each particle's acceleration has no component perpendicular to it. Comparing the two particles' perpendicular accelerations then forces $\ddot{\theta} = 0$: the string turns at a constant rate. After that, conservation of momentum finishes the job.

Solution

Throughout, the particles move on a smooth horizontal plane, the string is light and inextensible, and (as we may assume) it stays taut, so $AB = r$ at all times.

Part (i)

The vector $\overrightarrow{AB}$ has length r. Since θ is measured *clockwise* from the positive x-axis, the vector points below the x-direction when $0 < \theta < \frac{\pi}{2}$, as in the diagram.



Going from A to B we move $r\cos\theta$ in the x-direction and $r\sin\theta$ in the *negative* y-direction, so

$$X = x + r\cos\theta, \qquad Y = y - r\sin\theta.$$

The same expressions hold for every θ, because $(\cos\theta, -\sin\theta)$ is the unit vector at angle θ clockwise from the x-axis.

Part (ii)

Differentiating with respect to t (with r constant, and θ a function of t):

$$\dot{X} = \dot{x} - r\dot{\theta}\sin\theta, \qquad \dot{Y} = \dot{y} - r\dot{\theta}\cos\theta,$$
$$\ddot{X} = \ddot{x} - r\ddot{\theta}\sin\theta - r\dot{\theta}^2\cos\theta, \qquad \ddot{Y} = \ddot{y} - r\ddot{\theta}\cos\theta + r\dot{\theta}^2\sin\theta.$$

Part (iii)

The tension T pulls A towards B, along the unit vector $(\cos\theta, -\sin\theta)$, and pulls B towards A, along $(-\cos\theta, \sin\theta)$. By Newton's second law,

$$m\ddot{x} = T\cos\theta, \quad m\ddot{y} = -T\sin\theta, \qquad M\ddot{X} = -T\cos\theta, \quad M\ddot{Y} = T\sin\theta.$$

Hence

$$m(\ddot{x}\sin\theta + \ddot{y}\cos\theta) = T\cos\theta\sin\theta - T\sin\theta\cos\theta = 0,$$

and in exactly the same way $M(\ddot{X}\sin\theta + \ddot{Y}\cos\theta) = 0$. Dividing by m and M gives both required results. (Geometrically: $(\sin\theta, \cos\theta)$ is perpendicular to the string, and neither particle accelerates in that direction.)

> **Ask yourself** We have two equations saying "no perpendicular acceleration". How do they tell us anything about θ?

Subtract the first from the second and substitute the expressions from part (ii) for $\ddot{X} - \ddot{x}$ and $\ddot{Y} - \ddot{y}$:

$$\begin{aligned} 0 &= (\ddot{X} - \ddot{x})\sin\theta + (\ddot{Y} - \ddot{y})\cos\theta \\ &= \left(-r\ddot{\theta}\sin\theta - r\dot{\theta}^2\cos\theta\right)\sin\theta + \left(-r\ddot{\theta}\cos\theta + r\dot{\theta}^2\sin\theta\right)\cos\theta \\ &= -r\ddot{\theta}\left(\sin^2\theta + \cos^2\theta\right) = -r\ddot{\theta}. \end{aligned}$$

Since $r \neq 0$, $\ddot{\theta} = 0$, so $\dot{\theta}$ is constant.

To find the constant, use the initial conditions. At $t = 0$, B is at $(r, 0)$ relative to A, so $\theta(0) = 0$. Also $\dot{y}(0) = u$ and $\dot{Y}(0) = 0$, so the expression for $\dot{Y}$ in part (ii) gives

$$0 = u - r\dot{\theta}(0)\cos 0 \quad \Longrightarrow \quad \dot{\theta}(0) = \frac{u}{r}.$$

(The $\dot{X}$ equation is consistent: $0 = 0 - r\dot{\theta}(0)\sin 0$.) So $\dot{\theta} = u/r$ for all t, and integrating with $\theta(0) = 0$,

$$\theta = \frac{ut}{r},$$

as required.

Part (iv)

Adding the equations of motion from part (iii) in pairs,

$$m\ddot{x} + M\ddot{X} = T\cos\theta - T\cos\theta = 0, \qquad m\ddot{y} + M\ddot{Y} = -T\sin\theta + T\sin\theta = 0.$$

This is conservation of momentum: the tension is internal to the system. Integrating the second equation once, $m\dot{y} + M\dot{Y}$ is constant, equal to its initial value $mu + M \cdot 0 = mu$. Integrating again, with $y(0) = Y(0) = 0$,

$$my + MY = mut.$$

Part (v)

Substitute $Y = y - r\sin\theta$ from part (i) and $\theta = ut/r$ from part (iii):

$$my + M\Big(y - r\sin\frac{ut}{r}\Big) = mut \quad \Longrightarrow \quad y = \frac{1}{m+M}\Big(mut + Mr\sin\frac{ut}{r}\Big),$$

which is the required result.

Part (vi)

Differentiating the result of part (v),

$$\dot{y} = \frac{1}{m+M}\Big(mu + Mu\cos\frac{ut}{r}\Big) = \frac{u}{m+M}\Big(m + M\cos\frac{ut}{r}\Big).$$

At the time $t = \pi r/u$, when the string has turned through half a revolution, $\cos(ut/r) = -1$ and

$$\dot{y} = \frac{u(m-M)}{m+M},$$

which is negative when $M > m$ (as $u > 0$). So A's velocity has a negative y component at that moment, and by continuity on an interval of times around it.

Checks

- At $t = 0$ the formula in part (v) gives $\dot{y} = u(m+M)/(m+M) = u$, matching the initial velocity of A.
- The same method in the x-direction gives $mx + MX = Mr$, so $x = \dfrac{Mr}{m+M}(1 - \cos\theta)$. Together with part (v), the centre of mass moves in a straight line at speed $mu/(m+M)$ in the y-direction while both particles circle it at angular speed u/r, which is the familiar picture. The tension is then the constant $\dfrac{mM}{m+M}\dfrac{u^2}{r} > 0$, so the string really does stay taut.
- If $M \to \infty$, B stays fixed and part (v) gives $y = r\sin(ut/r)$: A moves in a circle of radius r about B, as it should.

- With $m = 1$, $M = 3$, $u = r = 1$, a computer simulation of the two particles joined by a very stiff spring gave $y \approx 0.7854$ and $\dot{y} \approx -0.5000$ at $t = \pi$, matching $\frac{\pi}{4}$ and $\frac{1-3}{1+3} = -\frac{1}{2}$.

Common slips

- Sign errors from the clockwise convention: $Y = y - r\sin\theta$, so $\dot{Y}$ has $-r\dot{\theta}\cos\theta$, not $+$.
- Forgetting to justify the constant u/r: it comes from $\dot{Y}(0) = 0$, not from A's speed alone.
- In part (vi), claiming $\dot{y} < 0$ "eventually" without exhibiting a time; $t = \pi r/u$ gives a definite one.

Beyond STEP

This is the two-body problem in its simplest form: the motion splits into uniform motion of the centre of mass plus motion relative to it, governed by the reduced mass $mM/(m+M)$. The same splitting underlies planetary orbits and the vibrations of diatomic molecules. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops centre-of-mass coordinates and reduced mass carefully.

WORKED SOLUTION

STEP III 2023, Question 10

Mechanics • Equilibrium of a Beam

STEP III 2023, Question 10

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

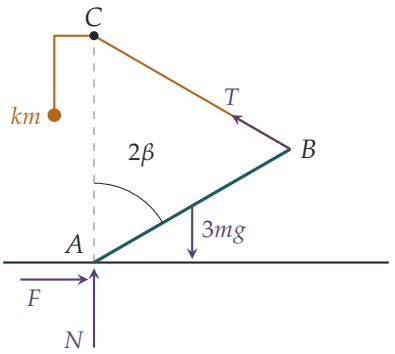
www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

The pulley sits exactly a beam-length above A, so A, B and C form an isosceles triangle with $AB = AC = 2h$. That geometry hands us the direction of the string and its perpendicular distance from A for free. Then everything follows from the usual recipe for a rod in equilibrium: take moments about A (which removes both unknown contact forces), resolve, and impose $F \leq \mu N$.

Solution

We model the beam as a uniform rod, the particles as point masses, and the string as light and inextensible over a smooth pulley, so the tension is the same throughout. Since the hanging particle is in equilibrium, the tension is $T = kmg$.



Put A at the origin with the x-axis horizontal (towards the side of B) and the y-axis vertically up. Then $C = (0, 2h)$ and $B = 2h(\sin 2\beta, \cos 2\beta)$.

Direction of the string. The vector from B to C is

$$2h\big(-\sin 2\beta,\ 1-\cos 2\beta\big) = 4h\sin\beta\,(-\cos\beta,\ \sin\beta),$$

using $1 - \cos 2\beta = 2\sin^2\beta$. So the tension on the beam has horizontal component $T\cos\beta$ in the $-x$ direction (back towards the line AC) and vertical component $T\sin\beta$ upwards. Geometrically: in the isosceles triangle ABC the base angles are $90^\circ - \beta$, so BC makes angle β with the horizontal.

Moment arm. The perpendicular from A to BC bisects the apex angle 2β of the isosceles triangle, so its length is $2h\cos\beta$.

Part (i)

Moments about A. The weight $3mg$ acts at the midpoint of AB, a horizontal distance $h\sin 2\beta$ from A. Balancing it against the tension:

$$kmg \cdot 2h\cos\beta = 3mg \cdot h\sin 2\beta = 6mgh\sin\beta\cos\beta.$$

Since $0 < 2\beta < 180^\circ$, $\cos\beta \neq 0$, and we may divide by it:

$$\boxed{k = 3\sin\beta.}$$

(This is consistent with $k < 3$.)

Resolving. Vertically, $N + T\sin\beta = 3mg$, so

$$N = 3mg - 3mg\sin^2\beta = 3mg\cos^2\beta.$$

Horizontally, friction must balance the horizontal pull of the string, so it acts in the $+x$ direction with

$$F = T\cos\beta = 3mg\sin\beta\cos\beta.$$

Hence $F/N = \tan\beta$.

Friction condition. Equilibrium requires $F \leq \mu N$, that is $\tan\beta \leq \mu$. Now

$$k^2 = 9\sin^2\beta = \frac{9\tan^2\beta}{1+\tan^2\beta}.$$

The function $t \mapsto \dfrac{t}{1+t} = 1 - \dfrac{1}{1+t}$ is increasing for $t \geq 0$, and $0 \leq \tan^2\beta \leq \mu^2$, so

$$k^2 \leq \frac{9\mu^2}{\mu^2+1},$$

as required.

Part (ii)

Now $T = 2mg$, and the extra particle of mass m sits a horizontal distance $xh\sin 2\beta$ from A. The geometry of the string is unchanged.

Moments about A:

$$2mg \cdot 2h\cos\beta = 3mg\,h\sin 2\beta + mg\,xh\sin 2\beta = 2mgh(3+x)\sin\beta\cos\beta,$$

and dividing by $2mgh\cos\beta \neq 0$ gives

$$\sin\beta = \frac{2}{3+x}.$$

For $0 \leq x \leq 2$ this lies between $\frac{2}{5}$ and $\frac{2}{3}$, so such a β exists.

Resolving:

$$N = 4mg - 2mg\sin\beta, \qquad F = 2mg\cos\beta,$$

so

$$\frac{F^2}{N^2} = \frac{4\cos^2\beta}{(4-2\sin\beta)^2} = \frac{\cos^2\beta}{(2-\sin\beta)^2}.$$

Substituting $\sin\beta = 2/(3+x)$:

$$\cos^2\beta = 1 - \frac{4}{(3+x)^2} = \frac{(3+x-2)(3+x+2)}{(3+x)^2} = \frac{(x+1)(x+5)}{(3+x)^2},$$
$$2-\sin\beta = \frac{2(3+x)-2}{3+x} = \frac{2(x+2)}{3+x}.$$

The factors of $(3+x)^2$ cancel, leaving

$$\frac{F^2}{N^2} = \frac{(x+1)(x+5)}{4(x+2)^2} = \frac{x^2+6x+5}{4(x+2)^2},$$

which is the required result.

Ask yourself How do we turn "whatever the value of x" into a condition on μ?

For each x we need $F \leq \mu N$, i.e. $\mu \geq F/N$. To cover every $x \in [0,2]$, μ must be at least the *largest* value of F/N over that interval. So we want the maximum of F^2/N^2.

Following the hint,

$$\frac{1}{3} - \frac{F^2}{N^2} = \frac{4(x+2)^2 - 3(x^2+6x+5)}{12(x+2)^2} = \frac{x^2-2x+1}{12(x+2)^2} = \frac{(x-1)^2}{12(x+2)^2} \geq 0,$$

with equality exactly when $x = 1$, which is in the allowed range. So $F^2/N^2 \leq \frac{1}{3}$ for all $x \in [0,2]$, and the bound is attained at $x = 1$. Hence F/N has maximum value $1/\sqrt{3}$, and the beam can be in equilibrium for every x if and only if $\mu \geq 1/\sqrt{3}$. The minimum value is

$$\boxed{\mu = \frac{1}{\sqrt{3}}}.$$

Checks

- In part (i), try the beam lying horizontal, $2\beta = 90^\circ$. Then $B = (2h, 0)$, the string BC is at 45° and its distance from A is $\sqrt{2}\,h$. Moments give $T\sqrt{2}\,h = 3mgh$, so $T = 3mg/\sqrt{2}$, matching $k = 3\sin 45^\circ$. Also $N = F = \frac{3}{2}mg$, so $F/N = 1 = \tan 45^\circ$.

- At $x = 0$ in part (ii) the formula gives $5/16$, and directly $\sin\beta = \frac{2}{3}$, $\cos^2\beta = \frac{5}{9}$, $(2 - \frac{2}{3})^2 = \frac{16}{9}$, giving $\frac{5}{16}$ again.
- At $x = 1$: $\sin\beta = \frac{1}{2}$, so $\beta = 30^\circ$ and the beam is at 60° to the vertical; $F/N = \cos 30^\circ / (2 - \frac{1}{2}) = 1/\sqrt{3} \approx 0.5774$. A computer check of the moments and resolved forces for x across $[0, 2]$ confirmed the formula and this maximum.

Common slips

- Forgetting that the string pulls along BC, not vertically: the pulley is at C, so the tension has a horizontal component, and that is what friction has to resist.
- Taking the minimum of F/N (or its value at an end point) instead of its maximum over $0 \le x \le 2$. The worst case is in the interior, at $x = 1$.
- Mixing up the moment arms: the weight's arm is horizontal ($h\sin 2\beta$), while the tension's arm is the perpendicular distance to BC ($2h\cos\beta$).

Beyond STEP

Taking moments and resolving are the two halves of the equilibrium conditions for a rigid body: zero total force and zero total torque. At university these become vector statements, $\sum \mathbf{F} = \mathbf{0}$ and $\sum \mathbf{r} \times \mathbf{F} = \mathbf{0}$, and the "worst case over a parameter" reasoning of part (ii) is a small taste of how engineers design for safety margins. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops statics and torque carefully in this vector language.

WORKED SOLUTION

STEP III 2023, Question 11

Probability and Statistics • Conditional Expectation

STEP III 2023, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

The opening series splits into two familiar exponential series once we write $k+1 = k+1$ and cancel the k against $k!$. In part (i) the distribution of D involves a sum over every value of Y that could have produced $D = d$; swapping the order of the double sum makes the inner sum the easy $1+2+\cdots+k$, and the factor $\frac{1}{2}(k+1)$ that appears is exactly what the opening series handles. Part (ii) is then a matter of conditioning on the die.

Solution

The series

For $k \geq 1$ we have $\dfrac{k}{k!} = \dfrac{1}{(k-1)!}$, so, splitting the numerator,

$$\begin{aligned}\sum_{k=1}^{\infty} \frac{k+1}{k!}x^k &= \sum_{k=1}^{\infty} \frac{x^k}{(k-1)!} + \sum_{k=1}^{\infty} \frac{x^k}{k!} \\ &= x\sum_{j=0}^{\infty} \frac{x^j}{j!} + \left(\sum_{k=0}^{\infty} \frac{x^k}{k!} - 1\right) \\ &= xe^x + e^x - 1 = (x+1)e^x - 1,\end{aligned}$$

using $j = k-1$ in the first sum. Both series converge for every real x, so splitting is legitimate. This is the required identity.

Part (i)(a)

$D = 0$ exactly when $Y = 0$ (if $Y = k \geq 1$, the die shows at least 1), so

$$\mathrm{P}(D = 0) = \mathrm{P}(Y = 0) = e^{-n}.$$

Part (i)(b)

Fix $d \geq 1$. The die can show d only if it has at least d sides, that is, only if $Y = k$ with $k \geq d$; and given $Y = k \geq d$, the fair die shows d with probability $\frac{1}{k}$. By the law of total probability,

$$\mathrm{P}(D = d) = \sum_{k=d}^{\infty} \mathrm{P}(D = d \mid Y = k)\,\mathrm{P}(Y = k) = \sum_{k=d}^{\infty} \frac{1}{k} \cdot \frac{n^k}{k!}\mathrm{e}^{-n}.$$

By definition $\mathrm{E}(D) = \sum_{d=0}^{\infty} d\,\mathrm{P}(D = d)$, and the $d = 0$ term is zero, so

$$\mathrm{E}(D) = \sum_{d=1}^{\infty} \left[d \sum_{k=d}^{\infty} \left(\frac{1}{k} \cdot \frac{n^k}{k!}\mathrm{e}^{-n} \right) \right],$$

as required.

Ask yourself How do we turn a sum over d outside and $k \geq d$ inside into a sum over k outside?

The double sum runs over all pairs (d, k) of integers with $1 \leq d \leq k$. Every term $\frac{d}{k} \cdot \frac{n^k}{k!}\mathrm{e}^{-n}$ is non-negative, so we may add the terms in any order without changing the total. Grouping by k instead of by d: for each $k \geq 1$, d runs from 1 to k. Hence

$$\mathrm{E}(D) = \sum_{k=1}^{\infty} \left(\frac{1}{k} \cdot \frac{n^k}{k!}\mathrm{e}^{-n} \sum_{d=1}^{k} d \right),$$

which is the second required form.

Part (i)(c)

Using $\sum_{d=1}^{k} d = \frac{1}{2}k(k+1)$, the factor $\frac{1}{k}$ cancels:

$$\begin{aligned} \mathrm{E}(D) &= \sum_{k=1}^{\infty} \frac{1}{k} \cdot \frac{n^k}{k!}\mathrm{e}^{-n} \cdot \frac{k(k+1)}{2} = \frac{\mathrm{e}^{-n}}{2} \sum_{k=1}^{\infty} \frac{k+1}{k!} n^k \\ &= \frac{\mathrm{e}^{-n}}{2}\left((n+1)\mathrm{e}^{n} - 1 \right) = \tfrac{1}{2}\left(n + 1 - \mathrm{e}^{-n} \right), \end{aligned}$$

by the opening identity with $x = n$. This is the required result.

Part (ii)(a)

Let K be the number shown on the n-sided die, so $\mathrm{P}(K = k) = \frac{1}{n}$ for $k = 1, \ldots, n$, and given $K = k$ we have $Z = X_k$, which is Poisson with mean k. So $\mathrm{P}(Z = 0 \mid K = k) = \mathrm{e}^{-k}$, and by the law of total probability

$$\mathrm{P}(Z = 0) = \frac{1}{n} \sum_{k=1}^{n} \mathrm{e}^{-k}.$$

This is a geometric series with first term e^{-1} and ratio e^{-1}:

$$\mathrm{P}(Z=0) = \frac{1}{n}\cdot\frac{\mathrm{e}^{-1}\left(1-\mathrm{e}^{-n}\right)}{1-\mathrm{e}^{-1}} = \frac{1-\mathrm{e}^{-n}}{n(\mathrm{e}-1)},$$

multiplying top and bottom by e.

Part (ii)(b)

Conditioning on the die again, $\mathrm{E}(Z \mid K=k) = \mathrm{E}(X_k) = k$, so

$$\mathrm{E}(Z) = \sum_{k=1}^{n} \mathrm{E}(Z \mid K=k)\,\mathrm{P}(K=k) = \frac{1}{n}\sum_{k=1}^{n} k = \frac{1}{n}\cdot\frac{n(n+1)}{2} = \frac{n+1}{2}.$$

Comparing with part (i)(c),

$$\mathrm{E}(Z) - \mathrm{E}(D) = \frac{n+1}{2} - \frac{n+1-\mathrm{e}^{-n}}{2} = \tfrac{1}{2}\mathrm{e}^{-n} > 0,$$

so $\mathrm{E}(Z) > \mathrm{E}(D)$, as required.

Interestingly, without the zero case the two would agree: a fair die with a Poisson number Y of faces shows $\frac{1}{2}(Y+1)$ on average, which has mean $\frac{1}{2}(n+1)$. The only loss is that $Y=0$ gives $D=0$ rather than the $\frac{1}{2}$ that the formula $\frac{1}{2}(Y+1)$ would give, and that happens with probability e^{-n}, costing exactly $\frac{1}{2}\mathrm{e}^{-n}$.

Checks

- With $x=0$ the opening identity reads $0 = 1\cdot 1 - 1$, and with $x=1$ it gives $\sum_{k\geq 1}\frac{k+1}{k!} = 2\mathrm{e}-1 \approx 4.4366$, which the partial sums confirm.
- For $n=3$: $\mathrm{E}(D) = \frac{1}{2}(4-\mathrm{e}^{-3}) \approx 1.9751$, $\mathrm{E}(Z) = 2$ and $\mathrm{P}(Z=0) \approx 0.1843$. A computer simulation of 200 000 runs of each experiment gave 1.970, 2.002 and 0.185.
- For $n=1$, the die in part (ii) always shows 1, so $\mathrm{P}(Z=0) = \mathrm{e}^{-1}$; the formula gives $\frac{1-\mathrm{e}^{-1}}{\mathrm{e}-1} = \mathrm{e}^{-1}$.

Common slips

- Starting the inner sum in (i)(b) at $k=1$ rather than $k=d$: a k-sided die cannot show a number bigger than k.
- Forgetting the -1 in the opening identity, which comes from the missing $k=0$ term of e^{x}; it is the source of the e^{-n} in (i)(c).

Beyond STEP

Both parts are instances of the tower law $\mathrm{E}(Z) = \mathrm{E}\big(\mathrm{E}(Z \mid K)\big)$: in part (i) it gives $\mathrm{E}(D) = \mathrm{E}\big(\frac{1}{2}(Y+1)\mathbf{1}_{Y\geq 1}\big)$ in one line, and the swap of summation order we did by

hand is exactly what that law hides. The freedom to rearrange a double series of non-negative terms is a theorem (Tonelli's, in its general form) met in first-year analysis. Grimmett and Welsh, *Probability: An Introduction*, treats conditional expectation and the compound distributions built from it.

WORKED SOLUTION

STEP III 2023, Question 12

Probability and Statistics • Combinatorial Probability

STEP III 2023, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Although the two socks of a pair look alike, the selection is a uniformly random set of $2k$ of the $2n$ *physical* socks, so we label every sock and count sets. Then everything is a matter of building a selection in stages: first choose which pairs appear complete, then choose which other pairs contribute a single sock, then choose which sock of each such pair. Part (iii) is the familiar trick $r\binom{n}{r} = n\binom{n-1}{r-1}$, which turns a weighted sum into a sum of probabilities.

Solution

Label the socks so that all $2n$ are distinct. There are $\binom{2n}{2k}$ equally likely selections, and every probability below is (number of favourable selections)$/\binom{2n}{2k}$.

Part (i)

A selection with no complete pair contains at most one sock from each pair, so its $2k$ socks come from $2k$ different pairs. To build one, choose the $2k$ pairs that are represented, in $\binom{n}{2k}$ ways (possible since $2k \le n$), and then choose one of the two socks from each, in 2^{2k} ways. Each such choice gives a different set, and every set with no pair arises this way, so

$$\mathrm{P}(\text{no pair}) = \frac{\binom{n}{2k}\,2^{2k}}{\binom{2n}{2k}}.$$

Part (ii)

Fix r with $0 \le r \le k$. A selection with exactly r complete pairs consists of

- r pairs taken whole ($2r$ socks), and
- $2k - 2r = 2(k-r)$ further socks, all from different pairs (otherwise there would be

more than r pairs), and none from the r pairs already used.

So we choose the r complete pairs in $\binom{n}{r}$ ways; from the remaining $n-r$ pairs choose the $2(k-r)$ pairs that contribute one sock, in $\binom{n-r}{2(k-r)}$ ways; and pick one sock from each of these, in $2^{2(k-r)}$ ways. Again each set with exactly r pairs is produced exactly once, so

$$\mathrm{P}(X_{n,k}=r)=\frac{\binom{n}{r}\binom{n-r}{2(k-r)}\,2^{2(k-r)}}{\binom{2n}{2k}},$$

as required. (When $r=0$ this is part (i) again.)

Part (iii)

Let $1\le r\le k$. Since

$$r\binom{n}{r}=\frac{r\cdot n!}{r!\,(n-r)!}=n\cdot\frac{(n-1)!}{(r-1)!\,(n-r)!}=n\binom{n-1}{r-1},$$

part (ii) gives

$$r\,\mathrm{P}(X_{n,k}=r)=\frac{n\binom{n-1}{r-1}\binom{n-r}{2(k-r)}\,2^{2(k-r)}}{\binom{2n}{2k}}.$$

Now apply part (ii) with $n-1$ pairs, $2(k-1)$ socks drawn and $r-1$ pairs found. This is allowed, because $2(k-1)\le n-1$ follows from $2k\le n$, and $0\le r-1\le k-1$. Since $(n-1)-(r-1)=n-r$ and $(k-1)-(r-1)=k-r$,

$$\mathrm{P}(X_{n-1,k-1}=r-1)=\frac{\binom{n-1}{r-1}\binom{n-r}{2(k-r)}\,2^{2(k-r)}}{\binom{2n-2}{2k-2}}.$$

The numerators match apart from the factor n, so

$$\begin{aligned}\frac{r\,\mathrm{P}(X_{n,k}=r)}{\mathrm{P}(X_{n-1,k-1}=r-1)}&=\frac{n\binom{2n-2}{2k-2}}{\binom{2n}{2k}}=n\cdot\frac{(2n-2)!}{(2k-2)!\,(2n-2k)!}\cdot\frac{(2k)!\,(2n-2k)!}{(2n)!}\\&=n\cdot\frac{2k(2k-1)}{2n(2n-1)}=\frac{k(2k-1)}{2n-1}.\end{aligned}$$

(Dividing was safe: $n-r\ge 2k-r\ge 2(k-r)$, so every binomial coefficient here is positive.) Multiplying up gives the required result.

Ask yourself How does this identity help with $\mathrm{E}(X_{n,k})$?

The expectation is exactly a sum of the left-hand sides, and the right-hand sides add up to a whole probability distribution. The $r=0$ term of the expectation is zero, so

$$\begin{aligned}\mathrm{E}(X_{n,k})=\sum_{r=1}^{k}r\,\mathrm{P}(X_{n,k}=r)&=\frac{k(2k-1)}{2n-1}\sum_{r=1}^{k}\mathrm{P}(X_{n-1,k-1}=r-1)\\&=\frac{k(2k-1)}{2n-1}\sum_{s=0}^{k-1}\mathrm{P}(X_{n-1,k-1}=s).\end{aligned}$$

The variable $X_{n-1,k-1}$ counts pairs among $2(k-1)$ socks, so it takes only the values $0, 1, \ldots, k-1$, and the last sum is 1. Hence

$$\mathrm{E}(X_{n,k}) = \frac{k(2k-1)}{2n-1}.$$

Checks

- With $k = 1$ (two socks), the answer is $\frac{1}{2n-1}$: whichever sock comes first, exactly one of the remaining $2n-1$ socks is its partner.
- Indicators give the mean directly: pair i is complete with probability $\binom{2n-2}{2k-2}/\binom{2n}{2k} = \frac{2k(2k-1)}{2n(2n-1)}$, and multiplying by n pairs gives $\frac{k(2k-1)}{2n-1}$ again.
- For $n = 6$, $k = 3$: part (i) gives $\frac{64}{924} = \frac{16}{231}$ and the mean is $\frac{15}{11}$. A computer enumeration of all selections for every $n \leq 8$ confirmed the distribution in (ii), the identity in (iii) and the mean.

Common slips

- Treating the socks of a pair as truly identical and counting "colour patterns": those are not equally likely, so the counting must be over labelled socks.
- Forgetting the factor $2^{2(k-r)}$ for which sock of each odd pair is taken.
- In (iii), not noticing that $\sum_s \mathrm{P}(X_{n-1,k-1} = s) = 1$, and trying to evaluate the sum by brute force.

Beyond STEP

The second check is linearity of expectation with indicator variables, which finds a mean without ever knowing the distribution; it works even though the events "pair i is complete" are dependent. The identity in (iii) is a size-biasing relation: weighting the distribution by r gives a shifted copy of a smaller problem, a device that also yields the means of the binomial and hypergeometric distributions. Blitzstein and Hwang, *Introduction to Probability*, uses indicators and story proofs like these throughout.

Beyond STEP

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